[Research + Teaching + Practice Portfolio]

Xiwei Shen, Ph.D., CE

My Research and Teaching Philosophy:

- **Research Philosophy:** Develop evidence-based landscape designs that integrate art, science, and technology
- **Teaching Philosophy:** Encourage students to actively engage in the design process, fostering diverse approaches and perspectives in their work

Part 01:

Research Philosophy: Develop Data-Driven, Evidence-Based Strategies for Landscape Design

Key Concept:

“Technologies from Allied Disciplines”

+

“Experiment and Interactive Design

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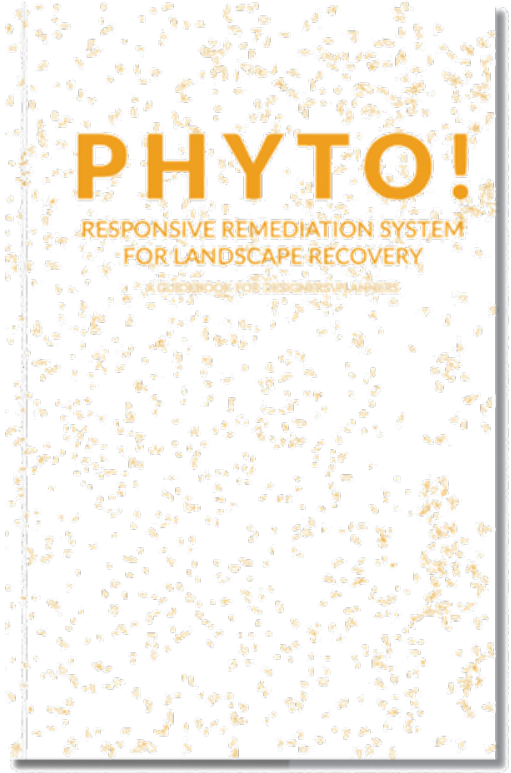
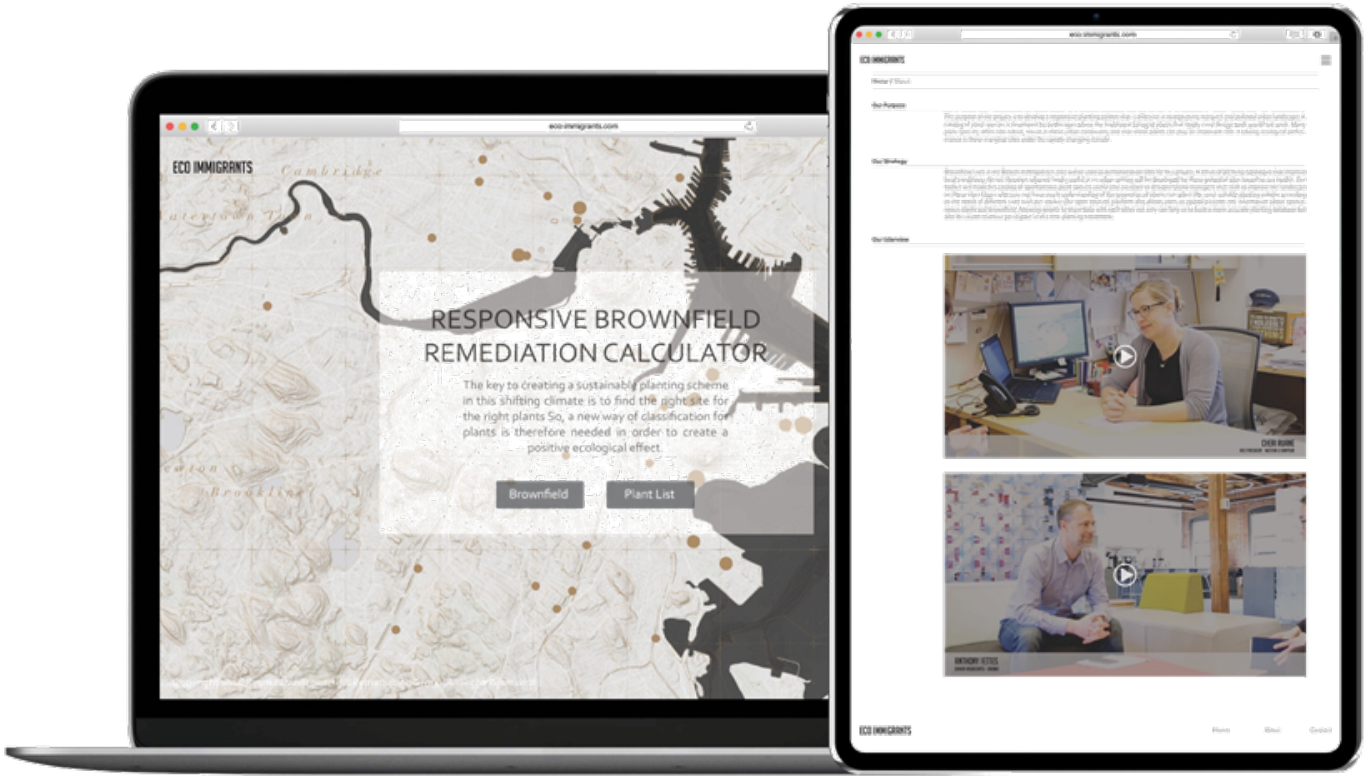
“Aim for Multiple Groups of Users”

Harvard Penny White Research Funded Project, $8,000

Advancing Environmental Design with Phytoremediation of Brownfield Soils Using Spontaneous Invasive Plants



FOR LANDSCAPE DESIGNERS / PLANNERS

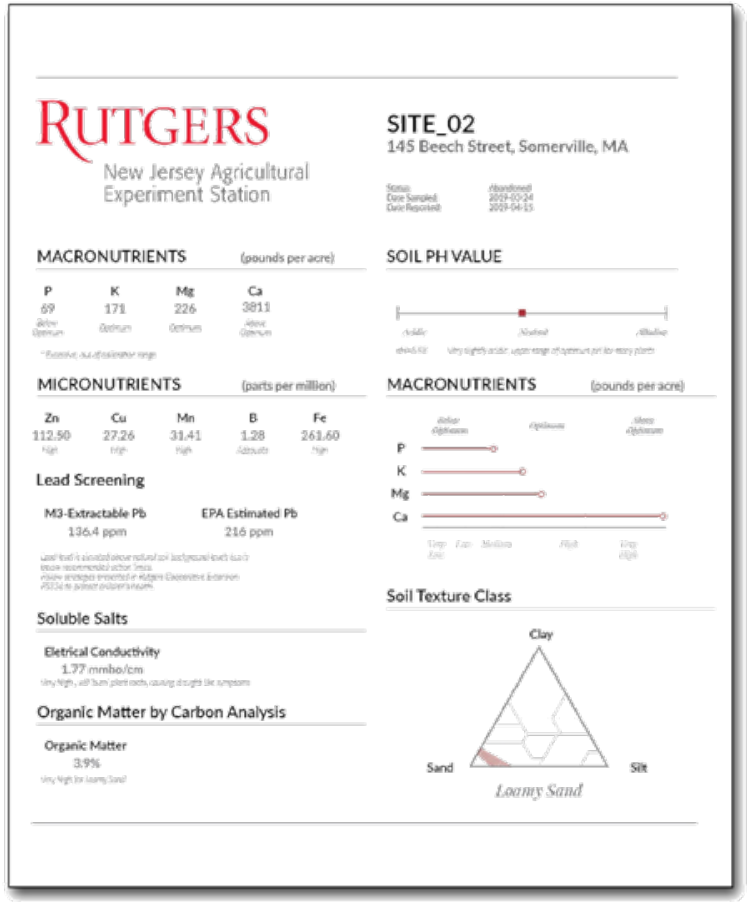


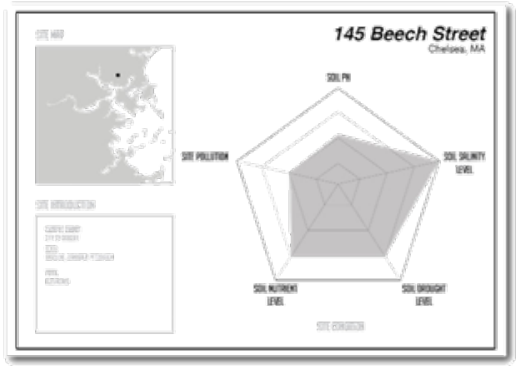
FOR BROWNFIELD LAND OWNER

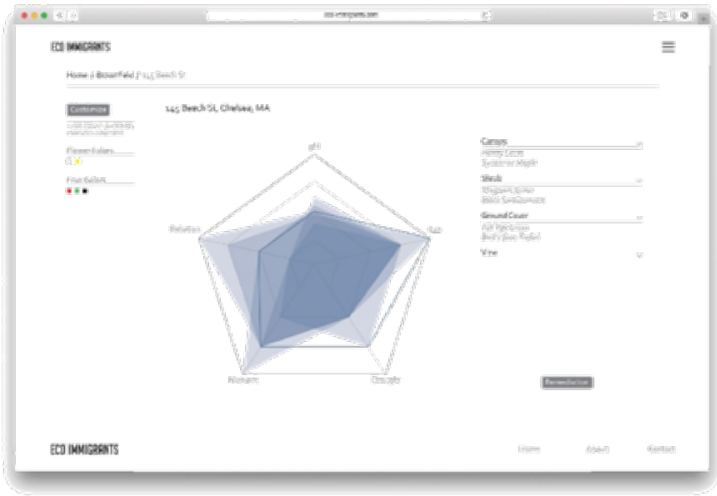
The goal of this research is to explore a responsive toolkits for designers to use and choose the appropriate plant species according to different conditions of the brownfield sites. The project was collaborating ad funded by Harvard GSD. The initial deliverable is a guidebook, a web-site based app. For the initiative project, the data base of plants were developed based on secondary data which from Dr. Peter De Tredici's book, who is a famous plant scholar from MIT. Later, the research has been further explored a more scientific method to examine the remediation ability of invasive plants.

Overview

STEP 1 : SOIL SAMPLING







STEP 2 : SOIL REPORT

STEP 3: SOIL DATA QUANTIFICATION
STEP 4: DATA OVERLAY / GENERATE PLANT LIST

The research use a site in Boston as an example. The concept of the research is to overlay the ph, soil salinity, soil drought, soil nutrient, soil metal pollution and the corresponding properties of local invasive plants. To identify the detail of the soil, the team collected the soil sample and send to Rutgers University's soil lab. Finally, the data will be quantified and code into the Python to be visualized.

Plant Database

	Plant Species	Common name	Type	Life Form	Height	Flower Color	Fruit Color	pH	Salt Tolerance	Drought Tolerance	Poor Soil Nutrition Tolerance	Pollution Tolerance	Soil Texture	Erosion Control	Wildlife Habitat/ Food	Metal Absorption	Soil Improvement
2	Acer negundo	Ash-Leaved Maple,Boxelder	Woody dicot	Canopy	60 ft	X	Brown	5 to 7.8	3	High	2	1	Clay Loam Sand	Y	Y	N	Y
3	Acer saccharinum	Silver Maple		Canopy	100 ft	Red/ Yellow	Brown	4 to 7.3	2	Low	1	3	Sand Silt Loam	Y	Y	Y	N
4	Alnus glutinosa	Common Alder		Canopy	35-60 ft	X	Brown	4.4 to 7.2	2	Medium	3	1	Clay Loam Sand	Y	Y	N	Y
5	Amorpha fruticosa	False Indigo-Bush		Shurb	12 ft	X	Brown	5 to 8.5	2	Medium	4	2	Loam Sand	Y	N	Y	Y
6	Betula nigra	River Birch		Canopy	30-60 ft	X	X	3 to 6.5	1	Low	1	1	Clay Loam Sand	Y	N	N	N
7	Betula populifolia	Gray Birch		Canopy	35 ft	X	X	3.5 to 6.5	3	Medium	4	1	Clay Loam Sand	Y	N	Y	N
8	Catalpa speciosa	Northern Catalpa		Canopy	80 ft	White	Brown	5.5 to 7.2	2	Medium	1	2	Loam Sand	Y	N	N	Y
9	Celtis occidentalis	Common Hackberry		Canopy	60 ft	X	X	6 to 7.8	2	High	4	2	Clay Loam Sand	Y	Y	N	N
10	Elaeagnus umbellata	Autumn-Olive		Shurb	20 ft	Yellow	Red	5 to 7.5	1	High	4	1	Clay Loam Sand	Y	Y	Y	Y
11	Gleditsia triacanthos	Honey-Locust		Canopy	70-90 ft	X	Green	4.8 to 8	3	High	3	3	Clay Loam	Y	Y	N	N
12	Malus species	Apple Tree		Canopy	15-40 ft	White/Pink	Red	5 to 7.5	1	Medium	2	1	Loam Sand	Y	Y	Y	N
13	Morus alba	White Mulberry		Canopy	30-60 ft	X	Purple	5 to 7	3	Medium	2	3	Clay Loam Sand	Y	Y	Y	N
14	Parthenocissus quinquefolia	Virginia-Creeper,Woodbine		Vine	60 ft	X	Blue	5 to 8	3	Medium	2	4	Clay Loam	N	Y	Y	N
15	Phellodendron amurense	Amur Corktree		Canopy	60 ft	Yellow	Black	5.0 to 8.2	2	Medium	1	2	Clay Loam Sand	Y	Y	N	N
16	Populus deltoides	Eastern Cottonwood,Necklace Poplar		Canopy	50-80 ft	X	White	4.6 to 6.5	3	Medium	1	1	Clay Loam Sand	Y	Y	Y	N
17	Populus tremuloides	Quaking Aspen,Quaking Poplar		Canopy	50 ft	X	White	4.3 to 9	3	Low	2	1	Clay Loam Sand	Y	Y	N	Y
18	Prunus serotina	Black Cherry		Canopy	40-75 ft	White	Black	4 to 7.5	3	Medium	1	2	Loam Sand	Y	Y	Y	N
19	Quercus palustris	Pin Oak		Canopy	60-70 ft	X	X	4.5 to 6.5	2	Low	1	2	Clay Loam	Y	Y	Y	N
20	Quercus rubra	Northern Red Oak		Canopy	60-90 ft	X	Brown	4.3 to 7.3	3	Low	1	2	Clay Loam Sand	Y	Y	Y	N
21	Rhamnus cathartica	European Buckthorn		Shrub/Canopy	10-25 ft	X	Black	6 to 8	2	Medium	1	1	Clay Loam Sand	Y	Y	N	N
22	Rhus typhina	Staghorn Sumac		Shurb	30 ft	Green	Red	4.5 to 7.5	3	Medium	3	3	Loam Silt	Y	Y	N	N
23	Robinia pseudoacacia	Black Locust		Canopy	30-70 ft	White	Black	4.6 to 8.2	3	High	4	3	Clay Loam Sand	Y	Y	Y	Y
24	Rubus allegheniensis	Common Blackberry		Shurb	3-8 ft	White	Black	4.6 to 7.5	1	High	1	2	Clay Loam	Y	Y	Y	N
25	Rubus flagellaris Willd	Northern Blackberry,Northern Dewberry		Vine	1 ft	White	Black	5 to 7	1	low	2	2	Clay Loam Sand	Y	Y	N	N
26	Rubus occidentalis	Black Raspberry		Shurb	4-6 ft	White	Purple	5.2 to 7.5	1	Medium	1	2	Clay Loam	Y	Y	N	N
27	Salix species	Willow		Shrub	10-20 ft	X	White	4.5 to 7.8	3	Low	2	4	Clay Loam Sand	Y	Y	Y	N
28	Solanum dulcamara	Climbing Nightshade		Vine	10 ft	Purple	Red	4.8 to 5.2	3	Low	1	1	Clay Loam	N	Y	N	N
29	Vitis riparia	River Grape		Vine	60 ft	X	X	6.1 to 8.5	1	Medium	1	1	Loam Sand	Y	Y	N	N
30	Achillea millifolium	Common Yarrow	Herbaceous dicots	Ground Cover	3 ft	White	X	6 to 8	2	Medium	3	2	Loam	Y	N	Y	N
31	Amaranthus retroflexus	Red-Rooted Amaranth		Ground Cover	6 ft	X	Black	4.2 to 9.1	2	High	1	1	Clay Loam	N	Y	Y	N
32	Ambrosia artemisiifolia	Common Ragweed		Ground Cover	4 ft	X	X	5 to 7.4	3	Medium	2	2	Clay Sand	N	Y	N	N
33	Asclepias syriaca	Common Milkweed		Ground Cover	3-4 ft	Tan	White	5.6 to 7.5	3	High	1	1	Clay Loam Sand	N	Y	N	N
34	Bidens frondosa	Devil'S Beggar-Ticks		Ground Cover	5 ft	X	Brown	5.2 to 7.2	1	Low	1	3	Loam Sand	Y	N	N	N
35	Cichorium intybus	Chicory		Ground Cover	5 ft	Blue	X	6 to 7.5	1	Medium	2	1	Clay Loam Sand	Y	Y	Y	Y
36	Cynanchum louisae	Black Swallowwort		Ground Cover	3-6 ft	Purple	X	4.7 to 8	4	Medium	3	2	Clay Loam Sand	Y	N	N	N
37	Daucus carota	Wild Carrot		Ground Cover	3-4 ft	White	X	4.2 to 8.7	2	Low	1	1	Clay Loam Sand	N	Y	Y	N
38	Hieracium sabaudum	Savoy Hawkweed		Ground Cover	3 ft	Yellow	White	3.1 to 6.5	2	Medium	2	2	Loam Sand	Y	Y	Y	N
39	Impatiens capensis	Jewelweed,Spotted Touch-Me-Not		Ground Cover	6 ft	X	X	6.4 to 7.4	1	Low	2	3	Clay Loam Sand	Y	Y	Y	N
40	Lactuca serriola	Prickly Lettuce		Ground Cover	6 ft	Yellow	Brown	4 to 7	2	Medium	2	2	Laom Sand	N	N	Y	N
41	Leontodon autumnalis	Fall-Dandelion		Ground Cover	2 ft	Yellow	White	5.5 to 7.5	4	Medium	1	2	Clay Loam Sand	N	Y	Y	N
42	Leucanthemum vulgare	Ox-Eye Daisy		Ground Cover	2 ft	White	Brown	5.2 to 7	1	Medium	1	1	Loam Sand	Y	Y	N	Y

Base on the work from Dr. Peter Del Tredici. The research extract the key information of aesthetic elements such like flower color. Fruit color, Height, etc. And some ecological performance such as erosion tolerance and wildlife habitat. Also, the salt tolerance, drought tolerance, soil nutrition tolerance, heavy metal pollution tolerance. These data will be used to not only remediate the site but also visualize the aesthetics factors to users. However, to explore the ecological properties of the invasive plants that out of the list, there needs to be a systematic assessment methods.

Research Method

Contents lists available at ScienceDirect

Science of the Total Environment

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Advancing environmental design with phytoremediation of brownfield soils using spontaneous invasive plants

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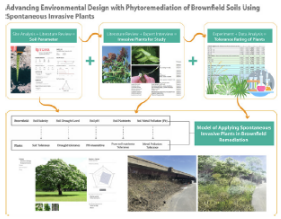
HIGHLIGHTS

- This research uses spontaneous plant species to assist brownfield phytoremediation.
- Soil conditions are measured in the lab and plant tolerances cataloged.
- A conceptual model is applied, selecting species for landscape design toolkits.
- The approach uses scientific data to enhance novel landscape restoration designs.

GRAPHICAL ABSTRACT



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ABSTRACT

Compared to traditional mechanical brownfield remediation strategies, phytoremediation as a sustainable and low-impact solution, yielding long-term soil chemical improvement. As a common part of many local plant communities, spontaneous invasive plants have advantages over native species in growth speed and resource-use efficiency and are many are effective on degrading or removing chemical soil pollutants. This research presents a methodology for using spontaneous invasive plants as the agent of phytoremediation for brownfield remediation is an innovative component of ecological restoration and design. This research explores s conceptual and applicable model of using spontaneous invasive plants in the phytoremediation of brownfield soil for environmental design practice. This research summarizes five parameters (Soil Drought Level, Soil Salinity, Soil Nutrients, Soil Metal Pollution, and Soil pH) and their classification standards. Based on the five parameters, a series of experiments were designed to examine 5 spontaneous invasive species' tolerance and performance to different soil conditions. Taking the research results as a data base, this research developed a conceptual model of selecting suitable spontaneous invasive plants for brownfield phytoremediation by overlaying the soil condition data and plants' tolerance data. Using a brownfield site in Boston metropolitan region as a case study, the research tested the feasibility and rationality of this model. The results propose a novel approach and materials for general environmental remediation of contaminated soil by involving spontaneous invasive plants. It also transforms the abstract phytoremediation knowledge and data to an applicable model which integrates and visualizes the requirements of scientific plant selection, design aesthetic, and ecosystem factors to help the environmental design process in brownfield remediation.

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but that can be anticipated to become invasive in poorly managed ecosystems inside the Commonwealth are referred to as "potentially invasive plants" (*MIPAG - Massachusetts Invasive Plant Advisory Group*, n.d.). Based on the work of MIPAG, "Wild Urban Plants of the Northeast: A Field Guide" (Del Tredici, 2020) describes urban plants including some species that can be defined as spontaneous invasive plants. Base on the definition of spontaneous invasive plants in these sources (Del Tredici, 2020; Massachusetts Invasive Plant Advisory Group, 2005) as well as data from USDA Natural Resources Conservation Service, local nurseries, and expert interviews, the research team selected 5 common spontaneous invasive plants in the Boston metropolitan area from the list provided by MIPAG (Massachusetts Invasive Plant Advisory Group, 2005). The studied species include *Lythrum salicaria, Lonicera periclymenum, Euonymus alatus, Berberis thunbergia, Phragmites australis.*

2.2.3. Experiment procedure

We designed a series of experiments testing invasive plants' tolerance to various soil parameters and metal pollution absorbing ability. The soil parameter classifications defined in the previous section serve as the variables in the experiments (Table 5).

To test an invasive plant's ability to tolerate soil drought levels, an experiment was conducted using four different levels of relative soil water content (R). To ensure consistency, 8 plants of each species were chosen based on their similar size and growth conditions and were then planted in 10 cm plastic pots. To simulate the soil drought levels of brownfields, eight pots of plants were irrigated with varying volumes of water, resulting in four distinct ranges of R ($R > 60$, $60 \geq R > 45$, $45 \geq R > 30$, $R \leq 30$), with two pots used for each range. After ten days of growth, one leaf was randomly selected from the 4th to 6th leaves from the top bud for assessment.

To test a plant's tolerance to Soil Salinity, an experiment with 4 levels of soil Electrical Conductivity (EC) was conducted. We selected 8 additional plants from each species and placed them in 10 cm pots. Sodium chloride solution was added to each pot and CE was measured. The CE of soil was controlled into 4 ranges (0–0.50, 0.51–1.25, 1.26–1.75, >1.75), 2 pots for each range. Again this experiment did not require the soil CE to be a particular value but only within the decided range. After 10 days of treatment, one leaf was randomly selected from the 4th to 6th leaves down from the top bud for assessment.

To test plant's tolerance to Soil Nutrients, a hydroponic method was adopted. We selected 8 equal sized and growth condition plants of each species for this study. The plants were removed from soil, washed and placed in 8 containers of distilled water for cultivation. The soil nutrients of a brownfield were simulated by controlling the Total Nitrogen (TN) into 4 ranges (0–0.3, 0.3–0.8, 0.8–2.0, >2.0) in 8 containers of distilled water (2 for each TN range). The distilled water was replaced every 2 days. This experiment did not require the TN of soil to be a specific value but only within the defined range. After 10 days of experimentation, one leaf was randomly selected from the 4th–6th leaves down from the top bud for assessment.

To test a plant's tolerance to Soil Metal Pollution, a hydroponic method was adopted and the Mehlich-3 Extractable Pb was controlled. We selected 8 equal-sized and growth condition plants of each species. The plants were removed from soil, washed and placed in 8 containers of distilled water for cultivation. 4 ranges of Mehlich-3 Extractable Pb concentration (0–47, 47–202, 202–278, >278) were set for 8 containers of distilled water (2 for each range). The distilled water was replaced every 2 days. This experiment did not require the Mehlich-3 Extractable Pb of distilled water to be a specific value but only within the decided range. After 10 days of experimentation, one leaf was randomly selected from the 4th–6th leaves down from the top bud for assessment.

Table 6
Relationship between MPE data and experimental settings (source: authors).

MPE data	MPE_1	MPE_2	MPE_3	MPE_4
Experimental settings/variable range	Range 1	Range 2	Range 3	Range 4

2.2.4. Assessment and data analysis

Chlorophyll fluorescence can be used to determine if stress on a plant is effecting the rate of photosynthesis. In this analysis the lighting strength was set at 1000 $\mu mol \cdot m^{-2} s^{-1}$, the temperature set to 25 °C, and the CO_2 gas source was the outdoor atmosphere CO_2. After the leaves were treated for 30 min, the experiments used a portable chlorophyll fluorescent measurement device. FLUORCAM. to measure the Maximal fluorescence (F_m) and Minimal fluorescence (F_o). Then, the Variable fluorescence (F_v) and the Maximal Photochemical Efficiency of PSII (MPE) are calculated to reflect the photosynthetic performance of the leaves.

$$MPE = F_v/F_m = (F_m - F_o)/F_m$$

Data of each studied plant was recorded 5 times, and the average value was selected as the final results. Because for each studied factor, 2 samples of a species were used for one factor range, the average value of two samples were decided as the final evaluation of this species.

Because this study aims to develop an applicable model of selecting spontaneous invasive plants for the phytoremediation of brownfield soil remediation for environmental design practice, the plants' tolerance should be represented in a straightforward and understandable form. Using the detailed data will complicate the follow-up application of an invasive plant's data, making it difficult for the experiment results to be conveniently and pragmatically used in environmental design practice. Therefore, based on the MPE data, the plant's tolerance to different soil parameters is rated as poor, fair, good, and excellent based on the experimental results. After obtaining the MPE data of all the studied plants (Appendix B), researchers compared the data of 4 ranges (MPE_1, MPE_2, MPE_3, MPE_4) under one soil parameter to decide the species' tolerance rating. Based on the experimental settings shown in Table 5, the relationship between MPE_1, MPE_2, MPE_3, MPE_4 and different variable ranges is explained in Table 6.

Existing literatures have shown that a MPE from 0.79 to 0.84 indicates a good photosynthetic performance of a plant while a value lower than 0.79 indicates plants stress (Maxwell and Johnson, 2000). Based on this range of MPE, the tolerance rating of plants (TR) can be characterized using the following criteria:

If $MPE_1 \geq 0.79$, $MPE_2 < 0.79$, then TR = Poor
If MPE_1 and $MPE_2 \geq 0.79$, $MPE_3 < 0.79$, then TR = Fair
If MPE_1, MPE_2, and $MPE_3 \geq 0.79$, MPE4 < 0.79 then T = Good
If MPE_1, MPE_2, MPE_3, and $MPE_4 \geq 0.79$, then T = Excellent

3. Results

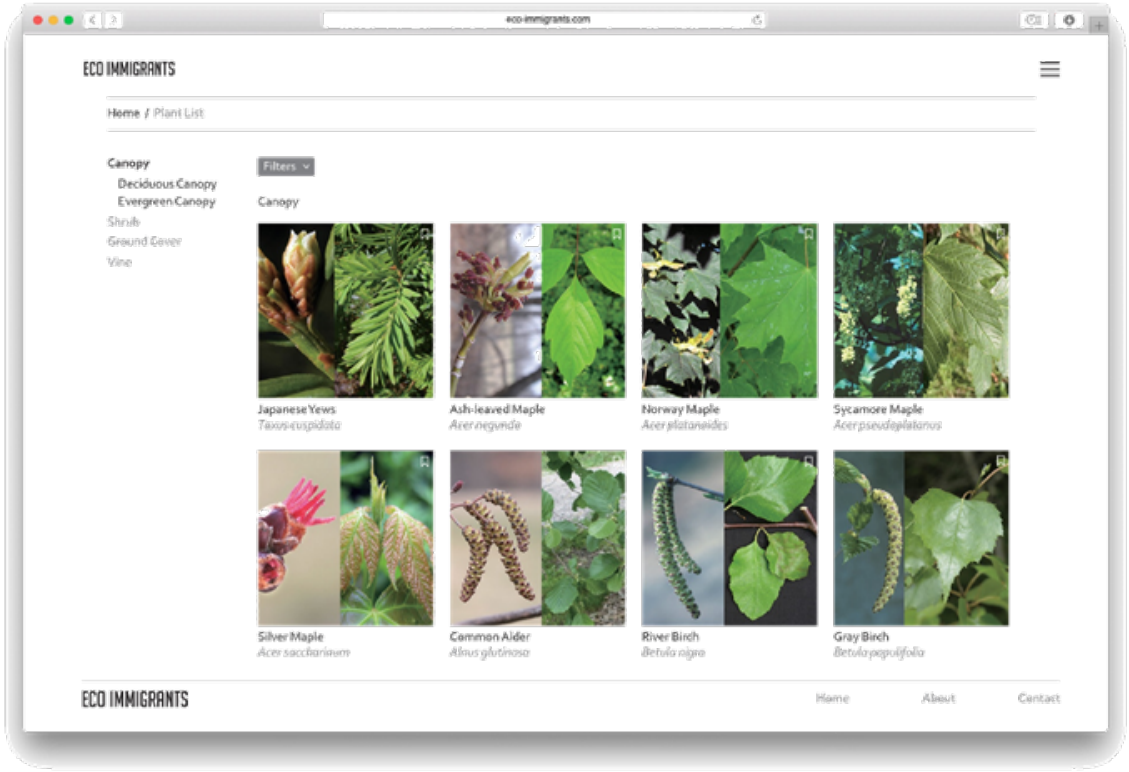
The above experiment procedure was applied to each of the 5 selected species to assess their tolerance to different parameters of brownfield soil.

Table 5
Experimental settings.

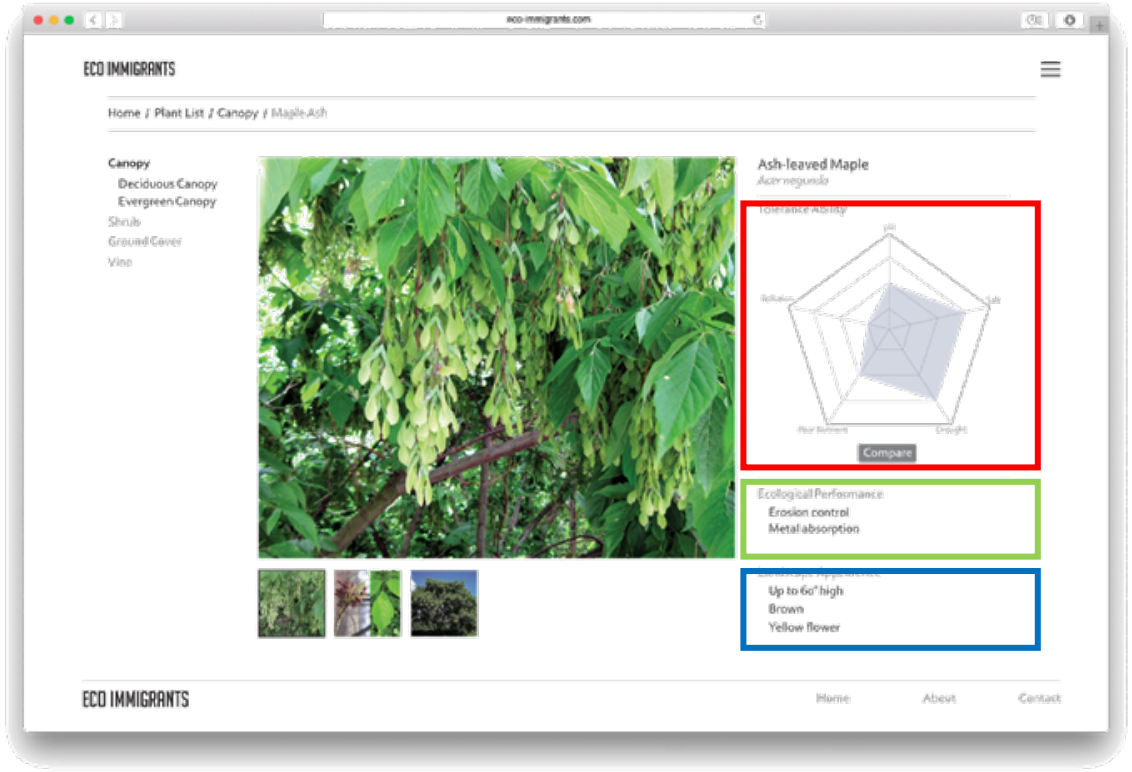
Soil parameter		Soil drought level	Soil salinity	Soil nutrients	Soil metal pollution
Measurement		Relative soil water content (R) (%)	Electrical conductivity (EC) (mmhos/cm)	Total nitrogen (TN) (%)	Mehlich-3 extractable Pb (ppm)
Experimental settings	Range 1	R > 60	0–0.50	TN ≥ 2.0	0–47
	Range 2	60 ≥ R > 45	0.51–1.25	0.8 ≤ TN < 2.0	47–202
	Range 3	45 ≥ R > 30	1.26–1.75	0.3 ≤ TN < 0.8	202–278
	Range 4	R ≤ 30	>1.75	TN < 0.3	>278

In order to assess the tolerance ability, A research team lead by me, collaborate with Harvard, rugters explored five method to test the plant tolerance on ph, soil salinity, heavy metal pollution, soil nutrient and soil drought. The overall concept are put them into different conditions and use chlorophyll fluorescence as an index to assess the ability.

Data Visualization



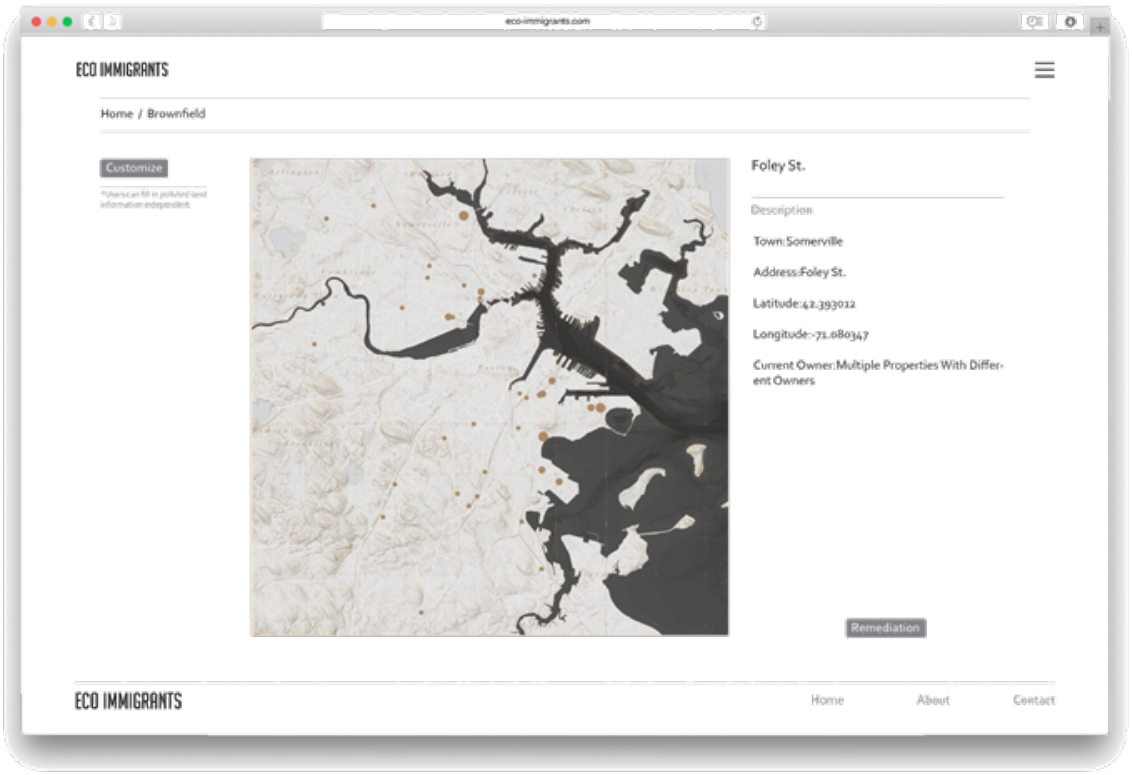
PLANT LIST



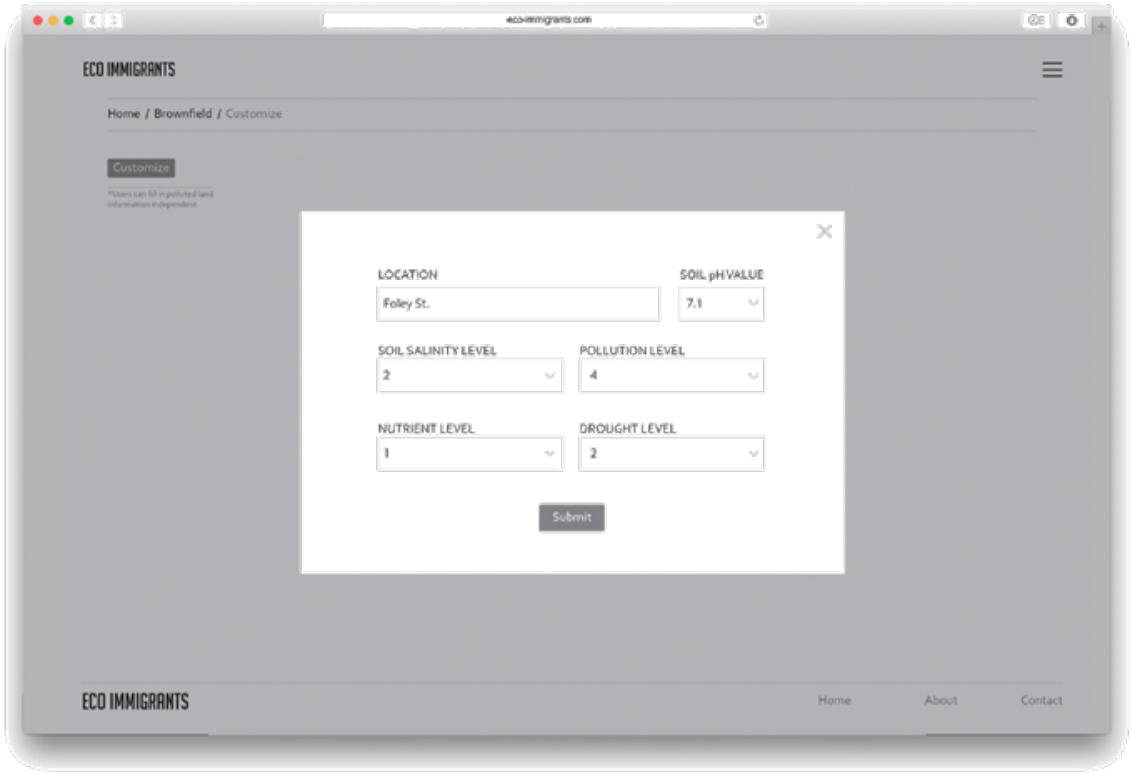
PLANT ID

By using the test method from last slides and Dr Peter Del Tredici'swork, each of the plant could be quantified their unique value from 1-5, see the red box area. The green box shows the significant ecological performance, such as erosion control, etc. the blue box shows the landscape appearances such as color of fruit, flowers, height. Etc.

Data Visualization



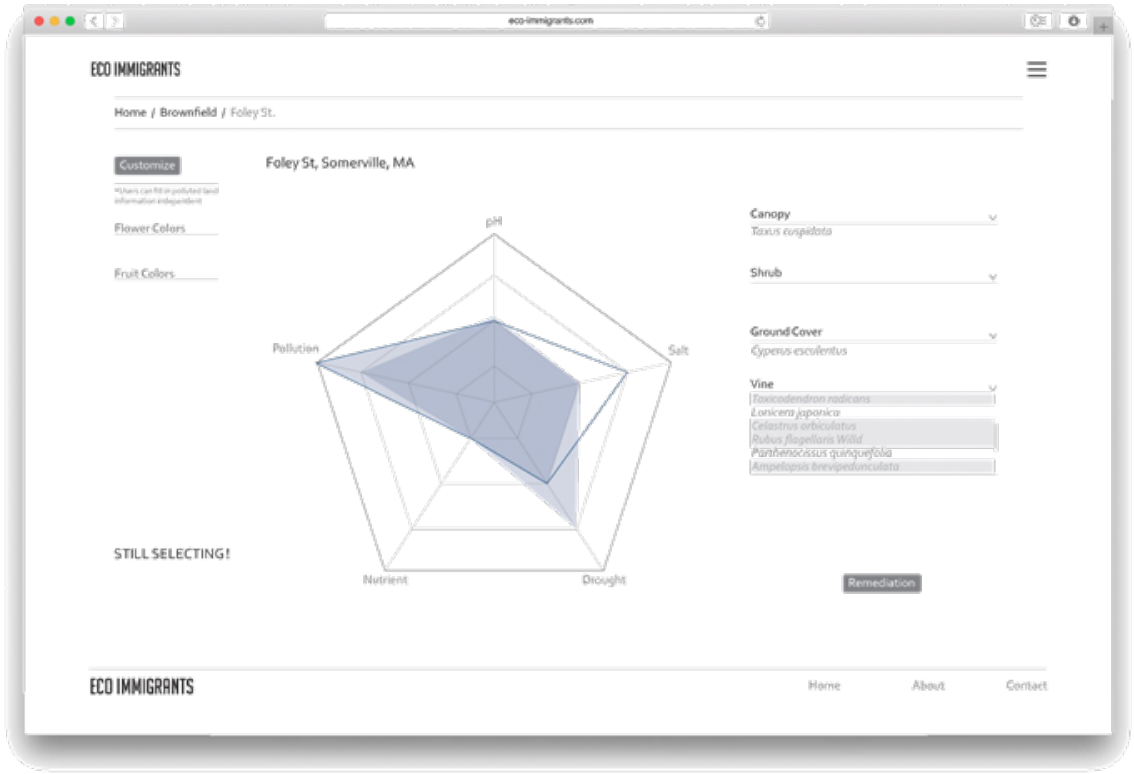
BROWNFIELD LIST



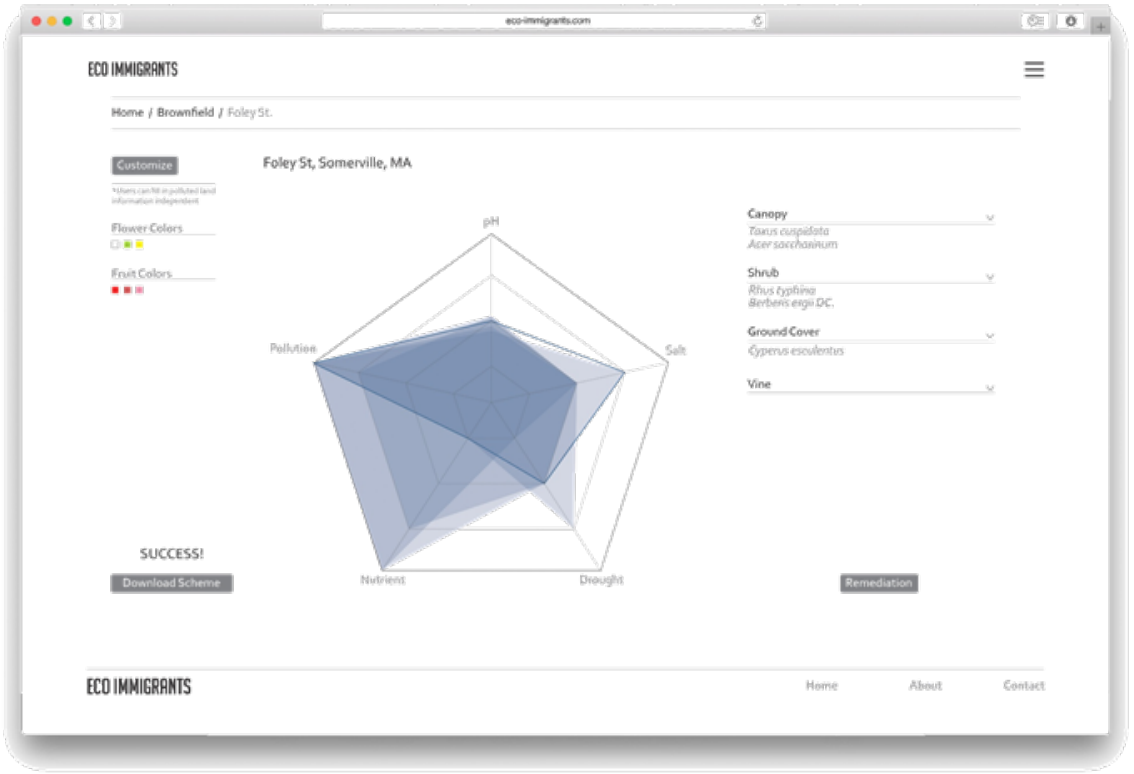
BROWNFIELD CUSTOMIZE FORM

Same concept to generate the radar diagram of plants, the system also allow the users to type in the quantified number of soil condition base on the soil report and defined parameters.

Application in Design



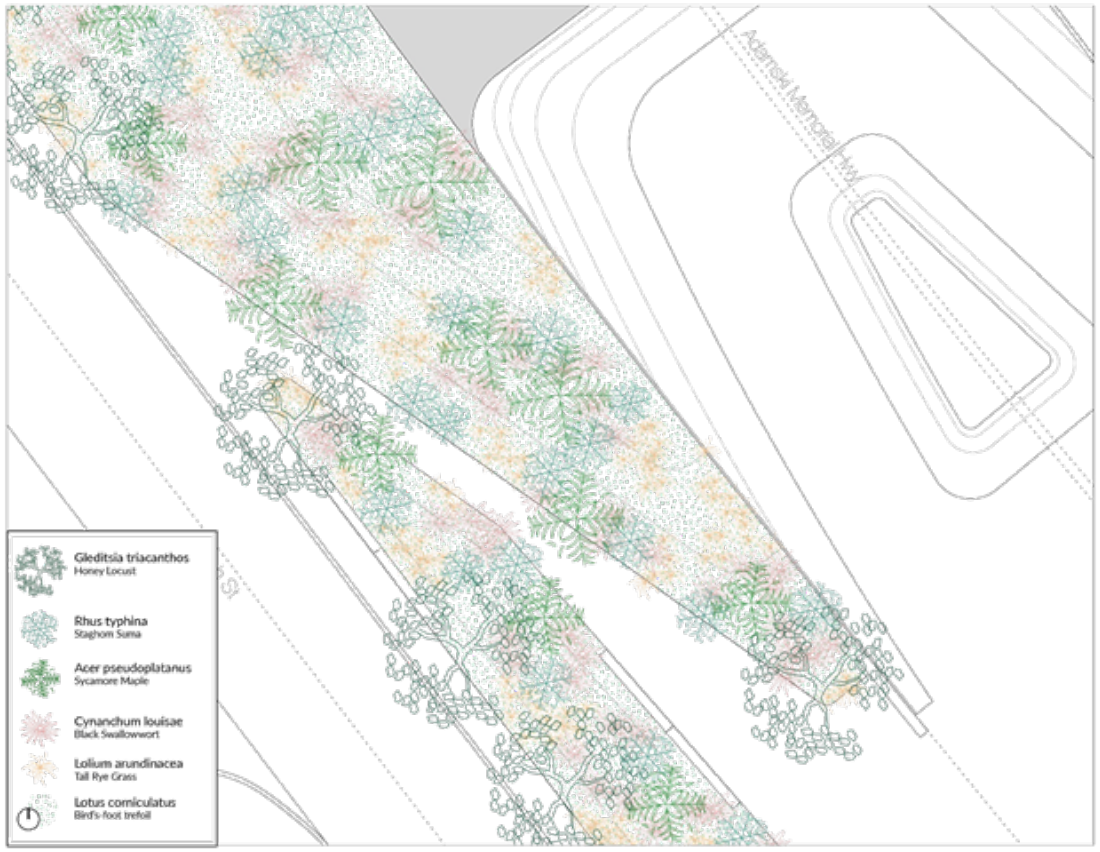
REMEDIATION FAIL



REMEDIATION SUCCEED

Eventually, the work would provide the new technology and toolkits for Landscape practitioners to use to choose the appropriate plant list scientifically and applied them in design process. Once the pattern of plants could cover the graph of the brownfield site, this refers to the generation of plant list successfully. The interface also could reflect the flower colors and fruit colors of the proposed plant design.

Application in Design



STEP 5 : DESIGN

By using the result, the designers could use it to better generate an eco- based planting design.

Key Concept:

“Crowdsource Data Analysis”

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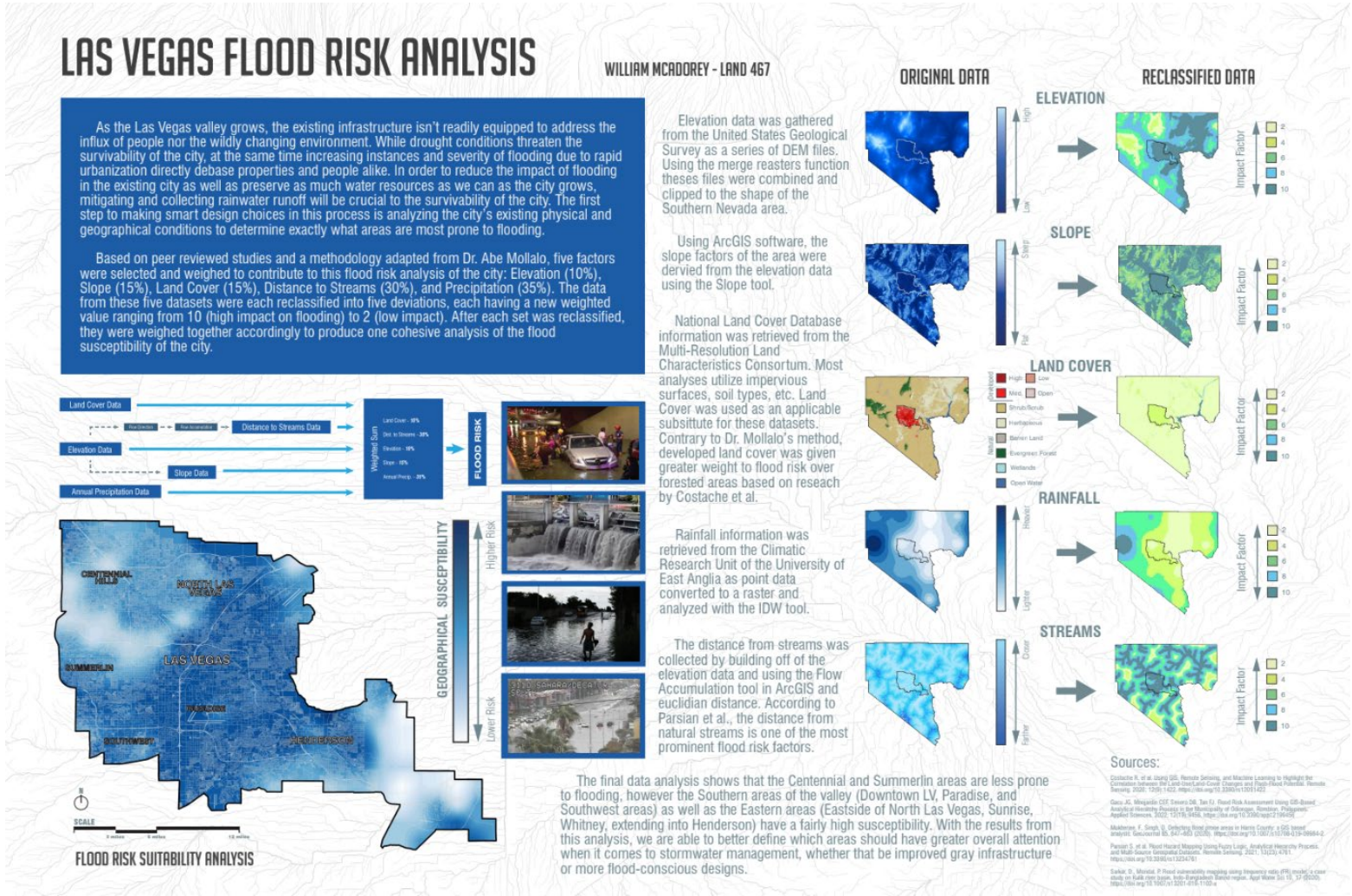
“Explore the Design Inquires”

Student Question: “How could we used those advanced technology to my studio work?

Student Work: UNLV LAND 467_ Introduction to GIS



All of the elective classes I taught are have a strong connection with design studio work. For example. This student's BLA capstone is to make strategies to prevent the flood. In my LAND 467, which is "Intro to GIS". Remember I did the suitability analysis in the last project I showed. By using the suitability analysis, he used it to find the places the suffered the flood most. Then become a toolkit to narrow down the boundary of the sites his going to design.

Student Work for LAND 467 Introduction to GIS

REMEDIATE THE URBAN FORESTRY *IN THE FASTEST-WARMING DESERT DRY CITY*

MARIE CASSANDRA TENORIO | LAND 467 | PROF. XIWEI SHEN | FALL 2023

PROJECT STATEMENT

The urbanization of the Las Vegas Valley is rapidly expanding and yet there is a visible gap between the growth of new and existing infrastructures vs the existing conditions of the vegetation and street trees. The limitation and lack of tree canopy and accessible corridors are highly stressed. This threatens longevity and survivability of the valley especially with the increase of the Urban Heat Islands (UHIs) in highly dense areas and areas with a lot of urban imperviousness. There is a need to solve this by strategically planting trees that will benefit us socially, economically, and environmentally.

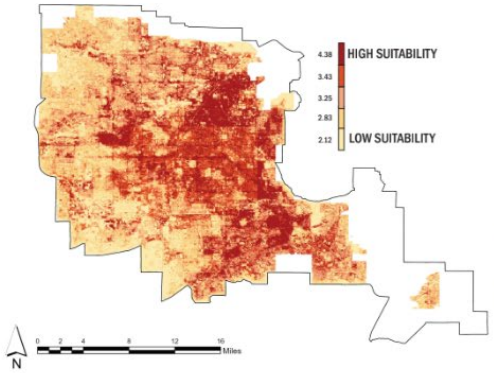
In a city so vast that is facing extreme temperatures and water scarcity, it is important to readdress the urban planning policies, tree, and landscaping policies to improve the current tree canopy of the whole valley.

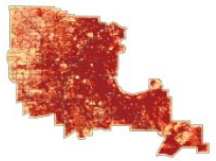
GIS DATASET WEIGHTED SUM

LAND SURFACE TEMPERATURE	→ 14%	→ SUITABILITY MAP
AFTERNOON TEMPERATURE	→ 14%	
LAND COVER	→ 14%	
URBAN IMPERVIOUSNESS	→ 14%	
PEOPLE IN POVERTY	→ 14%	
NORMALIZED DIFFERENCE VEGETATION INDEX	→ 14%	
TREE CANOPY	→ 16%	

SUITABILITY MAP OF THE LAS VEGAS VALLEY

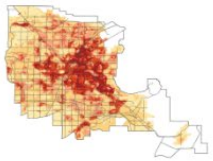
Based on the results of the suitability map, it shows that the center area and the eastern area of the Las Vegas Valley is suitable for improving its tree canopy coverage. Especially, in neighborhoods with low tree canopy and high poverty rate.





LAND SURFACE TEMPERATURE (LST)

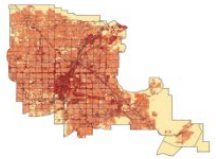
72 — 91



AFTERNOON HEAT TEMPERATURE

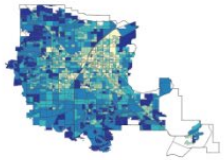
83 — 112

The temperature levels of the Las Vegas Valley experiences hot surface temperatures and peak afternoon heat temperatures. These maps represent the actual heat tempertaures the community experiences outside of their homes. The areas that feel the heat the most are located at the center of the valley where the Downtown Las Vegas is at. These two maps also show that the increase in the temperature levels can be due to the Urban Heat Islands (UHIs). Areas with high building density and the lack of shading (lack of trees) from the scorching heat can be the causes of UHIs.



LAND COVER AND URBAN IMPERVIOUSNESS

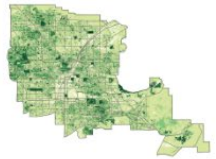
1.00 — 5.00



PEOPLE IN POVERTY

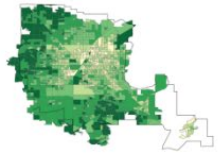
LOW — HIGH

One of the factors as to why there is UHIs and lack of trees are due to the impervious surfaces (such as asphalt roads and open parking lots) and the unequal distribution of trees within the neighborhoods and/or census block groups. Since, the urbanized valley is economically growing, it evidently shows that the land cover composes of more impervious surfaces than vegetation. This greatly contributes to the increase of the heat temperatures in the valley. Aside from that, it also shows communities far away from the center area such as the Summerlin area eceive better vegetation growth based on the NDVI map.



NORMALIZED DIFFERENCE VEGETATION INDEX (NDVI)

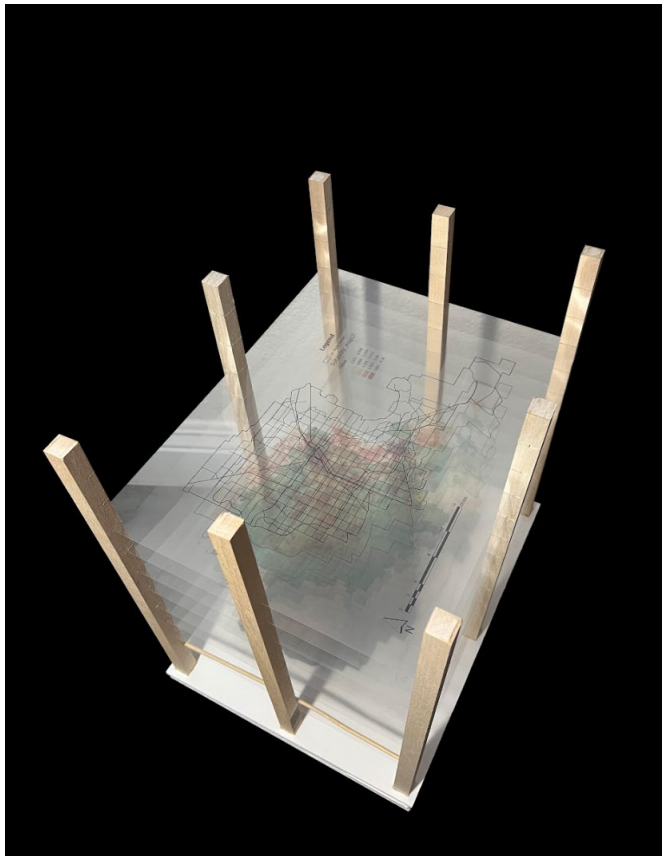
1.00 — 5.00



TREE CANOPY

LOW — HIGH

The people in poverty map and the tree canopy map correlate to each other as it depicts which neighborhoods and/or census block groups is in need of increasing its tree canopy coverage. This also connects to the temperature maps, land cover, urban imperviousness, and NDVI map. All of the maps tells us that the lack of trees and the limitation of the distribution of the trees all lead to the cause of UHIs and why it continues to get hotter and hotter in the Las Vegas Valley.



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Part 02:

Design Philosophy : Encourage students to actively engage in the design process, fostering diverse approaches and perspectives in their work

Key Concept I:

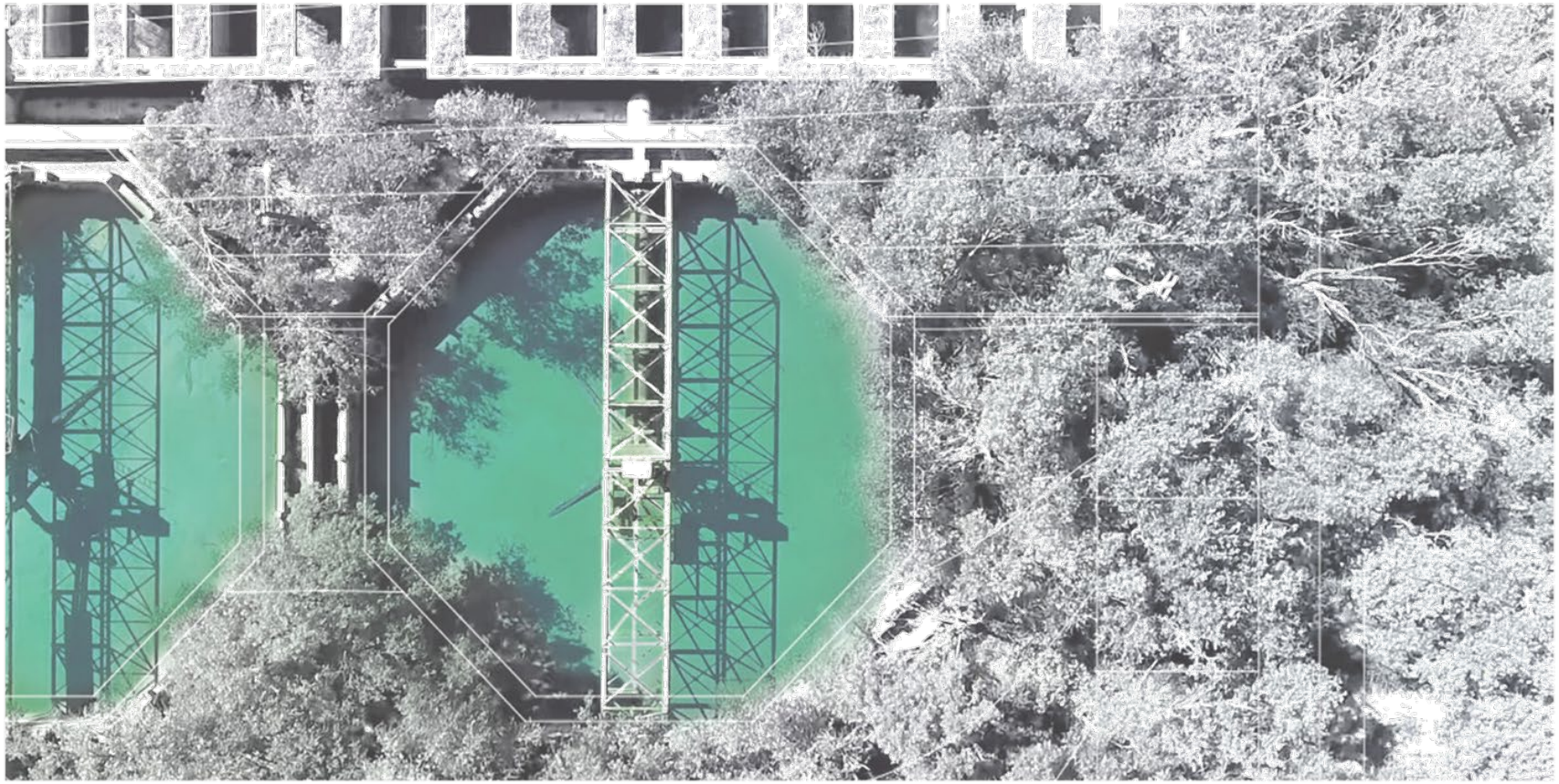
“Innovative Materials, Findings, Concepts from Students’ Interests”

+

“Make it Come True!”

BLOOM!

A Dynamic Landscape Biological System



01

How might algae be reconsidered as a spatial, material and temporal medium of landscape architecture design?

02

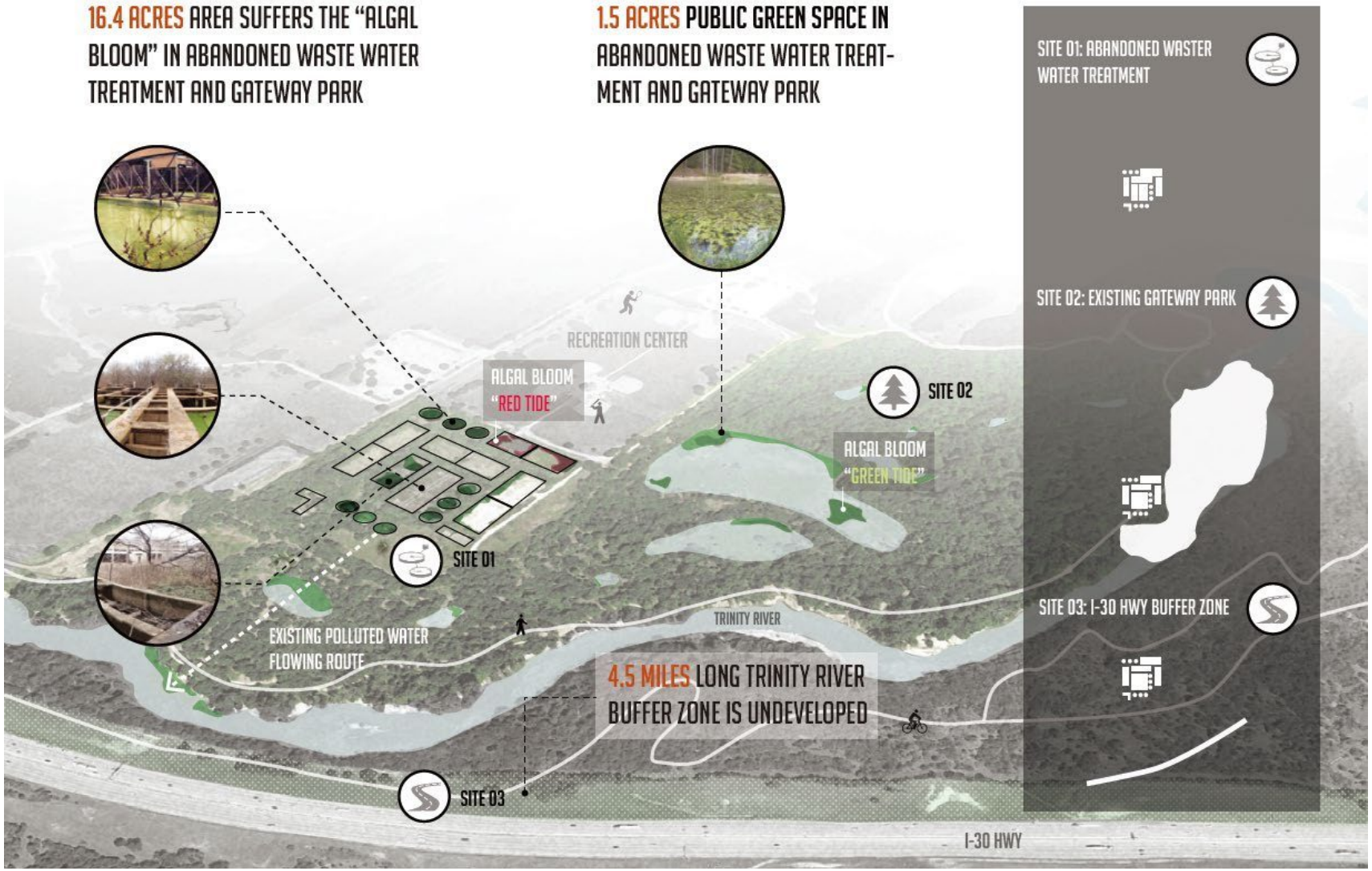
How might a systems-based solution to algal blooms be approached in a manner that contributes to ecological, social, public service issues and act as an agent to revitalize public space?

03

How might an algae Eco-system be applied as useful active agents in the urban landscape?

This project aspires to explore how problematic levels of algae (specifically Bioluminescent Algae, Blue Green Algae and Diatoms Algae) in waterways might be addressed through innovative landscape design strategies that move beyond reductive engineered solutions to productive spatial agents within urban environments. Healthy water bodies generally contain many kinds of algae that balance the ecological system, however, in certain conditions the balance can be disturbed by external environmental activities and weather events causing toxic algae blooms. The most common algal bloom types are the Green Tide and the Red Tide. These phenomena have the potential to be converted to improve Eco-system health and act as design drivers in the sustainable landscape. This study will focus on three case study types: the redundant industrial infrastructure (Site 01), the Gateway Park next to the redundant industrial infrastructure (Site 02) and the urban infrastructure close to I-30 Hwy (Site 03) on the site of Fort Worth, Texas, as a prototype for considering how this approach might be deployed nationally.

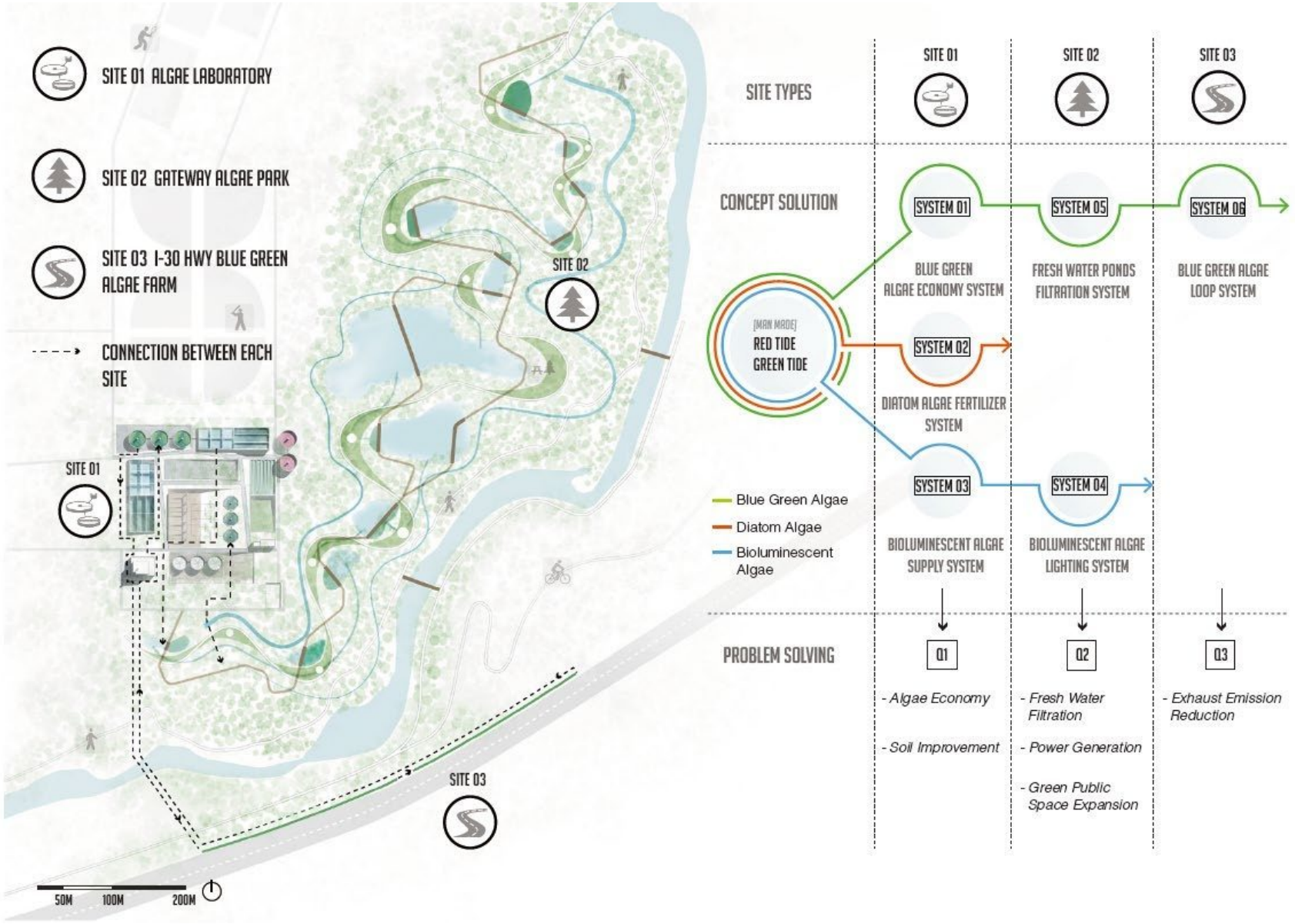
Site Inventory



The examined site is in the Fort Worth, the site are polluted by abandoned water treatment. The site contains two types of pollution. Red Tide and Green Tide. Since the site are huge, I decide to split the site into three scaled sites. Abandoned water treatment, nearby park and highway buffer zone

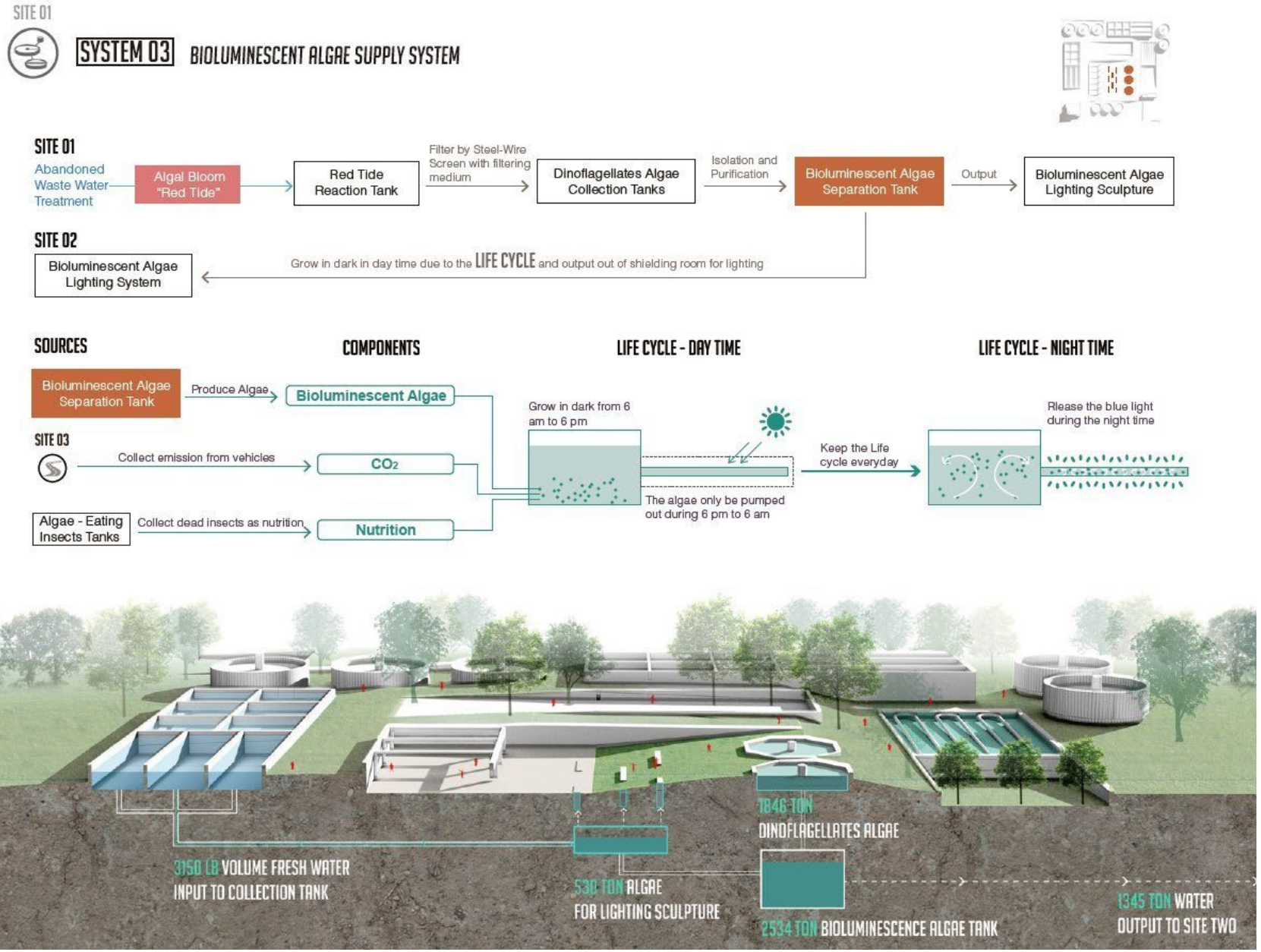
Framework

According to the report from Biological scientist, Diatom Algae is the main component of Red tide, and the bioluminescent and Blue Green algae are two of the major components of Green Tide. Thus, the design aims at applying these three types algae as media to renew the site. As the diagram shows, the design construct six systems base on the function of these three algae. The goal of the design is to incorporate these system through different scales of the site.



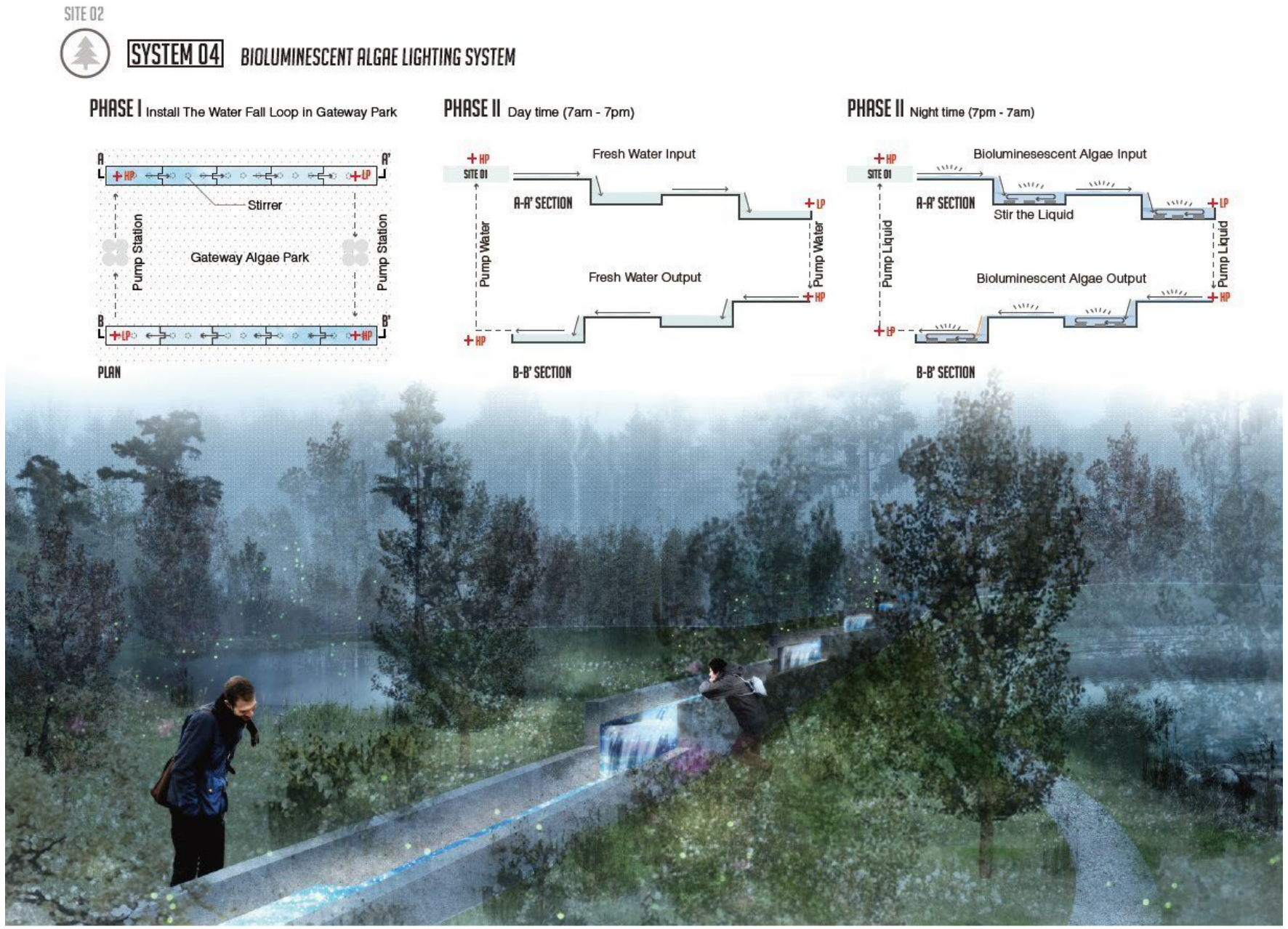
Typology

For the abandoned water treatment, the design reuses it as not only an innovated public park, but also the factory to educate the three types of algae. The diagrams above summarizes the key technologies from biology and used as parameters to consider how to reuse the abandoned infrastructure. To make the design more accurate, the design also simulate the quantity of how much algae they could grow



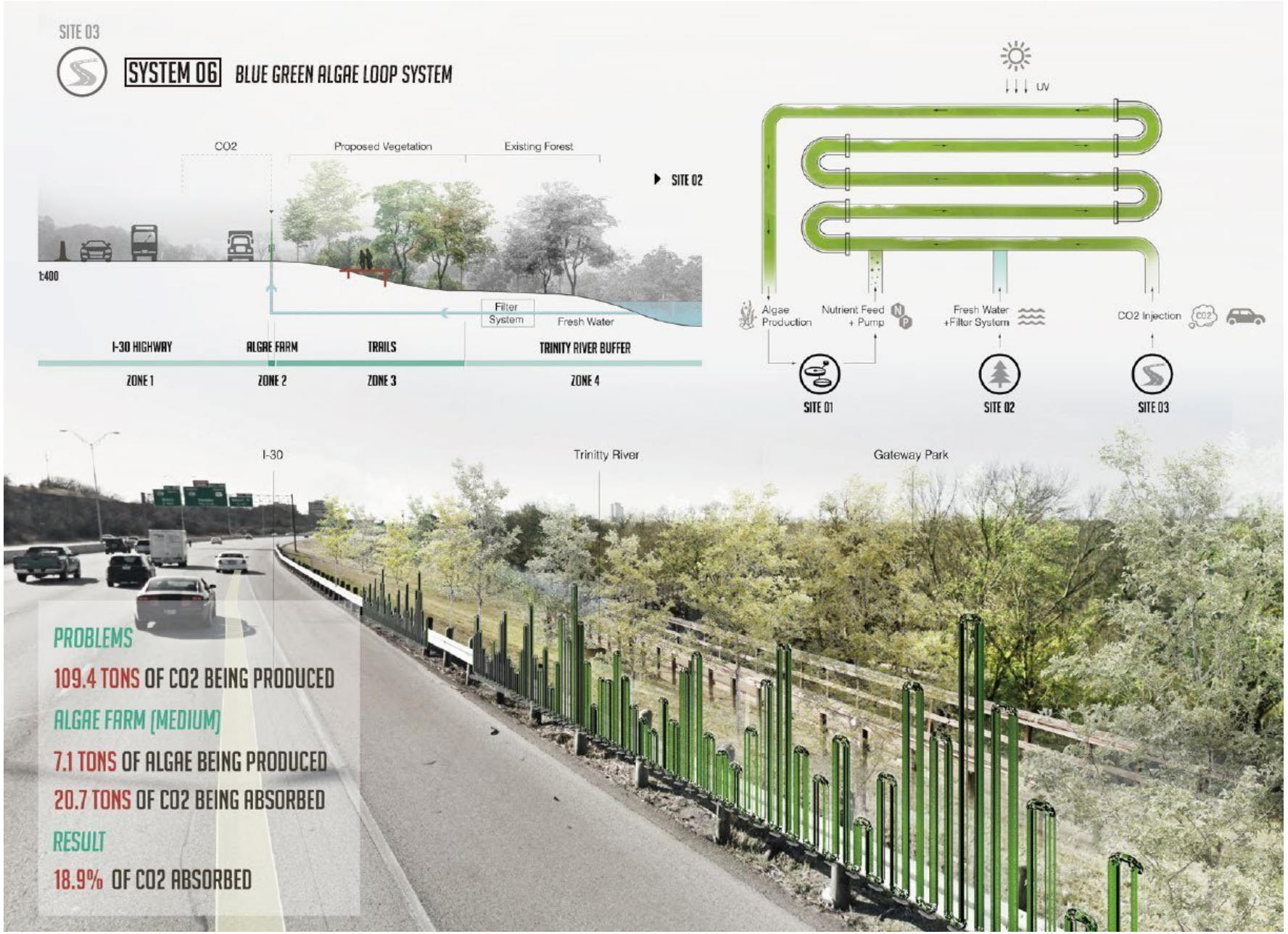
Typology

In the nearby park. the site has a natural slope. Consider the growth condition of bioluminescence lightened To grow it, the condition actually need to pump up the liquid in daytime so it could be lightened in the night. Thus, a nature slope ladder pond could actually fitin its growth condition. And this typology integrates the knowledge from biology with LA design



Typology

Considering the growth condition of Blue-green algae, it needs the CO2, fresh water. Thus, the Site 01 algae factory, Site 02 Nearby Park and Hwy could provide those growth elements. So, this typology could help absorb the CO2 from vehicles to improve the air quality.



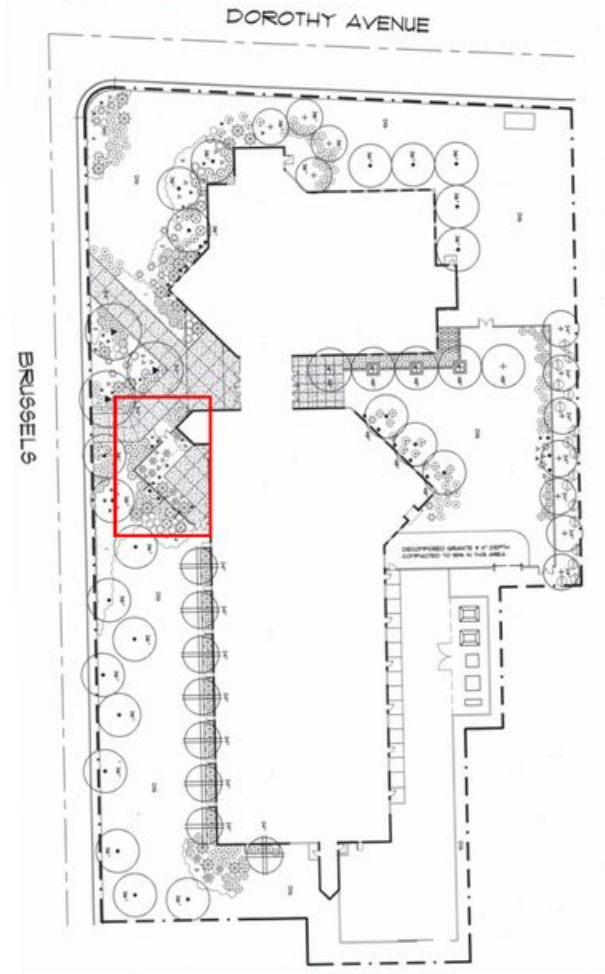
Student Question: “How could we examine the ‘dramatic’ concepts to real practice?

Student Work: UNLV BLA LAND 484_ OASIS Garden

Background | A Working and Learning Environment for Faculty and Students

The UNLV Landscape Architecture and Planning program vision is committed to creating a working and learning environment where faculty and students collaborate to promote the development of forward-thinking future leaders

The Project Site Original Plan



ORIGINAL LANDSCAPE PLANTING PLAN
BY SH ARCHITECTURE FORMALLY KNOWN
AS SWISHER & HALL (1997)

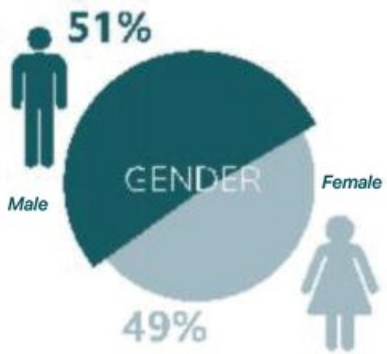
Project Timeline

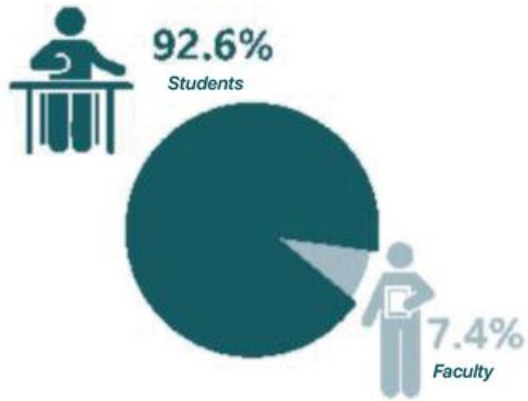
	JAN	FEB	MAR	APR	MAY
Initial Research					
Site Analysis					
Plant Selection					
Preliminart Design					
Community Outreach					
Design Development					
Construction Details					
Final Production					

Community Engagement and Synthesis

Staff and Student Feedback Regarding Design Choices + Project Goals

Overall Courtyard Survey



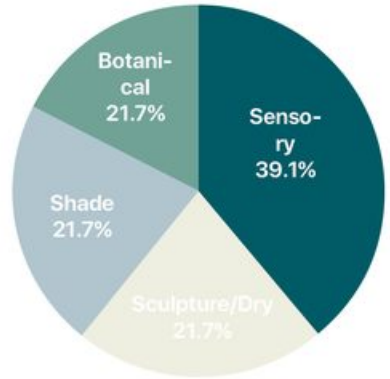


Have Visited 0 times

Have Visited 1-2 times

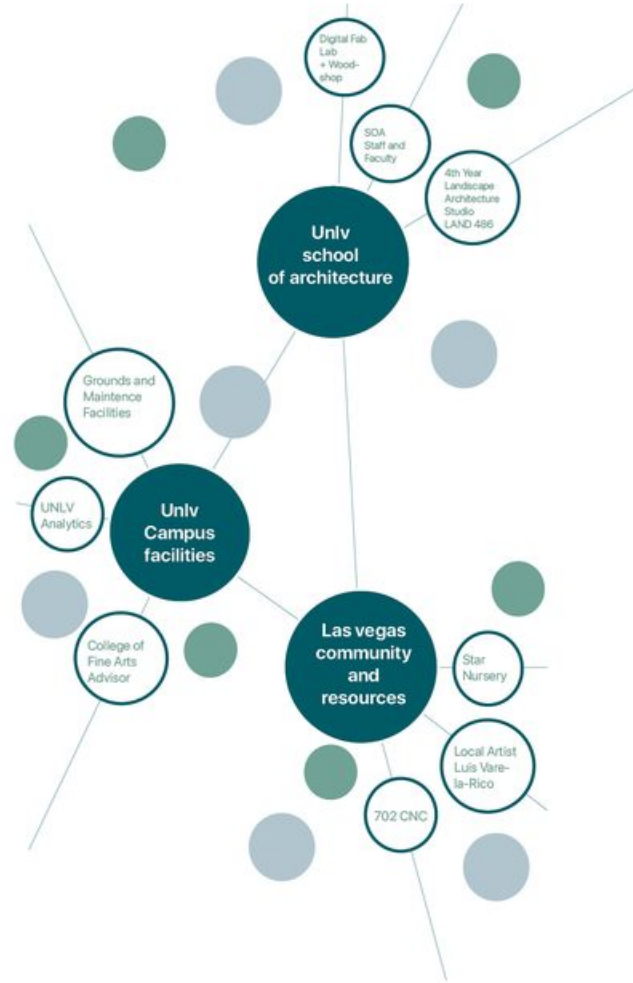
Have Visited 3+ times

Garden + Plant typology Survey



- Sensory - Engaging the senses. Allows connecting to nature, encourages you to become more aware of your surroundings.
- Sculpture/Dry - Garden to enjoy works of art; contains rocks and shrubs.
- Shade - Only certain plants are able to grow in shady conditions, creating a cooler microclimate.
- Botanical - Focuses on the idea of learning and conserving plants.

Synthesis



Design Goals:

Character + opportunity

Provide features that feature wood and natural stone materials. As well as take into account sculptural design; design with the idea to create an atmosphere that matches the student identity.

Student + Staff usage

Create a space where students and staff across campus can decompress, contemplate, or work during all operable school hours. Our main focus was designing our amenities with maneuverability and comfort in mind.

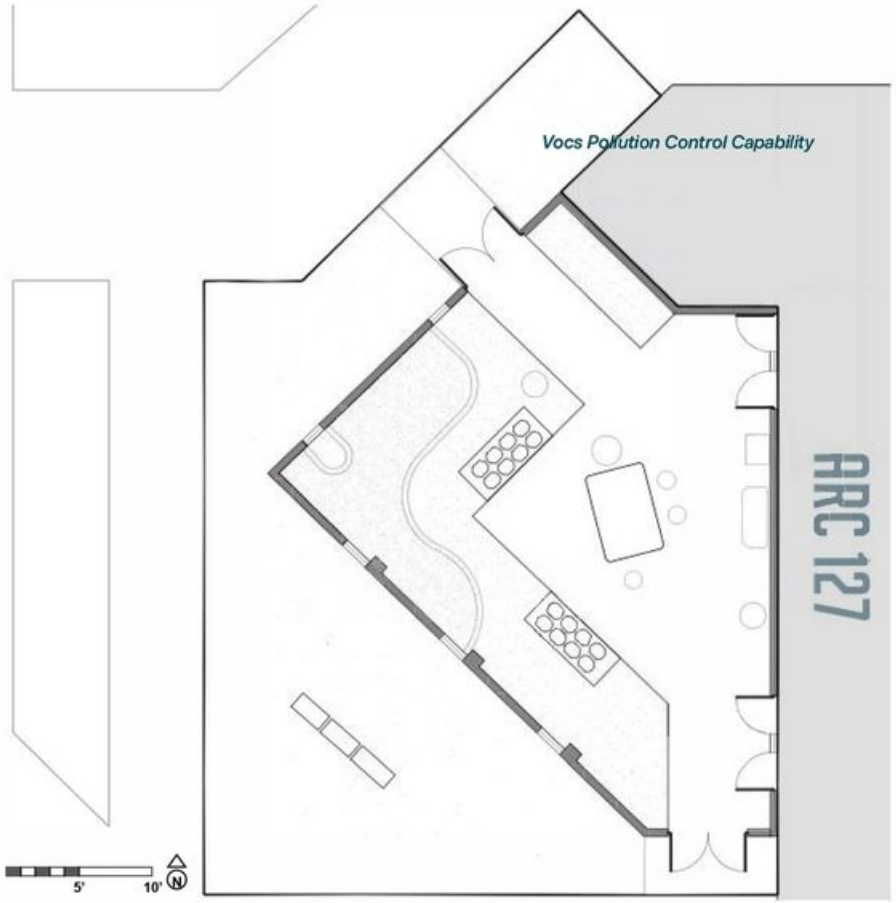
Beautification

Plant with native/adaptable vegetation; while keeping in mind sensory elements to allow users to interact with the space around them. Incorporate elements that make this space stand out from outside to inside.

SITE PLAN

SITE PLAN AND PROJECT STATEMENT

Our studio has concluded that the lack of utilization of this space can be attributed to the lack of planting narrativeand accomodation of amenities.



As landscape architecture students, we are acutely aware of the impact outdoor spaces has on the public from a psychological and physiological standpoint.

with the addition of student design and constructed amenities, we are continuing a precedent of student work being showcased and empowered through its functional existence within the school of architecture. additionally encouraging the curiosity and engagement of community members. In order to empower our community outward we are empowering our community inward.

PLANTING PLAN AND NARRATIVE

our goal with the planting design was to create a clear distinction between the sensory garden and desert landscape that surrounds the school of architecture. the sensory garden extends out from the courtyard to the main entrance of the school, drawing users to an oasis that softens the surrounding architecture and hardscape.

The sensory garden is grouped into different zones that stimulate the five senses. foxtail ferns are planted near the main entrance and interior couryard where they can be easily accessed for tactile play. to provide a pop of color, purple heart plants and regal mist grass are spread throughout the courtyard. the remaining plants were chosen to stimulate taste and smell.

Decorative white rock brightens the space and 2-4" chunky rock is used to create swales, ensuring proper site drainage.

Overall, the new sensory garden provides a much needed space for students to spend time outdoors for learning and recreation.

GROUNDCOVERS:

Foxtail Fern
Asparagus densiflorus 'Myersii'
Height X Width: ... 2' X 4'
Sun: ... Partial
Water: ... Medium
Habit: ... Evergreen

Variegated Flax Lily
Dianella tasmanica 'Variegata'
Height X Width: ... 2' X 3'
Sun: ... Partial
Water: ... Medium - Low
Habit: ... Evergreen

Yellow Butterfly Lily
Dietes bicolor
Height X Width: ... 2' X 2'
Sun: ... Full - Partial
Water: ... Medium
Habit: ... Evergreen

Purple Heart Plant
Tradescantia pallida
Height X Width: ... 2' X 3'
Sun: ... Full - Partial
Water: ... Medium - Low
Habit: ... Evergreen

Creeping Thyme
Thymus spp.
Height X Width: ... 1' X 2'
Sun: ... Full - Partial
Water: ... Low
Habit: ... Evergreen

GROUNDCOVERS CONTINUED:

Huntington Carpet Rosemary
Rosmarinus officianalis 'Huntington Carpet'
Height X Width: ... 2' X 8'
Sun: ... Full
Water: ... Low
Habit: ... Evergreen

Cat's Claw Vine
Dalichandra unguis-cati
Height X Width: ... 25' X 25'
Sun: ... Full - Partial
Water: ... Medium - Low
Habit: ... Evergreen

SHRUBS:

Pink Muhlygrass
Muhlenbergia capillaris 'Regal Mist'
Height X Width: ... 4' X 3'
Sun: ... Full - Partial
Water: ... Medium
Habit: ... Evergreen

Dwarf Mock Orange
Pittosporum tobira 'Wheeler's Dwarf'
Height X Width: ... 3' X 4'
Sun: ... Full - Partial
Water: ... Medium
Habit: ... Evergreen

Spanish Lavender
Lavandula stoechas
Height X Width: ... 3' X 3'
Sun: ... Full - Partial
Water: ... Medium
Habit: ... Evergreen

TREES:

Meyer Lemon
Citrus x meyeri
Height X Width: ... 20' X 15'
Sun: ... Full
Water: ... Medium
Habit: ... Evergreen

Twisted Myrtle
Myrtus communis 'Boetica'
Height X Width: ... 12' X 12'
Sun: ... Full - Partial
Water: ... Low
Habit: ... Evergreen

Housing Plants for Improving Air Quality

Philodendron hederaceum 'Brasil'

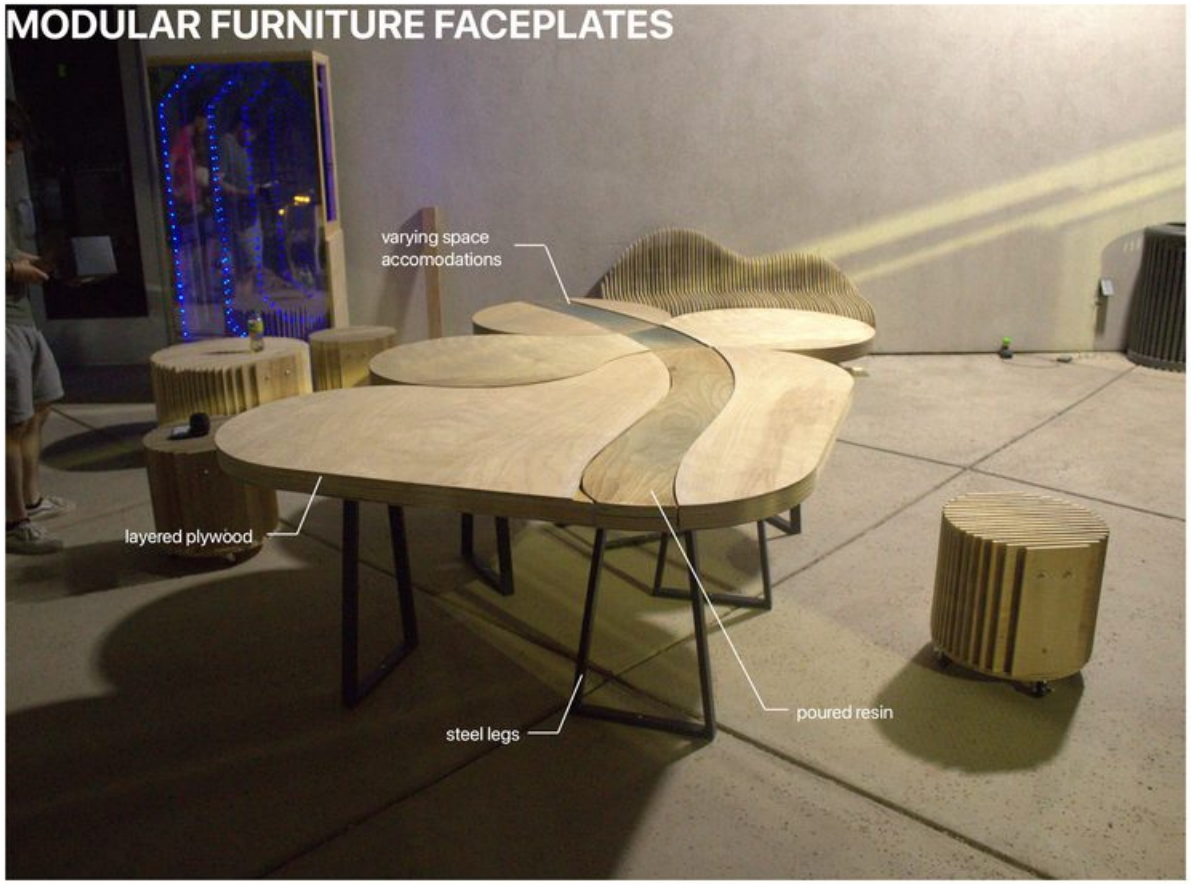
Adiontum sp.

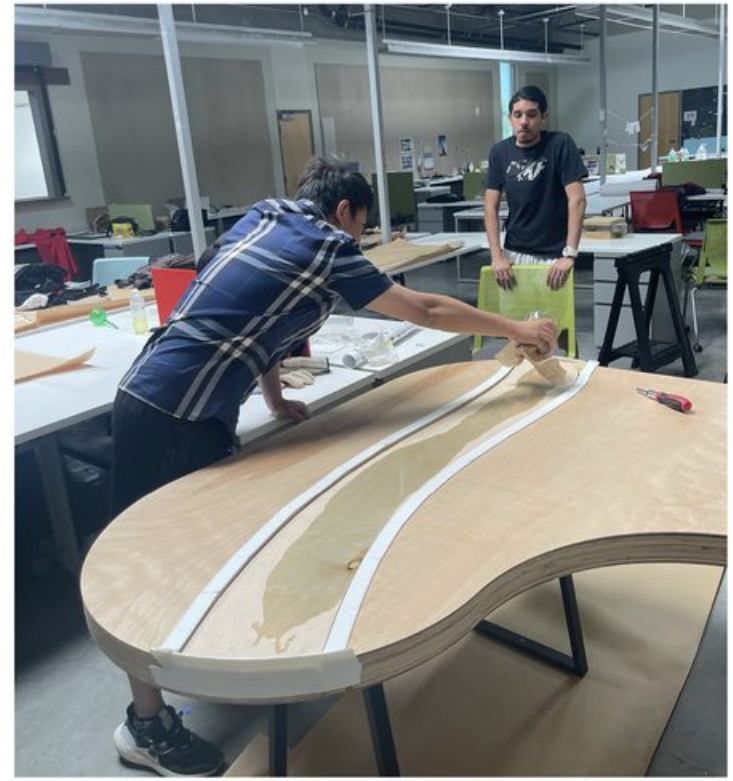
Maranta leuconeura

Dryopteris erythrosora

Davaltia fejeensis

MODULAR FURNITURE FACEPLATES



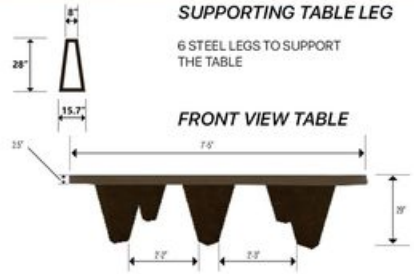


Complete Table

Plywood material for table top

Separated Table

Table is able to separate into 3 individual sets or come together as 1 full table

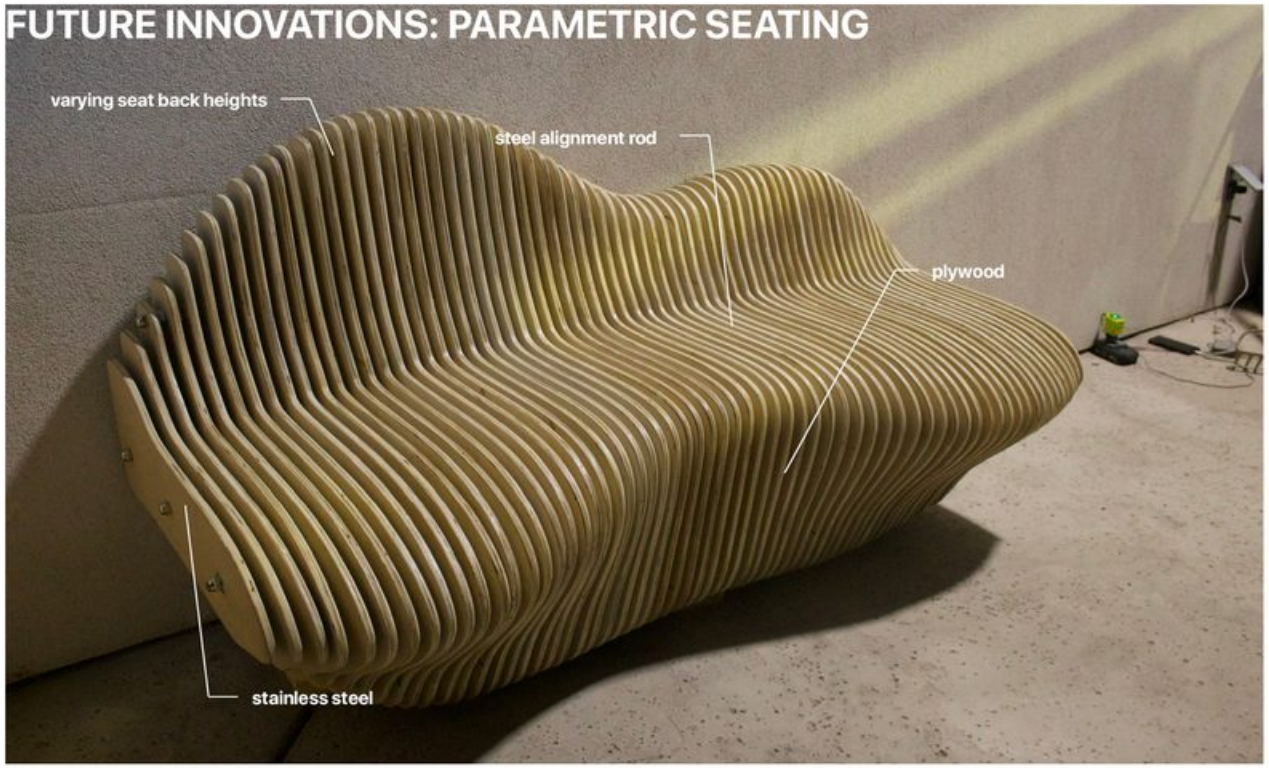


SUPPORTING TABLE LEG

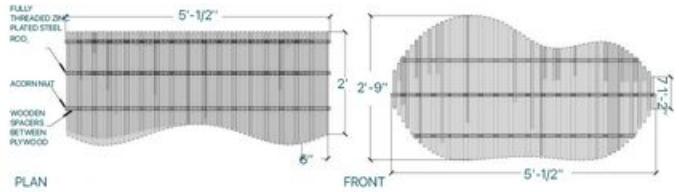
6 STEEL LEGS TO SUPPORT THE TABLE

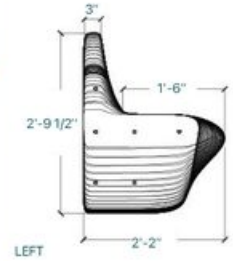
FRONT VIEW TABLE

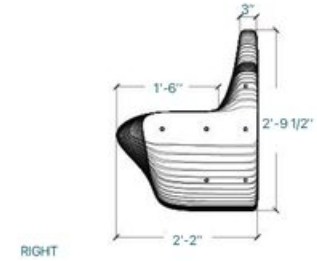
FUTURE INNOVATIONS: PARAMETRIC SEATING

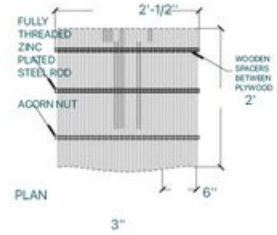


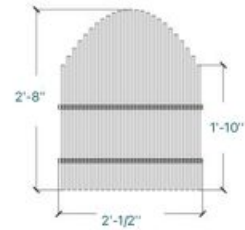
Product Prototype

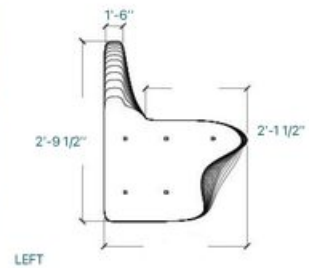


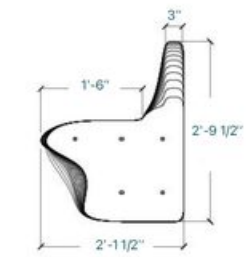






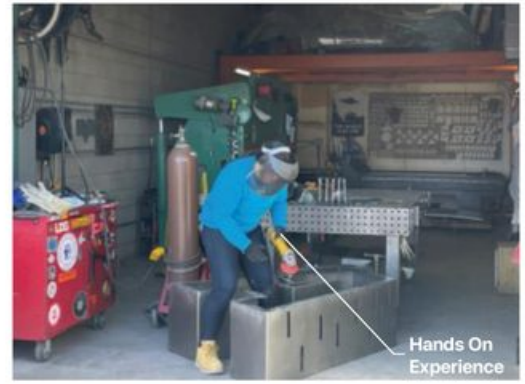


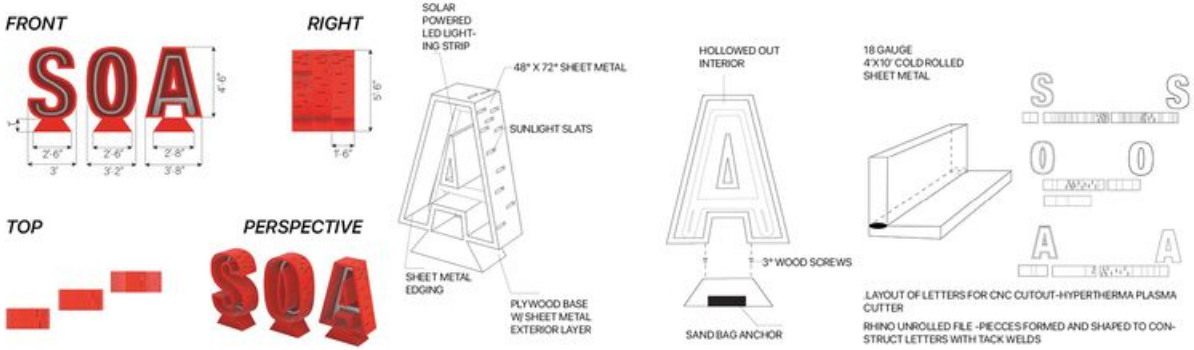




MARK INNOVATIONS-A REFLECTION OF US

Made from 7 sheets of 18 Gauge sheet metal, hand shaped, and tack welded the letters S O A gives a visual representation for the school of architecture. With slats on the right side Of the letters to Allow sunlight to Project into the letter's shell. The sign reflects the work of the spring 2023 4th year landscape Architecture studio And all their efforts.





FUTURE INNOVATIONS: BIOLUMINESCENT ALGAE

Interactive Art Installation Representing Applications of Bioluminescent Algae

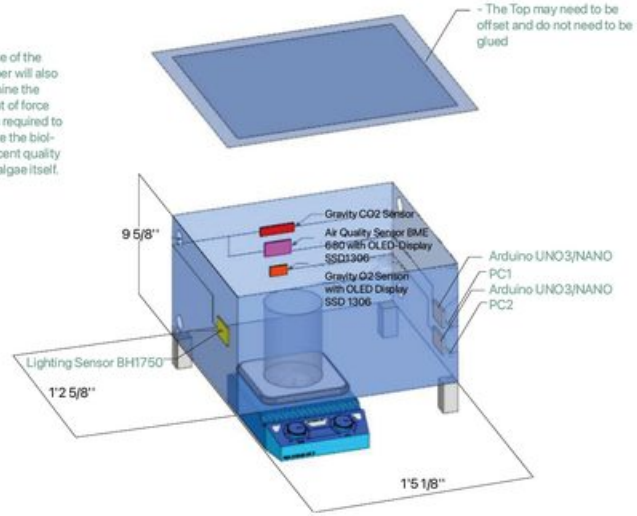
Utilizing bioluminescent algae creates an innovative opportunity in lighting amenitiy applications. Within this research, we aredeveloping a comparative analysis on the mitigation of CO2 levels and temperature control at the University ofLas Vegas, Nevada campus from different species of bioluminescent algae and methods of amenity production incorporating the medium.

As society embarks further into sustainable means of climatecontrol, our research centers the exploration of bioluminescent algae, which previous studies suggest to be more effective than raw plant material.

Technical Details

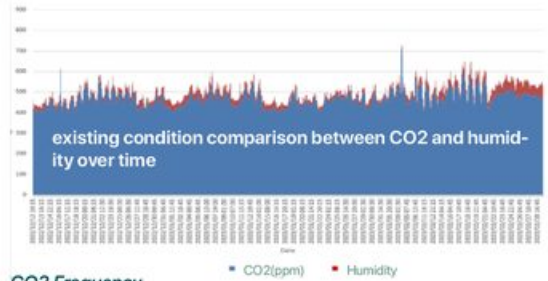
By measuring the algae in a controlled space, we will be able to conclude the difference between CO2, humidity, and temperature conditions and interior conditions of the courtyard site.

The use of the chamber will also determine the amount of force (ma) is required to activate the bioluminescent quality of the algae itself.

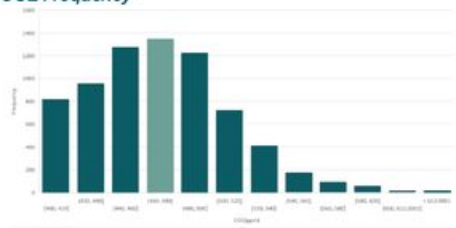


Comparison Of Environmental Parameters

At the moment, our conclusions remain theoretical, however, within the semester we were able to set the basis of our future research and applications that increase the aesthetic and interactive value of space.

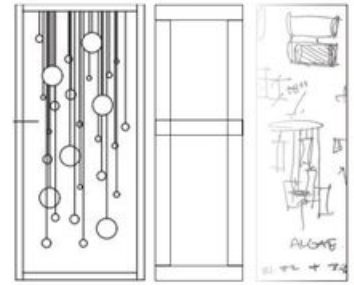


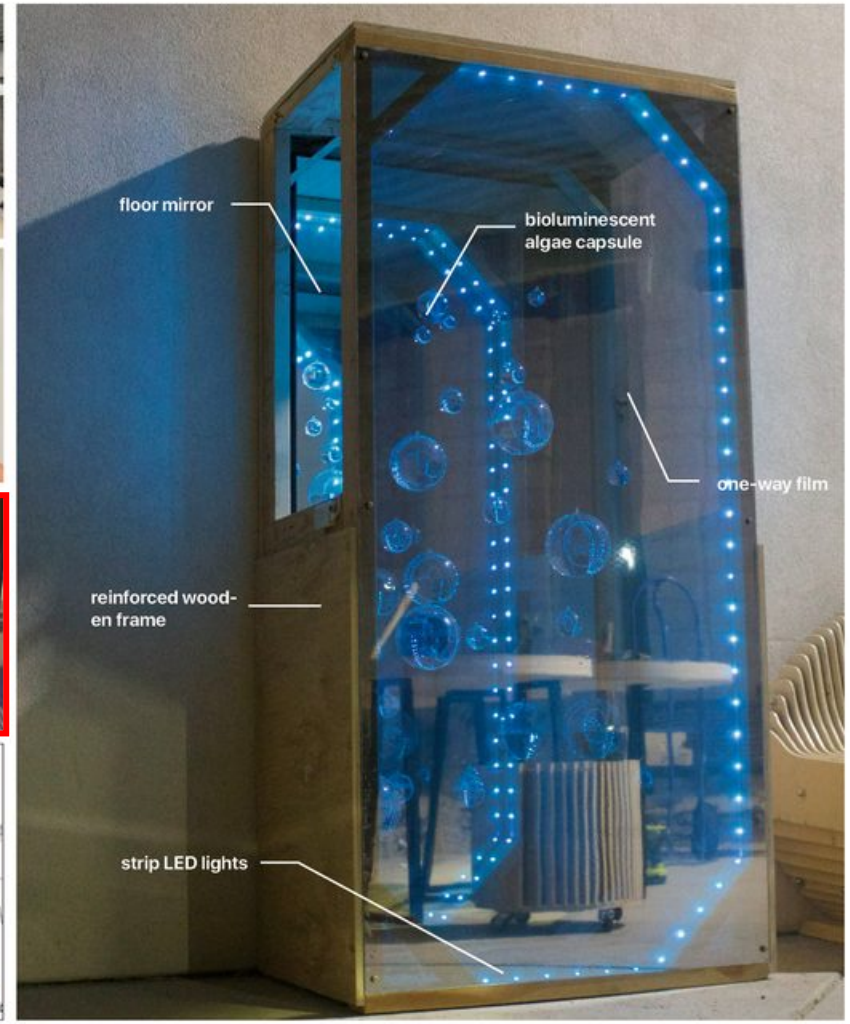
CO2 Frequency



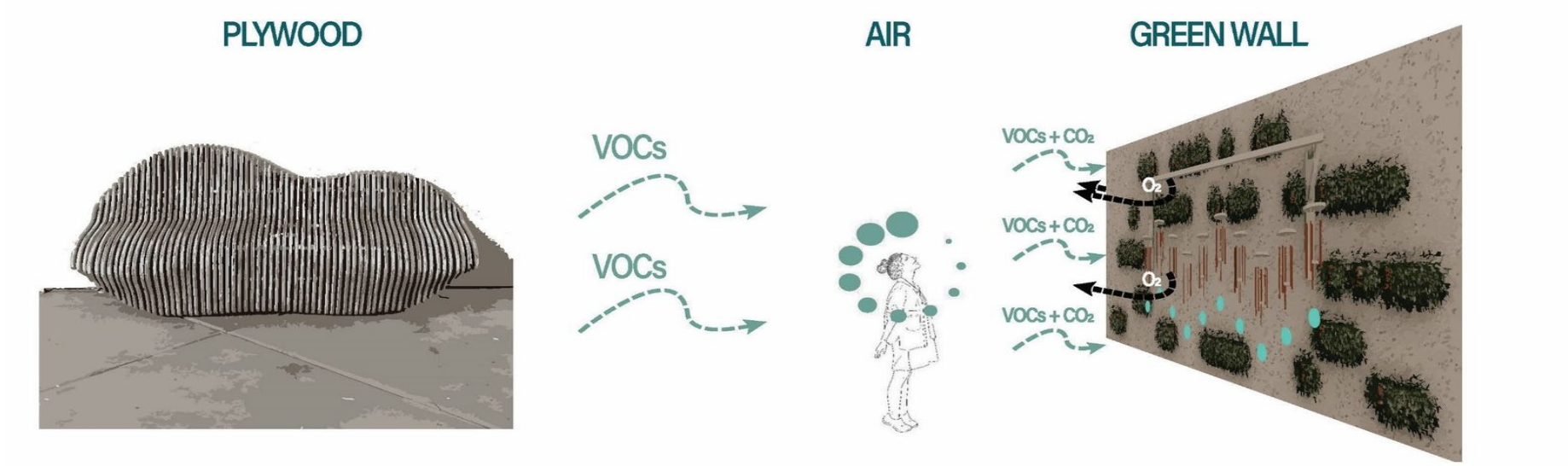
Installation Frame

1. Depicted ornament quantities are for representation purposes only
2. Acrylic screen in the front will have one way film along inside face
3. Backing mirror to be fastened to the frame by existing hardware

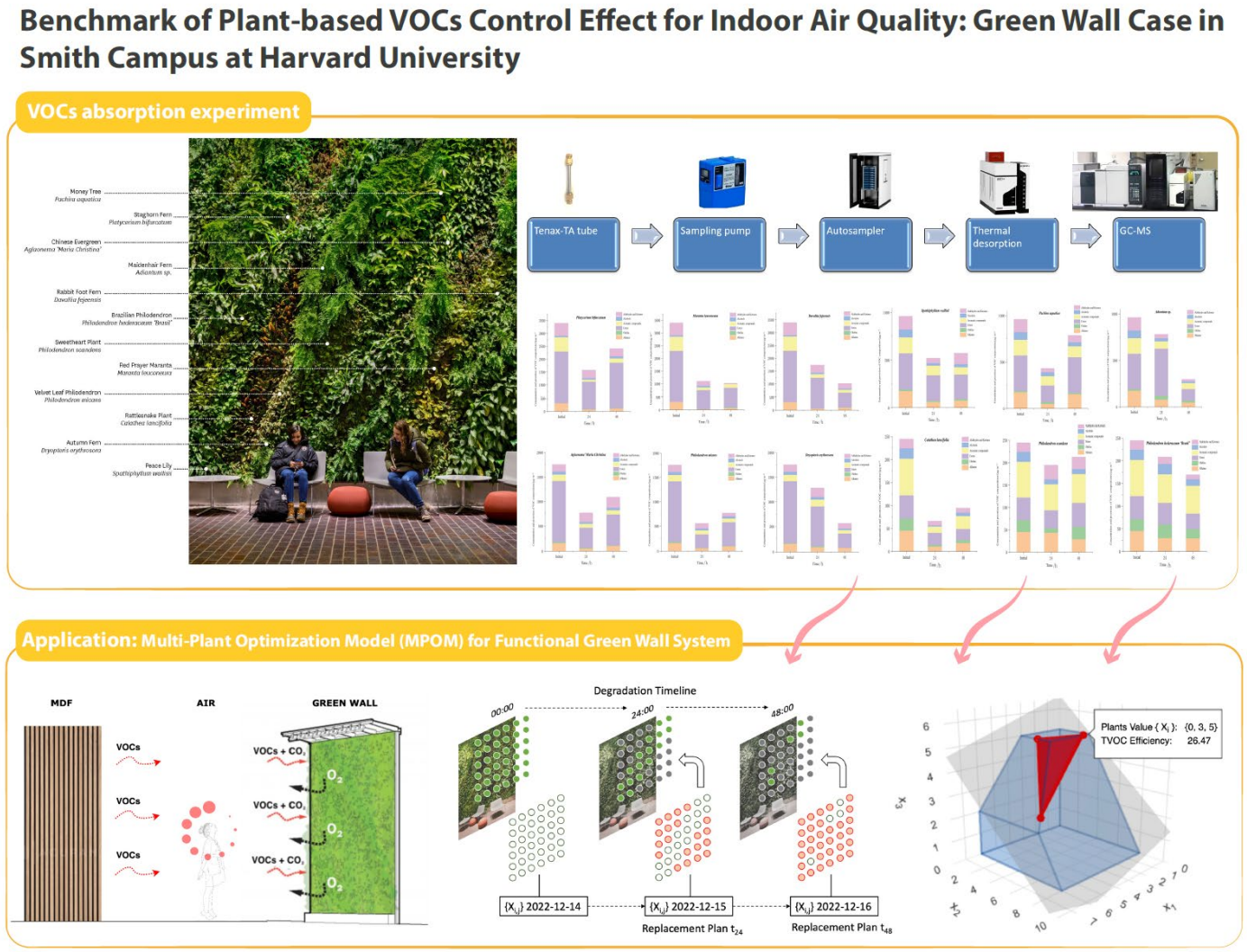




Student Question: “Our group wants to design a green wall to help absorb pollutants released by paint used on plywood. How can we determine if the green wall will actually be effective in doing this? In other words, is the role of a landscape designer only to focus on the aesthetic aspects of a design, or should we also consider its ecological functions?”



Grants for Climate Action with Shanghai Remade, $25,000



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Benchmark of plant-based VOCs control effect for indoor air quality: Green wall case in smith campus at Harvard University

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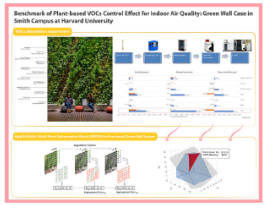
HIGHLIGHTS

- This study analyzes 12 green wall plants, measuring absorption/decomposition of VOCs.
- The study provides an algorithmic model to select green wall plants to manage VOCs.
- The findings suggest methods for designers to create healthier green infrastructure.
- The research integrates science and aesthetics into interior environmental design.

GRAPHICAL ABSTRACT



ARTICLE INFO

Editor: Pavlos Kassomenos

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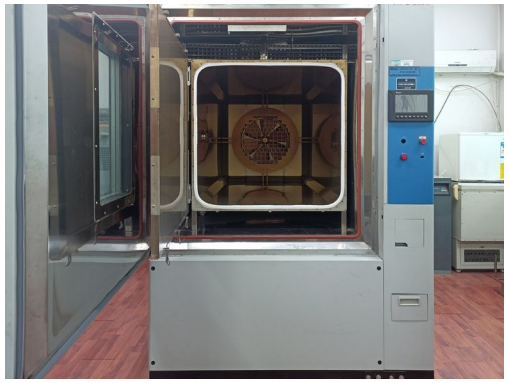
Environmental health
Green wall
Built environment
Air quality

ABSTRACT

The emission of volatile organic compounds (VOCs) from interior materials can significantly impact people's health and daily activities, necessitating effective management. In the construction of the interior built environment, plants serve as a suitable means to enhance air quality. They not only function as ecological living materials for air purification and VOCs removal but are also valued for their aesthetic appeal. However, often the emphasis in designing green infrastructure is placed more on the aesthetics of planting design rather than considering plants' ecological properties. This research examines the ability of 12 plant species used in the interior green wall design at Harvard University's Smith Center as a case study to decompose and absorb VOCs through experimental studies. By integrating ecological properties and key factors such as spatial and budget constraints into the design process, this research explores the potential of using an algorithmic model to select plant species capable of reducing interior VOC pollution in green wall design. The significance of this study lies in its contribution to indoor environmental health and environmental management practices through providing a potential plant selection model and suggesting a relevant workflow for interior planting design with the goal of controlling VOC emissions. By leveraging the knowledge gained from experiments on the VOC removal abilities

To investigate this, I applied for a grant from China and collaborated with several of my BLA students and faculty in the Architecture department to study the creation of functional green walls aimed at enhancing air quality. Our focus extended beyond aesthetics, such as plant shape and color, to the practical benefits these walls can provide. We began with a case study on the green walls at Harvard University's Smith Campus Center, a project completed by Michael Van Valkenburgh Associates. This is a renowned landscape architecture firm, where I also had the opportunity to work a few years ago. For those unfamiliar with Smith Campus Center, it serves as a student union with dining facilities. The kitchen releases volatile organic compounds (VOCs) during cooking, making it an ideal setting to test plants' VOC absorption capacity. We also chose this example to examine whether top-tier landscape design projects prioritize plant selection solely for aesthetic appeal or if they consider the plants' ecological functions as well.

Experiment Settings for Short-term VOCs Absorption Test



Tenax-TA tube

- Containing 2, 6-dibenzofuran porous polymer adsorbent 200 mg, can effectively adsorb and release VOCs

Sampling pump

- Constant flow Air sampling Pump, flow range 100 ~ 600 mL/min, error not more than ±5% of the set value

Autosampler

- ULTRA 100 bit thermal desorption automatic sampler，Markes UK

Thermal desorption

- The thermal desorption instrument of UNITY produced by Markes Company, UK, has a cold trap adsorption temperature of -15 ℃, helium carrier gas, carrier gas flow rate of 30 mL·min^{-1}, analytical temperature of 300 ℃, pipeline temperature of 180 ℃, thermal desorption and desorption of samples for 10 min, pre-purge for 1 min, and sample injection time for 1 min.

GC-MS

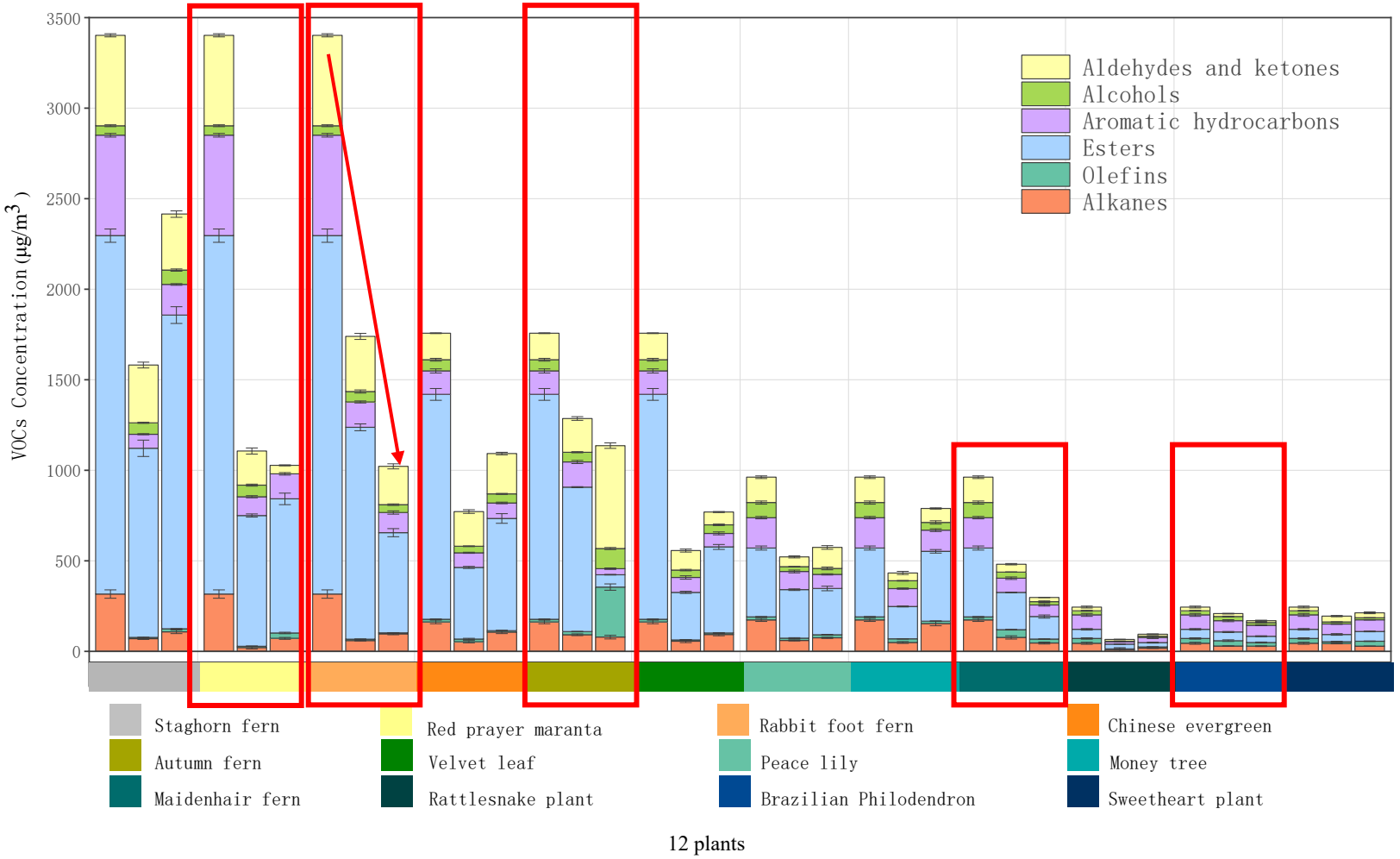
- Thermo produced DSQ single four-pole gas chromatography-mass spectrometry (GC/MS), instrument column type DB-5.

First, we collaborated with a local materials science lab to examine the VOC absorption capabilities of plants from the project's plant list. This approach is highly scientific and requires the use of costly equipment. In the study, MDF board coated with a clear finish served as the pollutant source. The process, illustrated in the diagrams, involves collecting air samples using Tenax-TA tubes and analyzing them with GC-MS. This method provides highly accurate results. However, because it was originally designed to test non-living materials, it doesn't replicate essential conditions for plant growth, like light and air exchange. As a result, it can only measure short-term changes over a 48-hour period.

Experiment Result

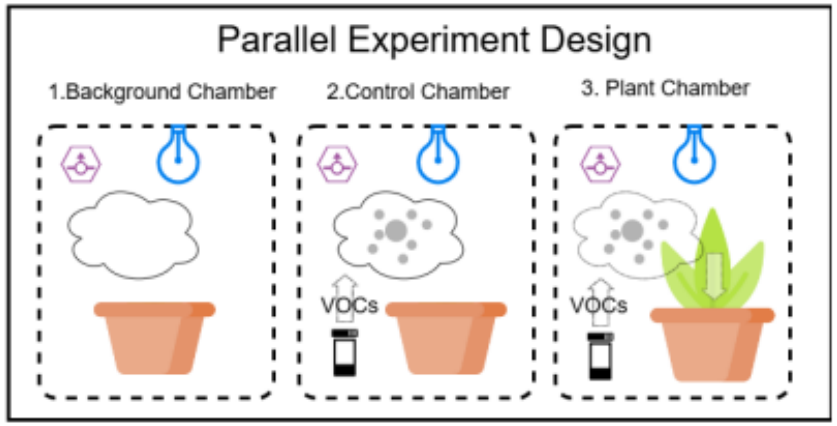
Mns`kkok`msr g`ud sgd
`akksx sn bnmshmt nt rkx
`arnqa UNBr nudq`rgnqs,
sdql odqhmc+rt bg`r 37
gnt qr-

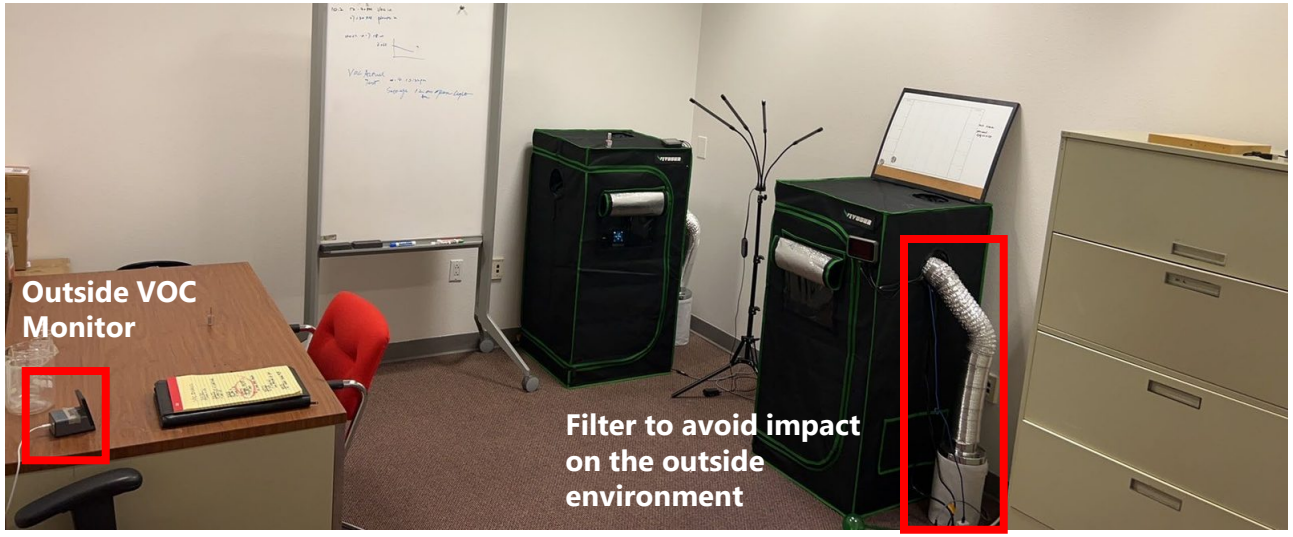
Gnv @ant s Knmf Sdql
'6,0/ c`xr(

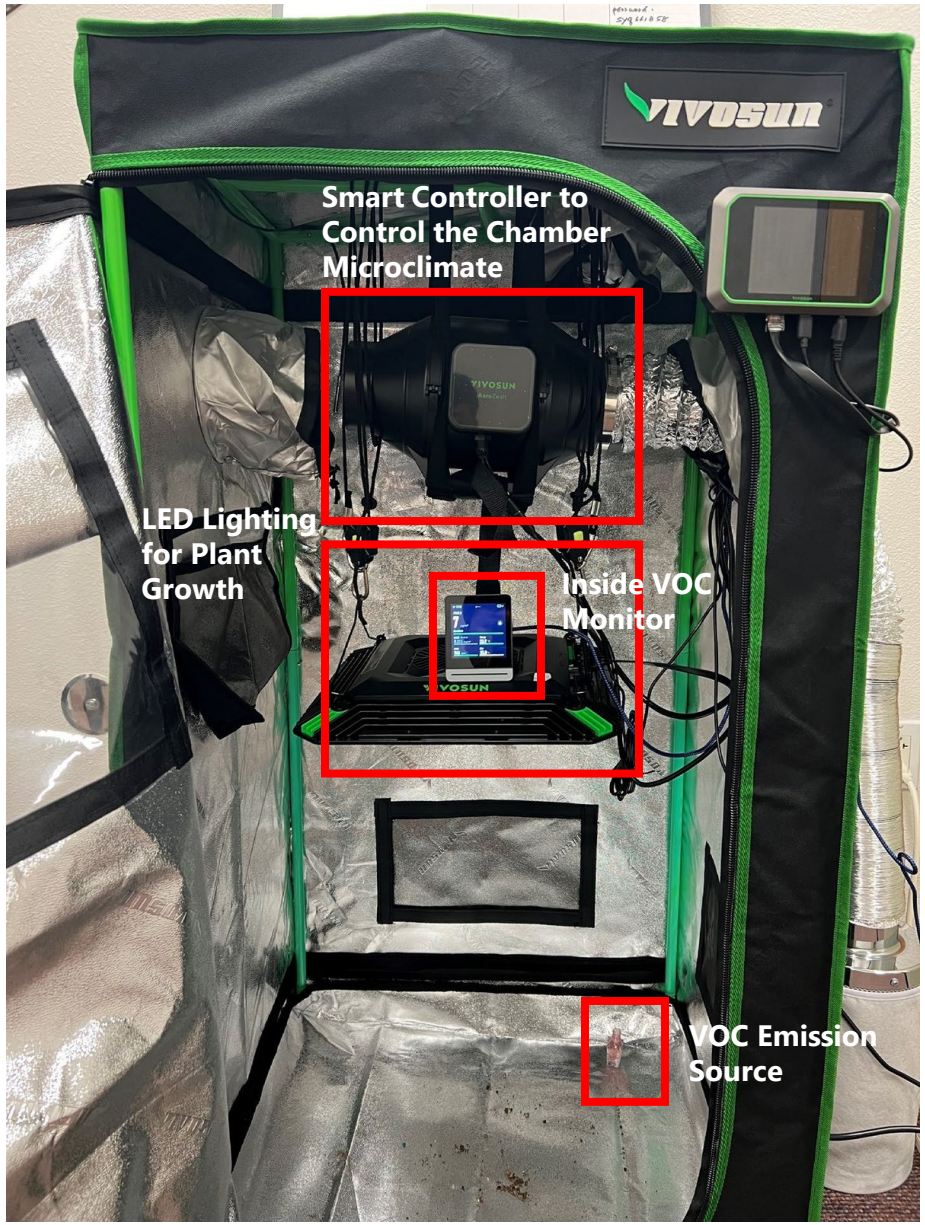


This diagram highlights some noteworthy findings. As shown within the red box, certain plants maintained their ability to absorb VOCs for 48 hours, while others lost this capability within the same timeframe. This indicates that not all plants can sustain VOC absorption over short periods, such as 48 hours. These results also reveal a knowledge gap in current landscape design practices, which often lack a strong ecological foundation. While most landscape design programs emphasize aesthetics, they sometimes overlook scientifically grounded approaches that could enhance the effectiveness and sustainability of designs. Additionally, compared to other fields, we face limited funding for expensive equipment, emphasizing the need for cost-effective solutions in our research. To further investigate these five plant species and assess their long-term VOC absorption potential, we extended the study period to 12 days and introduced automated controls for growth light cycles.

Experiment Settings for Long-term VOCs Absorption Test



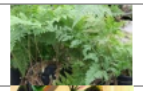




We use the rest of fund purchase the removable climate chamber with smart control of microclimate and air exchange motor and wi-fi supported the VOC monitor. Compared to the last one, this cannot analyze the components of each VOC but could test the Total VOC, still good enough for our research proposed. Each chamber only cost 2000 dollars.

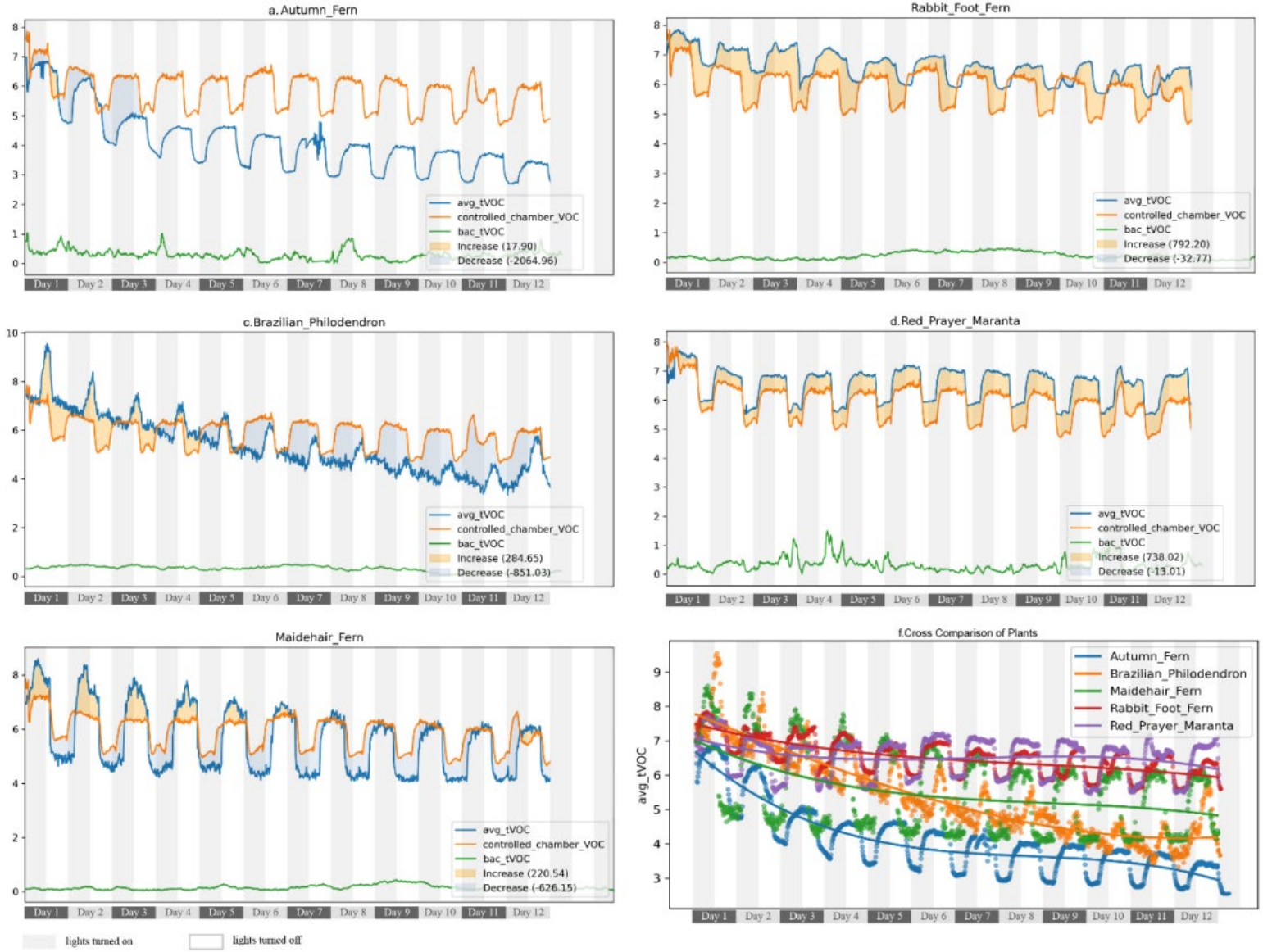
Experiment Result

Ok` ms Klrs Rdkdbsdc eqnl
Rgnqs,sdql Sdrs

Common Name	Latin Name	Samples
Rabbit foot fern	Davallia fejeensis	
Red prayer maranta	Maranta leuconeura	
Autumn fern	Dryopteris erythrosora	
Maidenhair fern	Adiantum sp.	
Brazilian philodendron	Philodendron hederaceum ‘Brasil’	

@t st l mEdqm+L `hc dmg`hq Edqm+`mc
Aq`ykkì mOghkn c dmc qn mr hf mhelb`msk x
qdc t bdc SUNB kdudkr nudq` 01,c`x
odqhnc-

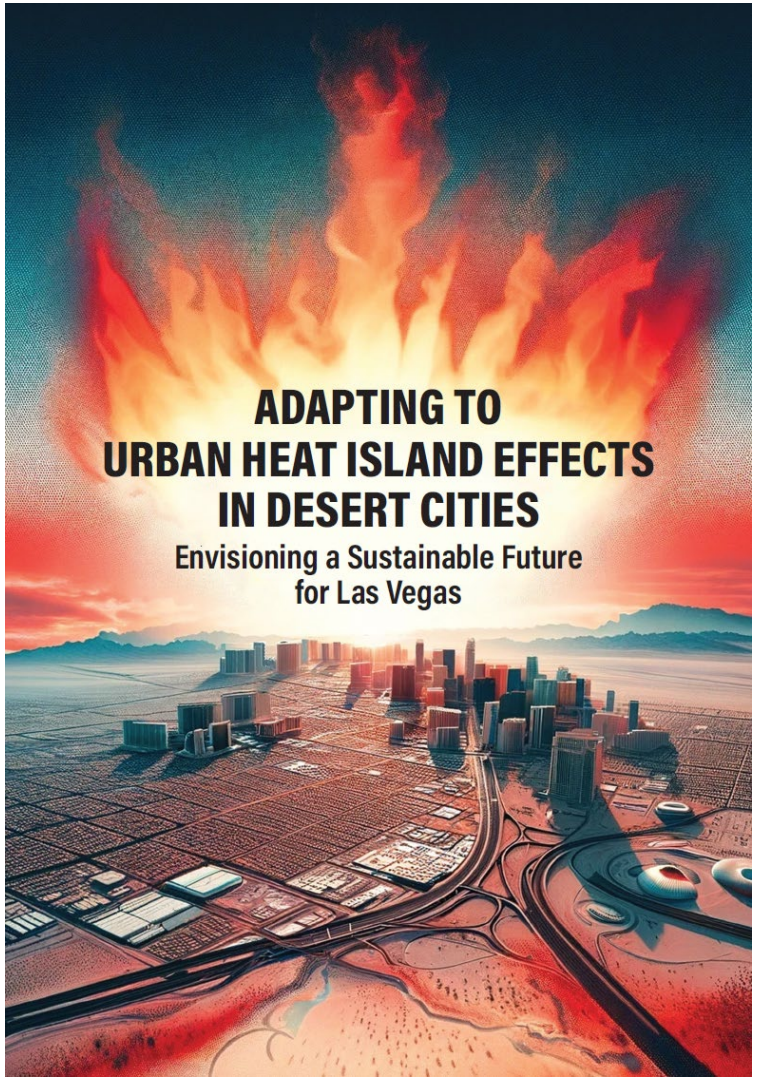
Sghr gdko sgd et st qd ok`ms r dkdbshnmr enq
f qddmhmeq`rsqt bst qd c drhf m-



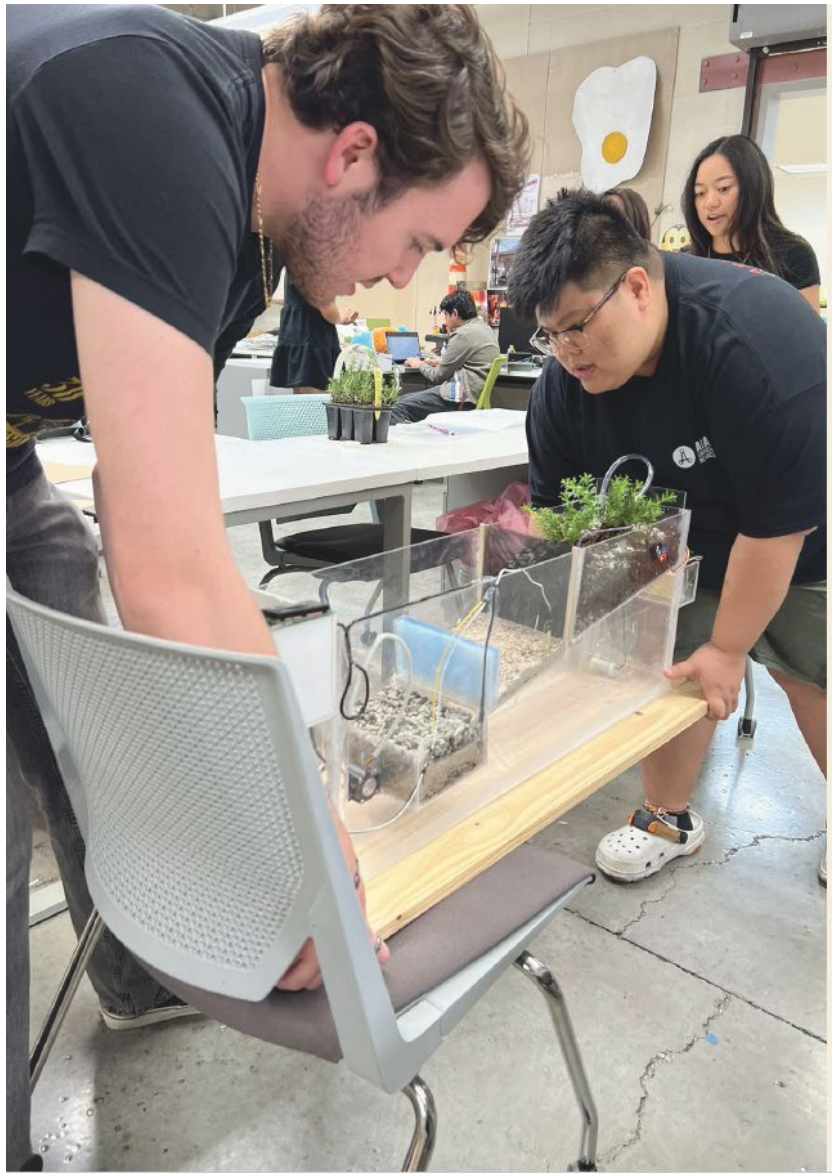
The results suggest that some plants have the potential for long-term VOC absorption, making them suitable materials for green infrastructure aimed at enhancing air quality in built environments. This affordable equipment setup is versatile, allowing us to test any suitably sized plants for their long-term VOC absorption capacity. With a growing database, designers could select plants based on both aesthetic and ecological performance, aligning with the principles in my earlier brownfield plant selection project.

Student Question: “How could we Achieve the Data for Constructing A Scientific Design Strategy?”

Student Work: LAND 330 Design With Climate

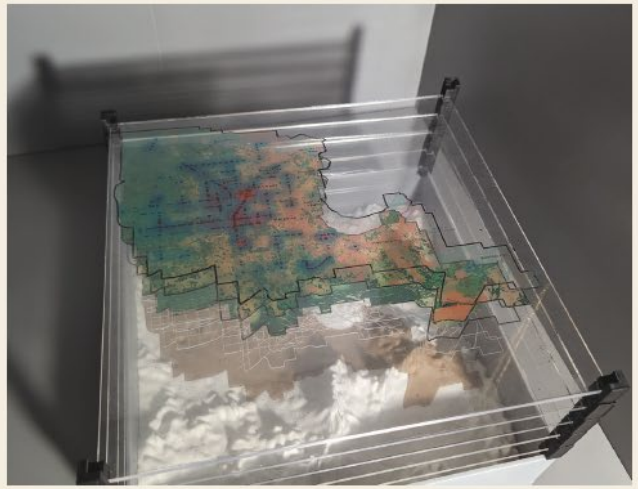


Booklet



4 DESIGN SOLUTIONS FOR UHI MITIGATION

This chapter explores the application of interactive design principles to mitigate the Urban Heat Island (UHI) effect, a phenomenon where urban areas experience significantly higher temperatures than their rural surroundings. Interactive design in this context involves the strategic use of materials, urban layout, and green infrastructure that respond dynamically to the urban environment and its inhabitants. By focusing on user-centered design, the chapter discusses how urban planners and architects can incorporate smart technologies and responsive systems to optimize shade, airflow, and green coverage in cities. These interventions are designed to reduce heat absorption and increase cooling, thus making urban areas more livable and energy-efficient. We will also look at various case studies that demonstrate successful implementation of interactive designs in cities around the world, analyzing the outcomes and lessons learned from each. This approach not only addresses the immediate challenges posed by UHI but also contributes to the broader goals of sustainable urban development.

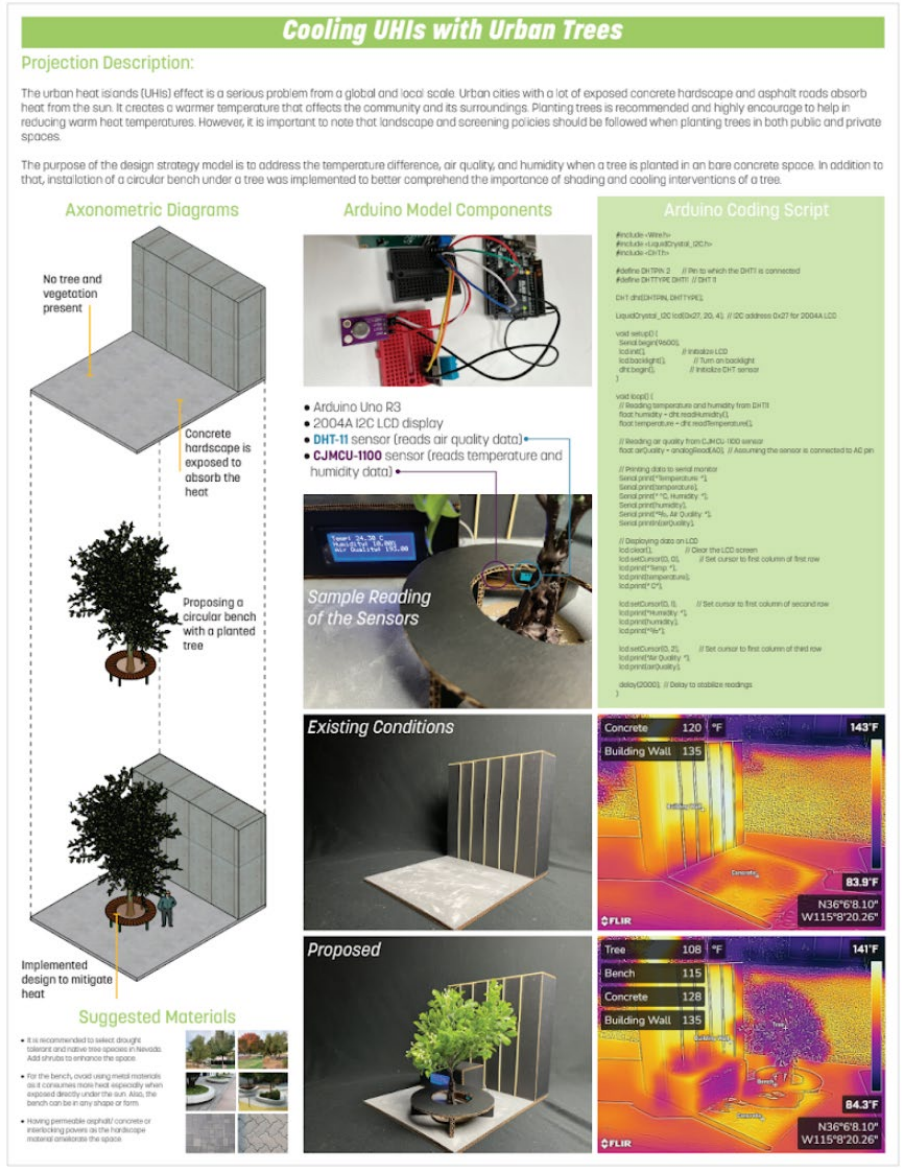


DESIGN SOLUTIONS

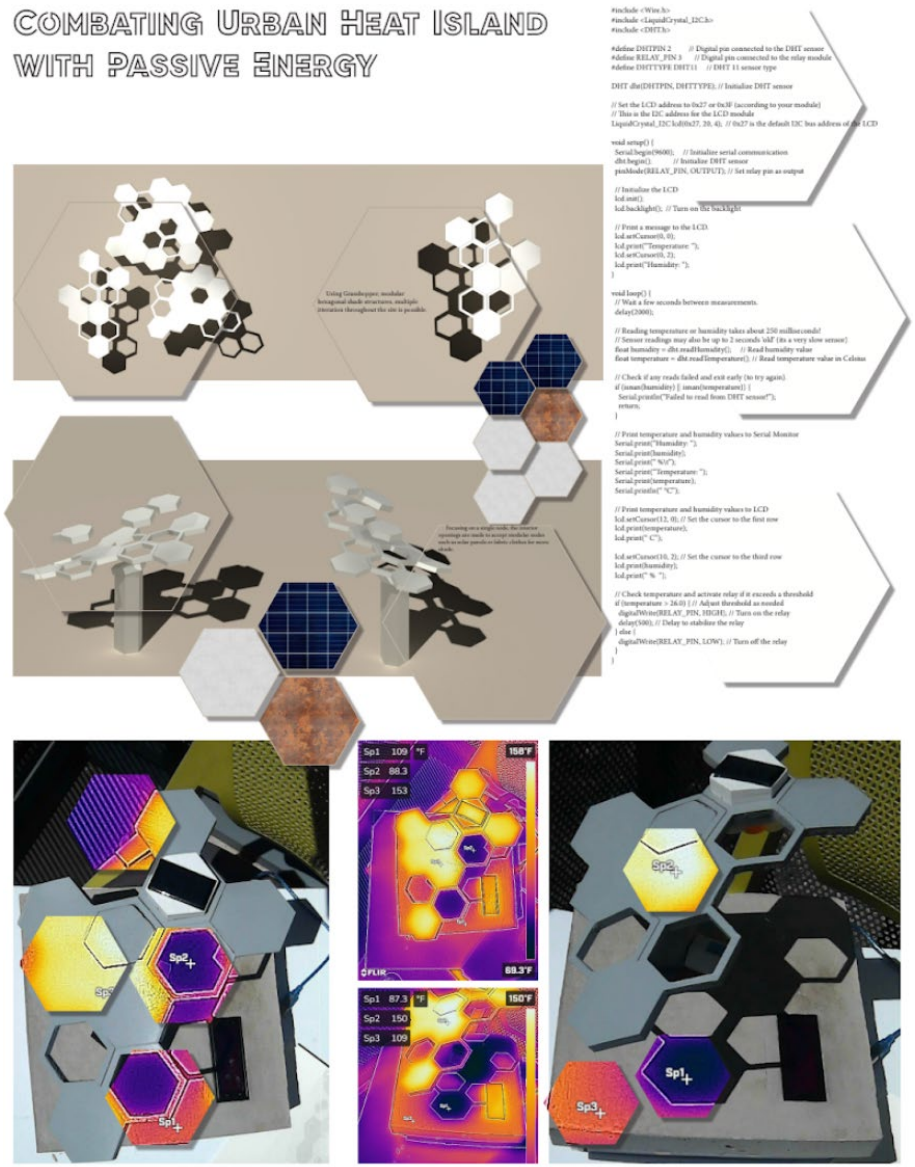
75

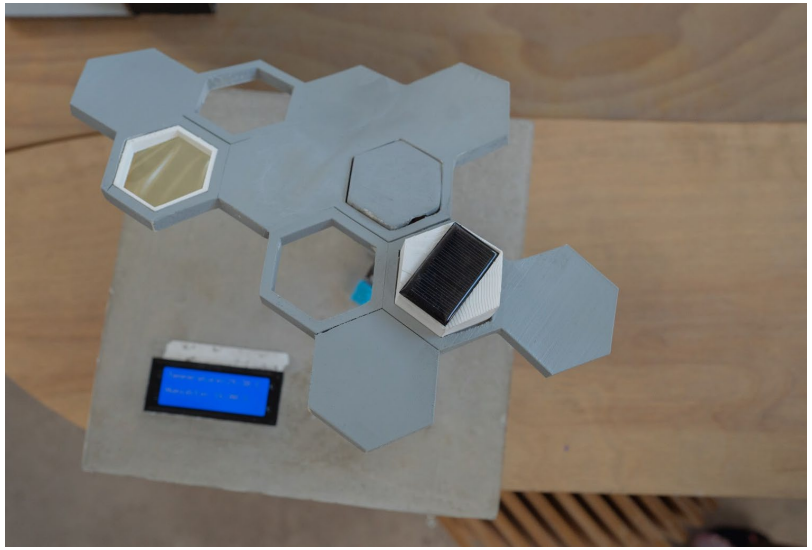
All students work eventually are collaborated as a booklet. The first three chapter are about the individual topics that caused the UHI in Las Vegas. I want to highlight the Chapter 4 which ask student using Arduino sensors, scale model, FLIR thermal camera to test if their design could really improve the UHI.

Interactive Models



As I mentioned, each student is assigned a unique topic related to UHI. They develop their projects by typing their inquiries into ChatGPT, and I assist them with debugging as their instructor. For example, one student is studying the impact of urban trees. She designed a bench with a tree to explore how much cooling the tree provides. Using a thermal camera, she took pictures to compare the temperature differences and employed environmental sensors to collect time-sequenced data to track changes from morning to night. This approach helps her gain a clearer understanding of the amount of vegetation needed to effectively mitigate UHI.





Same idea, this student design a solar panel based responsive climate fans. He used concrete and 3D print machine plus some sensors to construct this model. The fan will run once the temperature sensor test the different temperature.

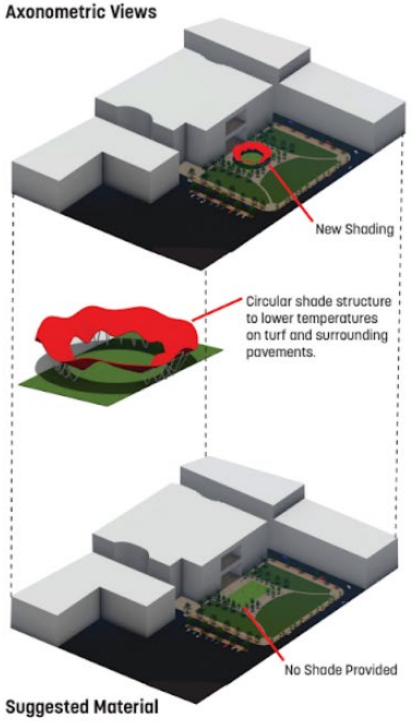
Pavements & UHI: Shade Design

Statement

Urban heat island effect is a consistent problem that continues to plague many cities across the globe. It has negative and uncomfortable effects towards pedestrians outdoors. To combat this, pavement materials and colors were researched to help combat and reduce UHI.

The focus area of Downtown Summerlin in Las Vegas, Nevada was selected to try and find a solution to reduce UHI effects on its pavements and users of the space. As a result, a new circular shade design was developed to provide shading, reduce temperatures, reduce light intensity, and further limit the impacts of UHI.

Axonometric Views



Suggested Material

Fabric would be an appropriate material to utilize for this shade structure design. It will not capture as much heat as other materials such as wood or metal, which will lead to reduced temperatures. Using colors such as red or white would also help lower temperatures captured.

Arduino Code

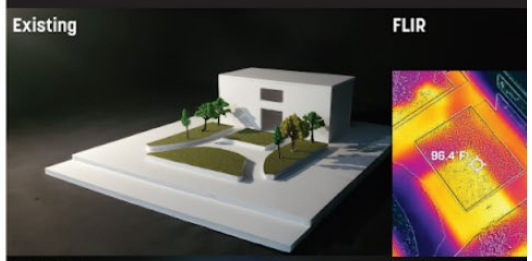
Part 1

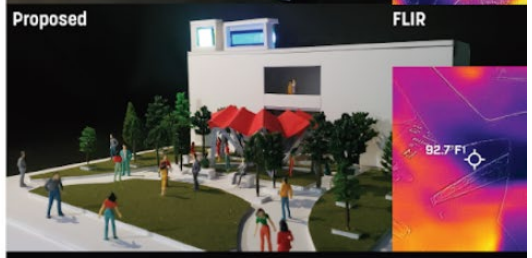
```
#include <Wire.h>
#include <Adafruit_AMG88xx.h>
#include <Adafruit_ST7735.h>
#include <LiquidCrystal_I2C.h>
#include <BH1750.h>

#define TFT_CS 10
#define TFT_RST 9
#define TFT_DC 8
#define BLOCK_SIZE 16

Adafruit_ST7735 tft = Adafruit_ST7735(TFT_CS, TFT_DC, TFT_RST);
Adafruit_AMG88xx amg;
LiquidCrystal_I2C lcd(0x27, 16, 2);
BH1750 lightMeter;

void setup() {
  Serial.begin(9600);
  tft.initR(INITR_BLACKTAB);
  tft.setRotation(1);
  lcd.init();
  lcd.backlight();
  if (!amg.begin()) while (1);
  lightMeter.begin();
  lightMeter.configure(0x10);
  delay(100);
}

void loop() {
  float pixels[AMG88xx_PIXEL_ARRAY_SIZE];
  amg.readPixels(pixels);
  float sum = 0.0;
  int count = 0;
  memset(pixels, 0, sizeof(pixels));
  for (int y = 0; y < 8; y++) {
    for (int x = 0; x < 8; x++) {
      int bx = x * BLOCK_SIZE;
      int by = y * BLOCK_SIZE;
      float temp = pixels[y * 8 + x];
```

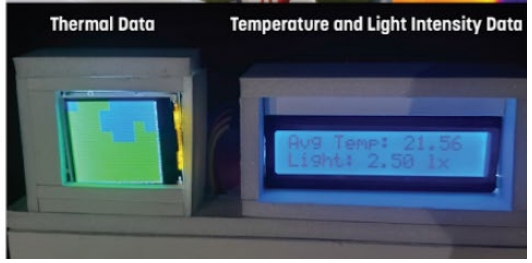
Part 2

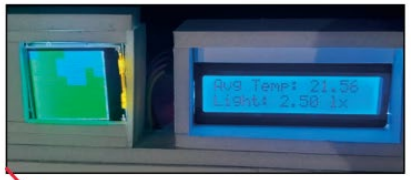
```
      int color = (temp <= 22) ? ST7735_GREEN : (temp
<= 24) ? ST7735_RED : (temp <= 26) ? ST7735_
ORANGE : (temp <= 28) ? ST7735_YELLOW : (temp
<= 30) ? ST7735_CYAN : ST7735_BLUE;
      for (int i = 0; i < BLOCK_SIZE; i++) {
        for (int j = 0; j < BLOCK_SIZE; j++) {
          tft.fillRect(bx + i, by + j, 1, 1, color);
        }
      }
      sum += temp;
      count++;
    }
  }
  float avgTemp = sum / count;
  float lightIntensity = lightMeter.readLightLevel();
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Avg Temp: ");
  lcd.print(avgTemp, 2);
  lcd.print((char)223);
  delay(1000);
  lcd.setCursor(0, 1);
  lcd.print("Light: ");
  lcd.print(lightIntensity);
  lcd.print(" lx");
  delay(1000);
}
```

Arduino Parts:
- -Uno R3
- -Bh1750 Sensor
- -Amg8833 Sensor
- -1602a Display
- -St7725 Display









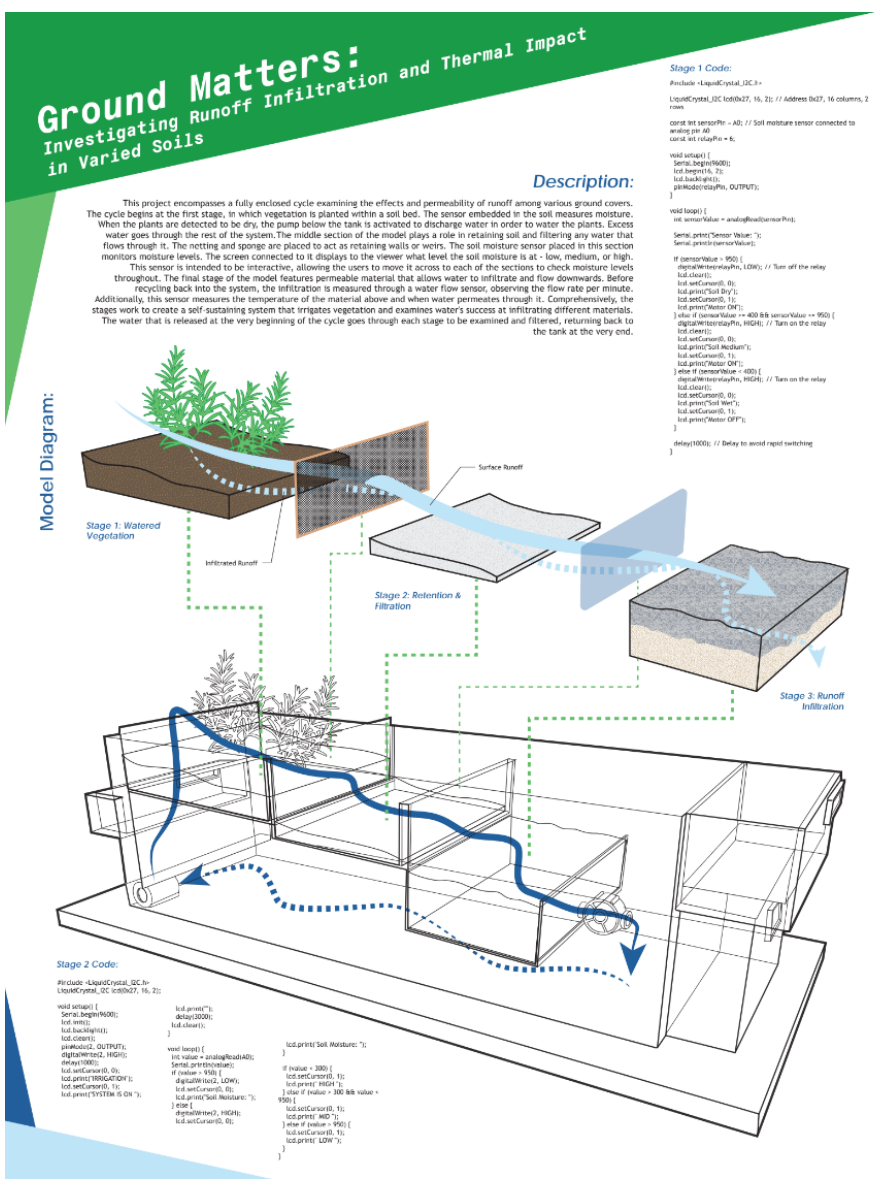
Lcd Data Displays:

A thermal Lcd display and a 2nd Lcd displaying live temperature and light intensity data provided by the sensors.

Sensors Collecting Data:

2 sensors are inside the shade design that captures light intensity and the other captures temperatures.

This student is interested in creating corridors in an anti-uhi area. By using the FLIR thermal camera, he tested different pavement ‘s heat absorption ability and build the corridor on the cooling area, he also designed the motion sensors (red box) which could catch up and record the movement of the wildlife animals.



The objective of this model is to examine water's infiltration through a number of soil conditions and materials. Within three interconnected stages, water runoff and moisture retention are measured. Stage one is a self-irrigating planter that tests the soil's water retention and allows water to filter to the next stage. Here, built components serve as walls and weirs retaining runoff and filtering it into the last stage. This last process collects all of the filtered runoff and examines the rate at which water infiltrates before returning to the tank to be recycled through the system again. All three of these can be compared to one another and existing urban conditions to examine the differences in water and heat retention.

Ground Matters:
Investigating Runoff Infiltration and Thermal Impact in Varied Soils
Description:
Progress Images
Model Images

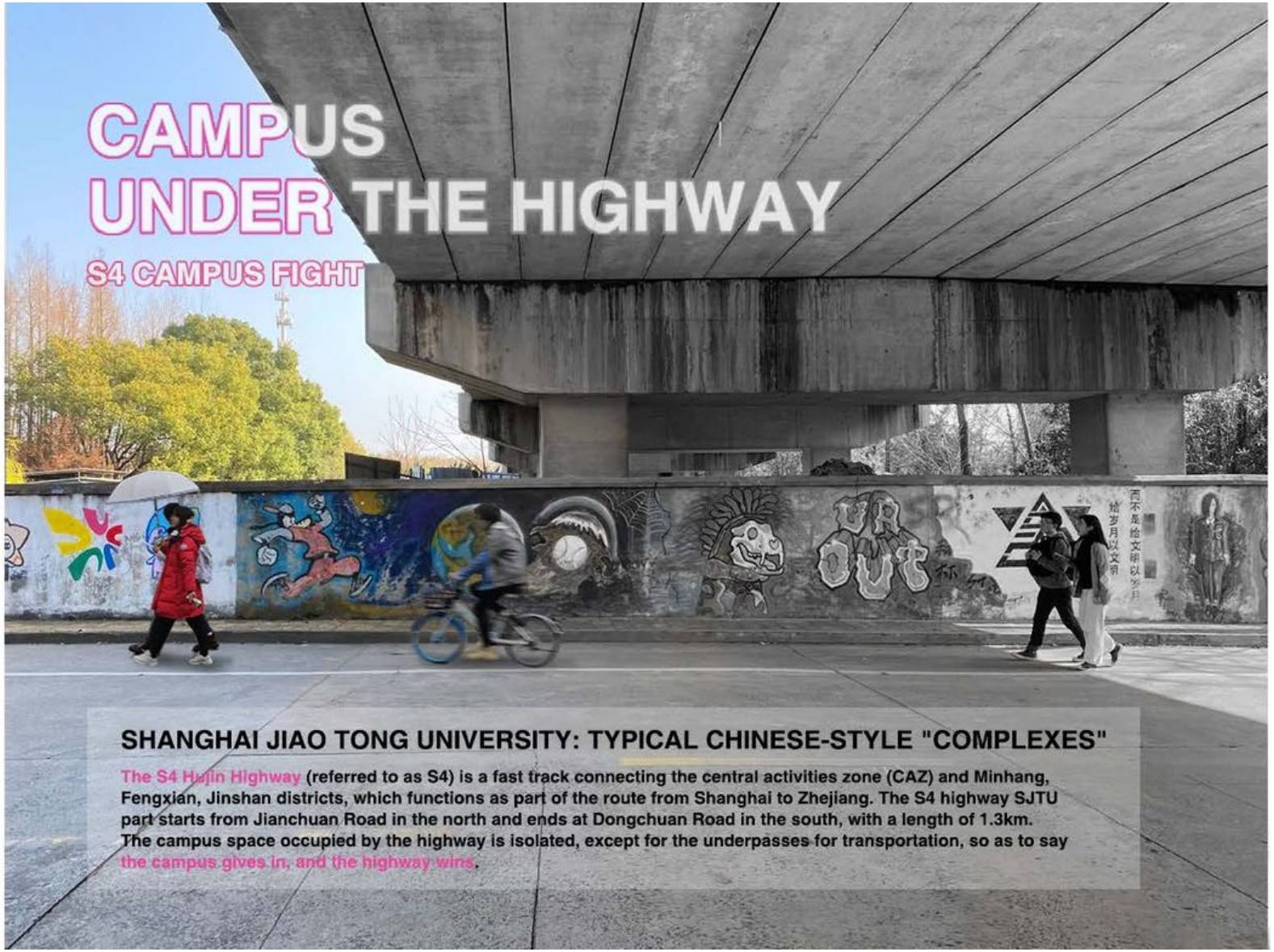
Key Concept II:

“Community Engagement”

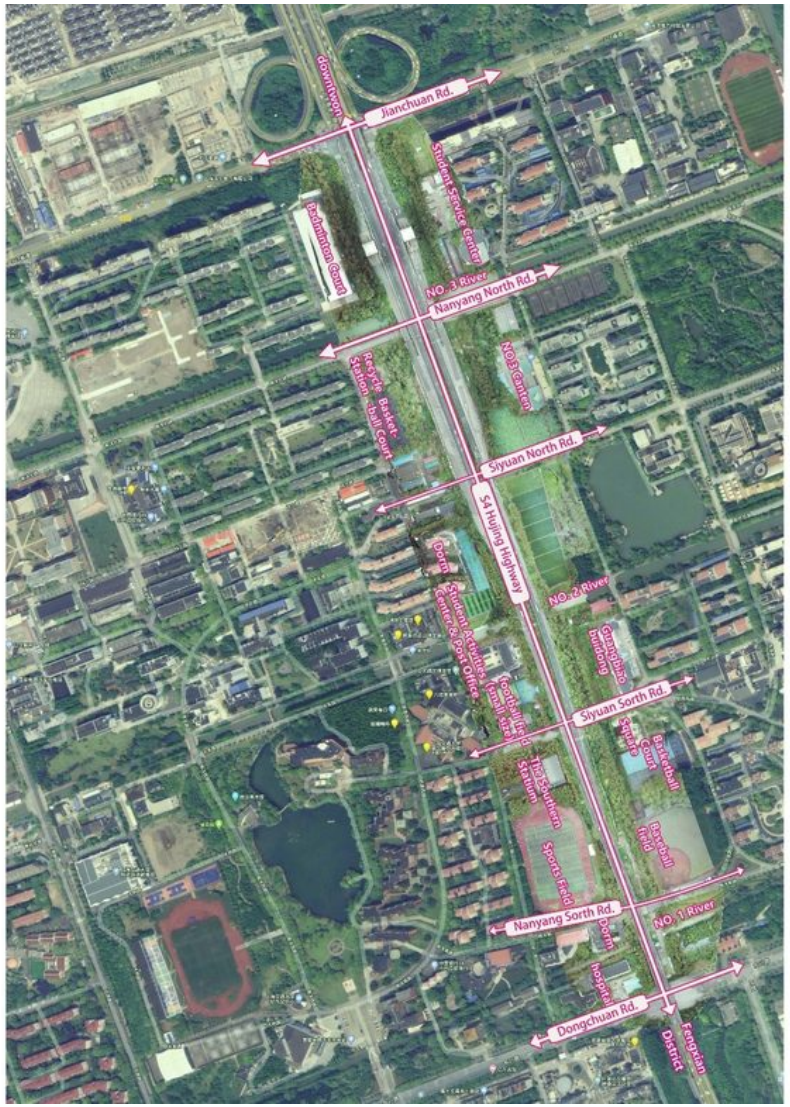
+

“Communication with Users!”

Student Work: STJU MLA Studio I Campus Fight!

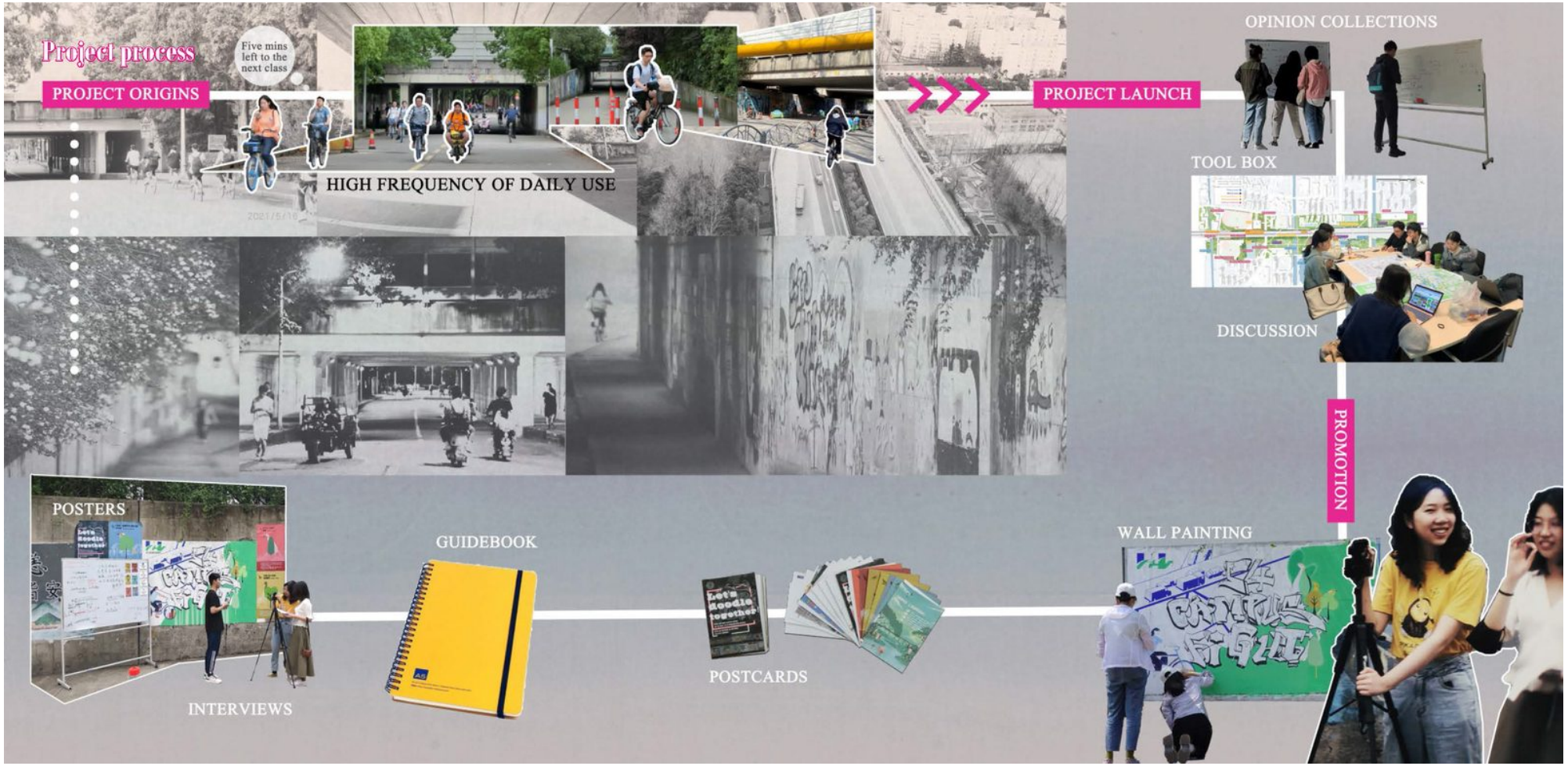


Site Inventory



There is a hwy that split up the campus. It becomes an invisible boundary that split up the campus. This boundary also make the students feel two different culture and atmosphere. Thus, the design aims at applying various system to dismiss this boundary and provide equity environment

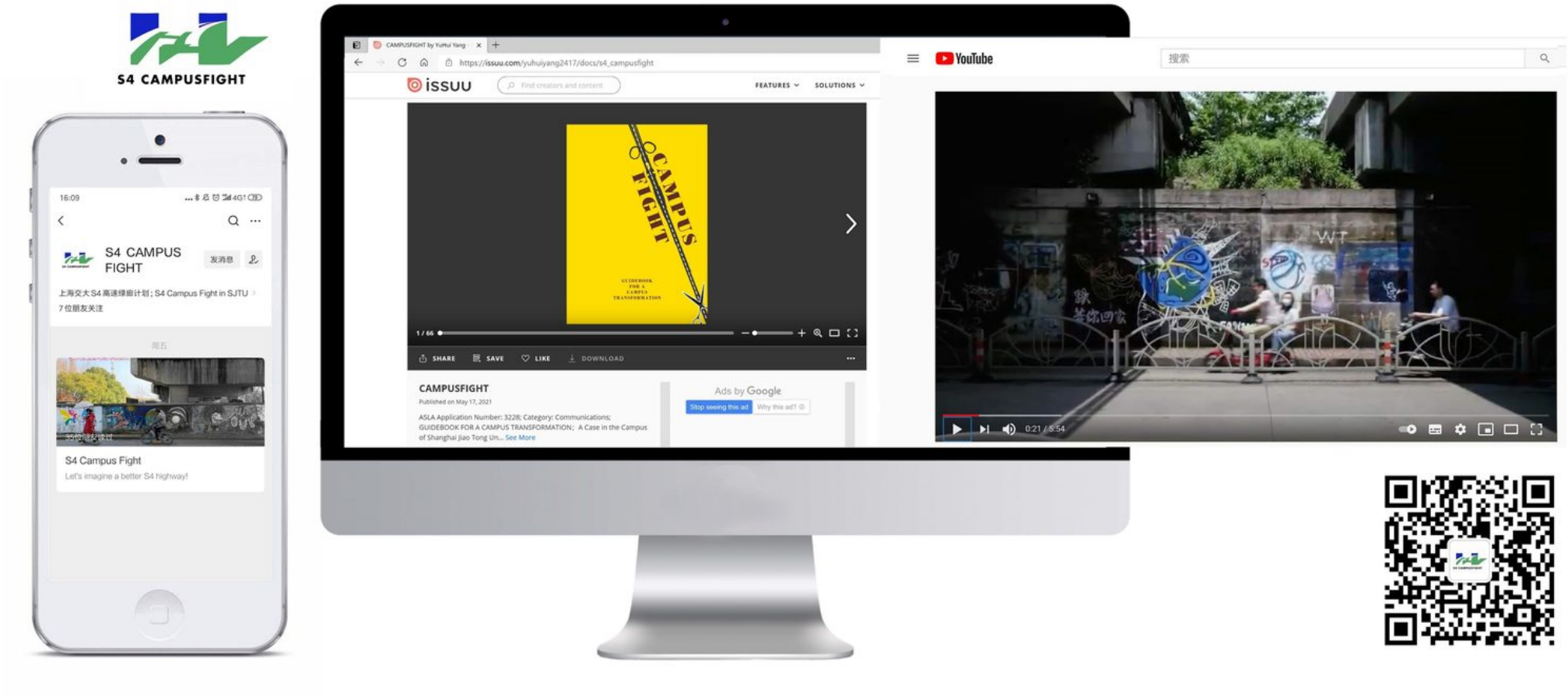
Theoretical Framework



The project is focus in interviewing the students, faculty to about their opinion about this issue. After gathering the primary data, the design strategies are focus on establishing the connection between human behavior such as landscape performance and the built environment. The research product include creating innovated programs in the places that receive less attention. The goal of the proposal is to make the balance environment for the spliced campus. The projects start from opinion collection from users, create toolkits, discuss and adjust them, promotion the ideas with audience, create guidebooks to campus planners and share, interview, exchange the ideas with users to discuss the feasibility of the design strategies.

Deliverables

Communications Platforms



The final deliverables are social media posts, hard copy and digital copy of guidebook, and a video to prompt the ideas to the public

CAMPUS
FIGHT
GUIDEBOOK
FOR A
CAMPUS
TRANSFORMATION

WHAT IS MISSING
FROM OUR CAMPUS?

CAMPUS
FIGHT

10 ROUTES
22 WATER ECOLOGY
40 REVOLUTION OF CULTURE
02 CAMPUS UNDER THE HIGHWAY
18 HIGHWAY ELEVATION
32 NOISE ECOLOGY
50 EDUCATION

Background
Heavy Traffic
Connections of Dorms, Teaching Blocks and Sports Areas
"S4 Effect"

Heatmap
Calculations Based on Wool-thread Model
Routes based on MCR
Greenway Routes

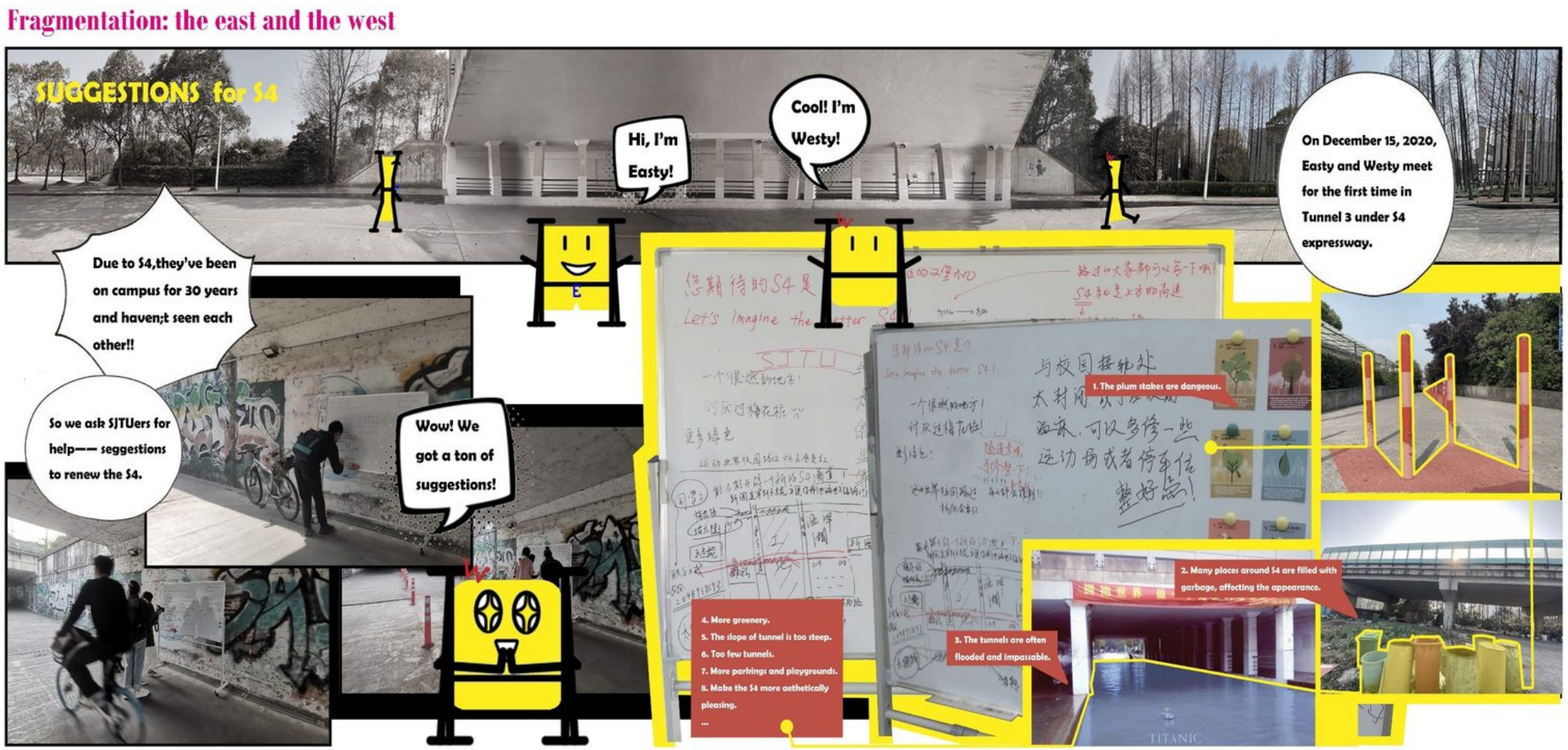
Vertical Properties
4 Toolkits

Wildlife Path
Case Study
Animal Channel Construction
Habitat Adaptability Elevation Matrix
Shoreline System

Plant Status
S4 Noise Status
Noise Distribution
Priciple of Noise Reduction
Planting Strategy

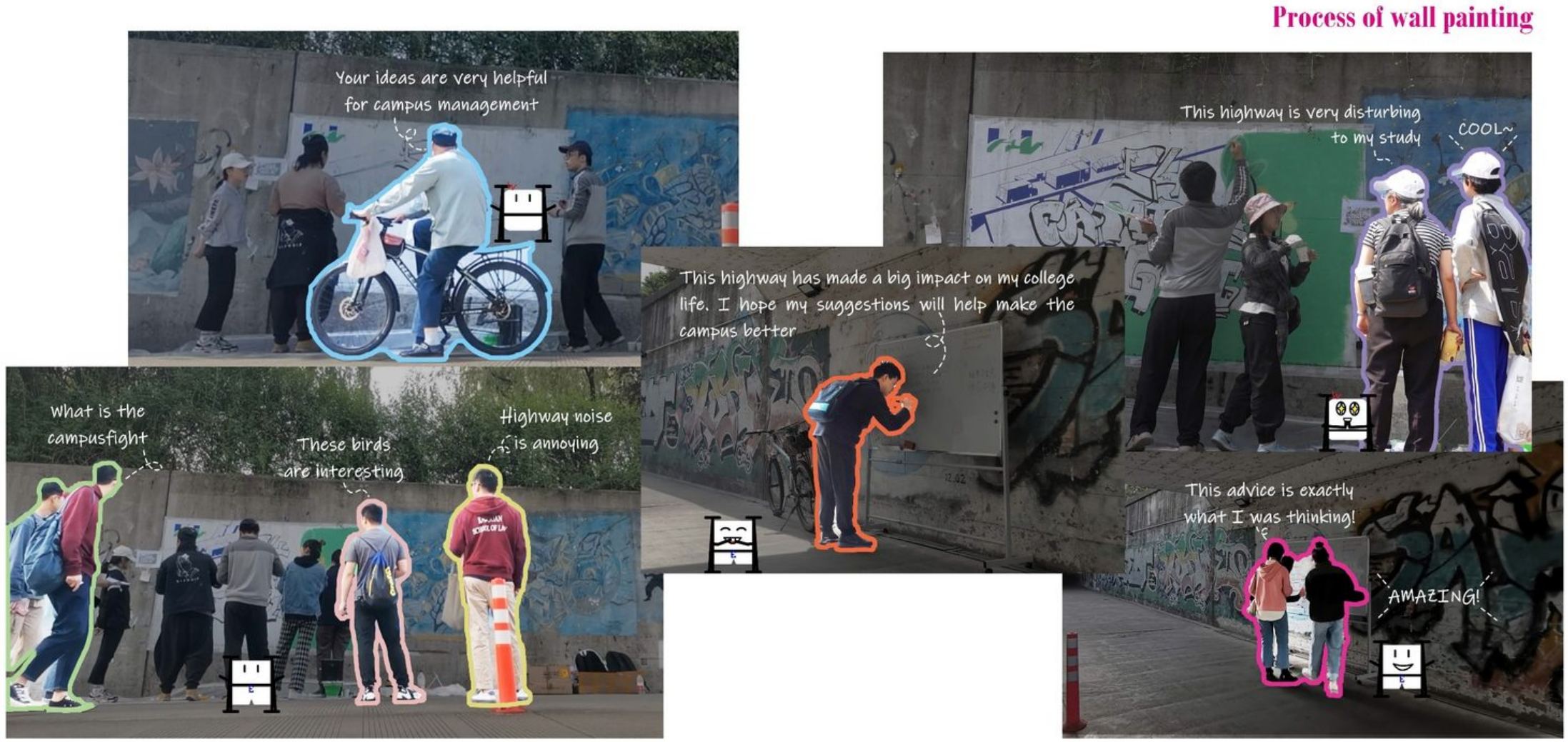
What is Missing from Our Campus
Campus Images
3 Steps
Toolkits

Data Collection



Many theories from behavior aspects of park, recreation and tourism management are being applied. Such as Leisure constraints and facilitator, environmental phycology. Theses are the key parameters to categorize the data from interviews. Here shows one significance spot under the HWY. the students write down their suggestions. Many students are faculties are participated.

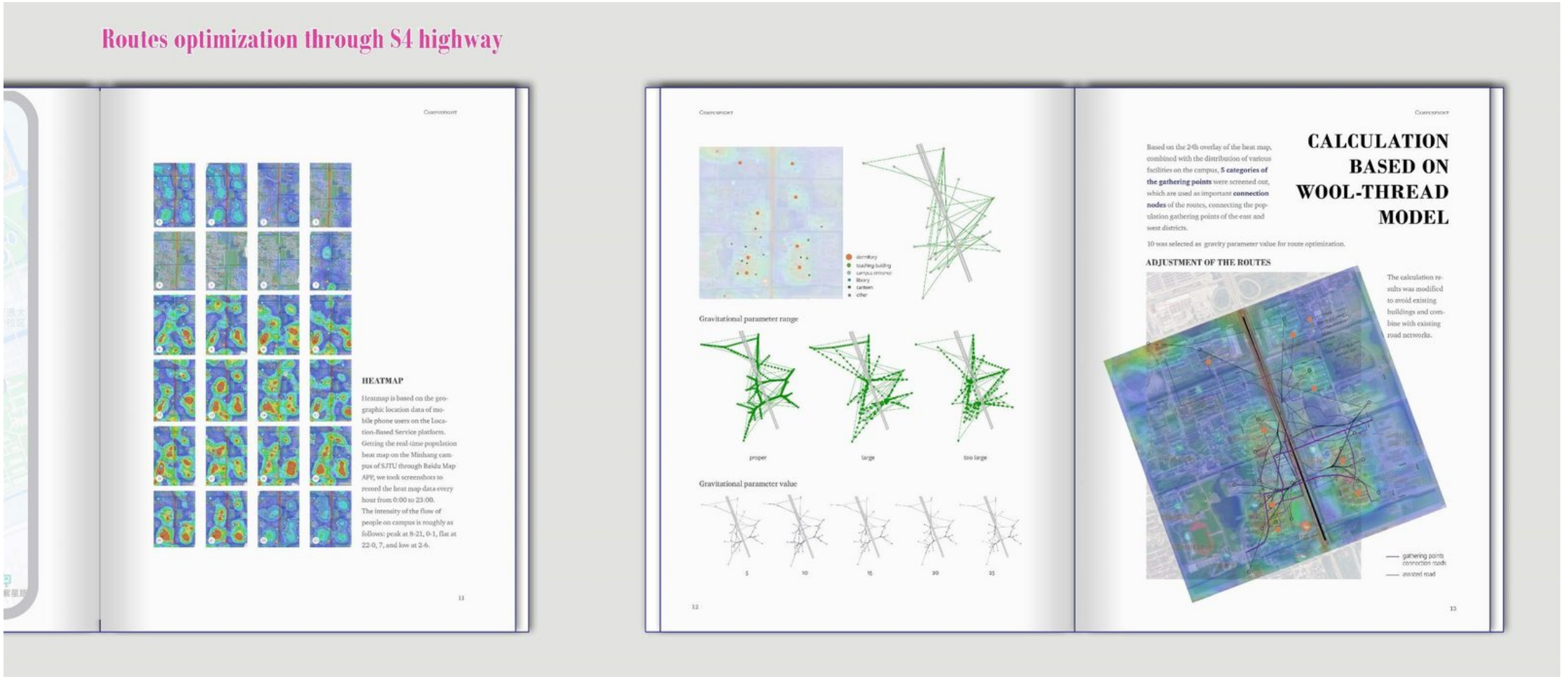
Data Collection



You can see the place has been a empty space for a long time. Many wall paintings. This is also a code of human behavior that shows it might be a interesting place to do the wall painting and attract the people then dismiss this boundary.

Typology

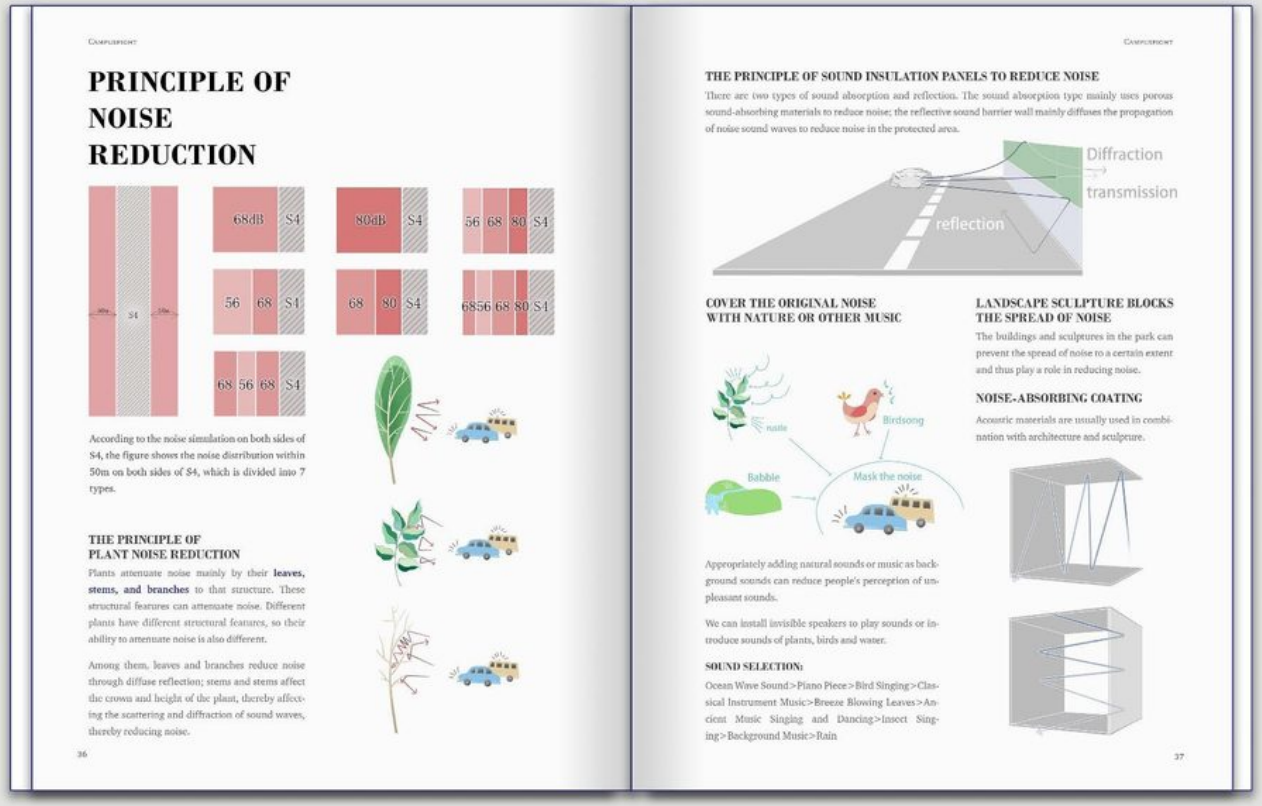
Routes optimization through S4 highway



The research also measure the heat map for the most popular spots in the campus. Based on these behaviors, the design proposes the new programs in the weak place to attract more population to come. The right pages simulates this and generate the new visiting route.

Typology

Noise reduction strategies



PRINCIPLE OF NOISE REDUCTION

According to the noise simulation on both sides of S4, the figure shows the noise distribution within 50m on both sides of S4, which is divided into 7 types.

THE PRINCIPLE OF PLANT NOISE REDUCTION

Plants attenuate noise mainly by their **leaves, stems, and branches** to that structure. These structural features can attenuate noise. Different plants have different structural features, so their ability to attenuate noise is also different.

Among them, leaves and branches reduce noise through diffuse reflection; stems and stems affect the crown and height of the plant, thereby affecting the scattering and diffraction of sound waves, thereby reducing noise.

36

THE PRINCIPLE OF SOUND INSULATION PANELS TO REDUCE NOISE

There are two types of sound absorption and reflection. The sound absorption type mainly uses porous sound-absorbing materials to reduce noise; the reflective sound barrier wall mainly diffuses the propagation of noise sound waves to reduce noise in the protected area.

COVER THE ORIGINAL NOISE WITH NATURE OR OTHER MUSIC

Appropriately adding natural sounds or music as background sounds can reduce people's perception of unpleasant sounds.

We can install invisible speakers to play sounds or introduce sounds of plants, birds and water.

SOUND SELECTION:

Ocean Wave Sound>Piano Piece>Bird Singing>Classical Instrument Music>Breeze Blowing Leaves>Ancient Music Singing and Dancing>Insect Singing>Background Music>Rain

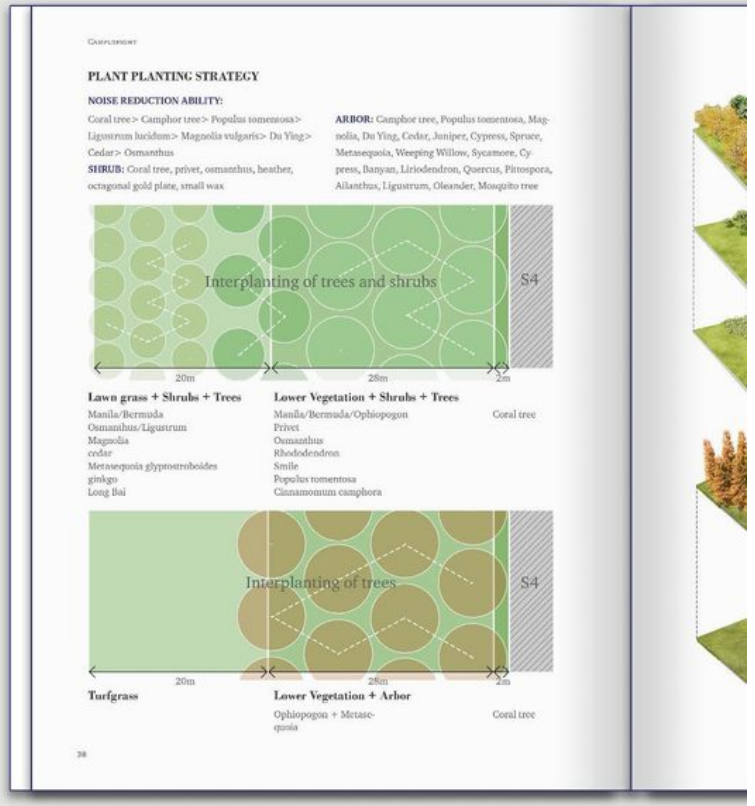
LANDSCAPE SCULPTURE BLOCKS THE SPREAD OF NOISE

The buildings and sculptures in the park can prevent the spread of noise to a certain extent and thus play a role in reducing noise.

NOISE-ABSORBING COATING

Acoustic materials are usually used in combination with architecture and sculpture.

37



PLANT PLANTING STRATEGY

NOISE REDUCTION ABILITY:

Coral tree> Camphor tree> Populus tomentosa> Ligustrum lucidum> Magnolia vulgaris> Du Ying> Cedar> Osmanthus

SHRUB: Coral tree, privet, osmanthus, heather, octagonal gold plate, small wax

ARBOR: Camphor tree, Populus tomentosa, Magnolia, Du Ying, Cedar, Juniper, Cypress, Spruce, Metasequoia, Weeping Willow, Sycamore, Cypress, Banyan, Liriodendron, Quercus, Pittospora, Ailanthus, Ligustrum, Oleander, Mosquito tree

Lawn grass + Shrubs + Trees

Manila/Bermuda
Osmanthus/Ligustrum
Magnolia
cedar
Metasequoia glyptostroboides
ginkgo
Long Bai

Lower Vegetation + Shrubs + Trees

Manila/Bermuda/Ophiopogon
Privet
Osmanthus
Rhododendron
Smile
Populus tomentosa
Cinnamomum camphora
Coral tree

Turfgrass

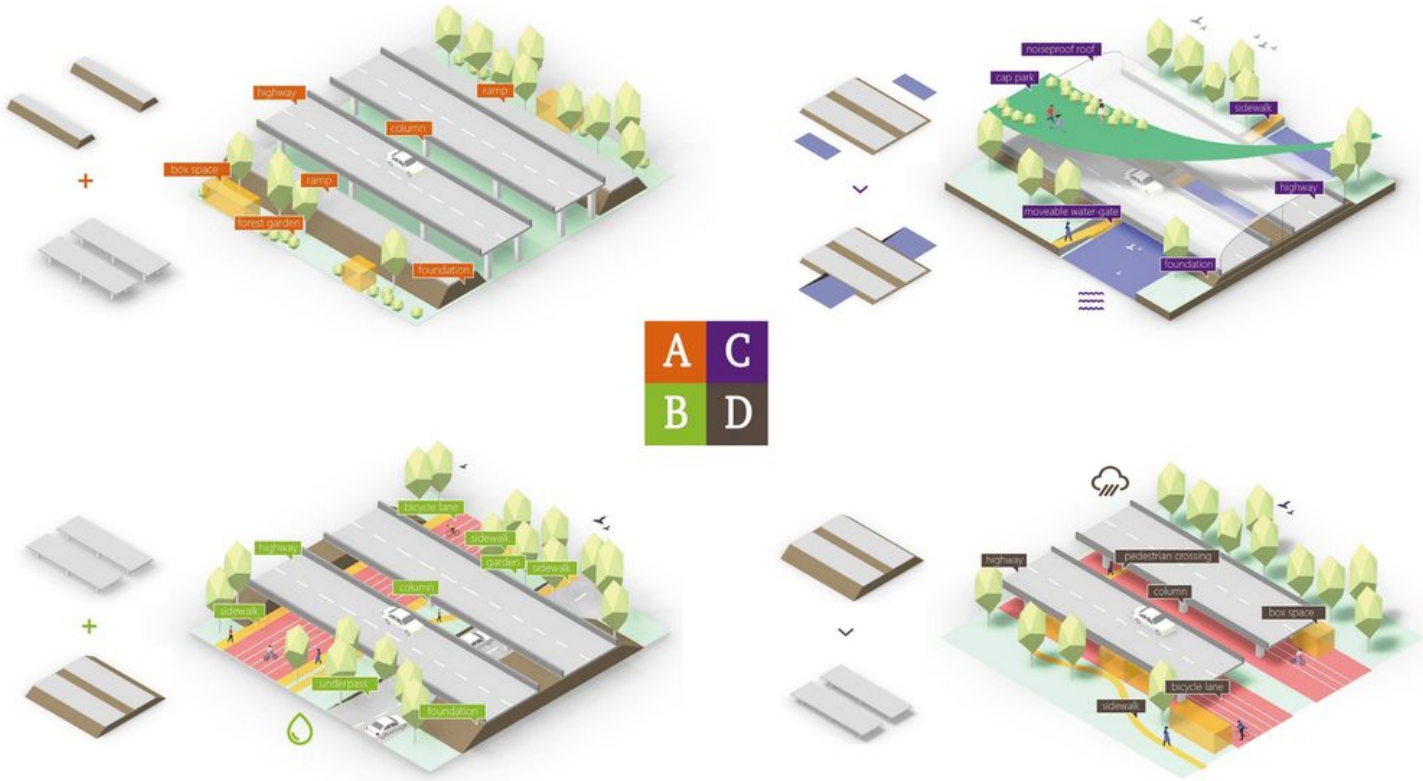
Lower Vegetation + Arbor

Ophiopogon + Metase-quoia
Coral tree

38

Typology

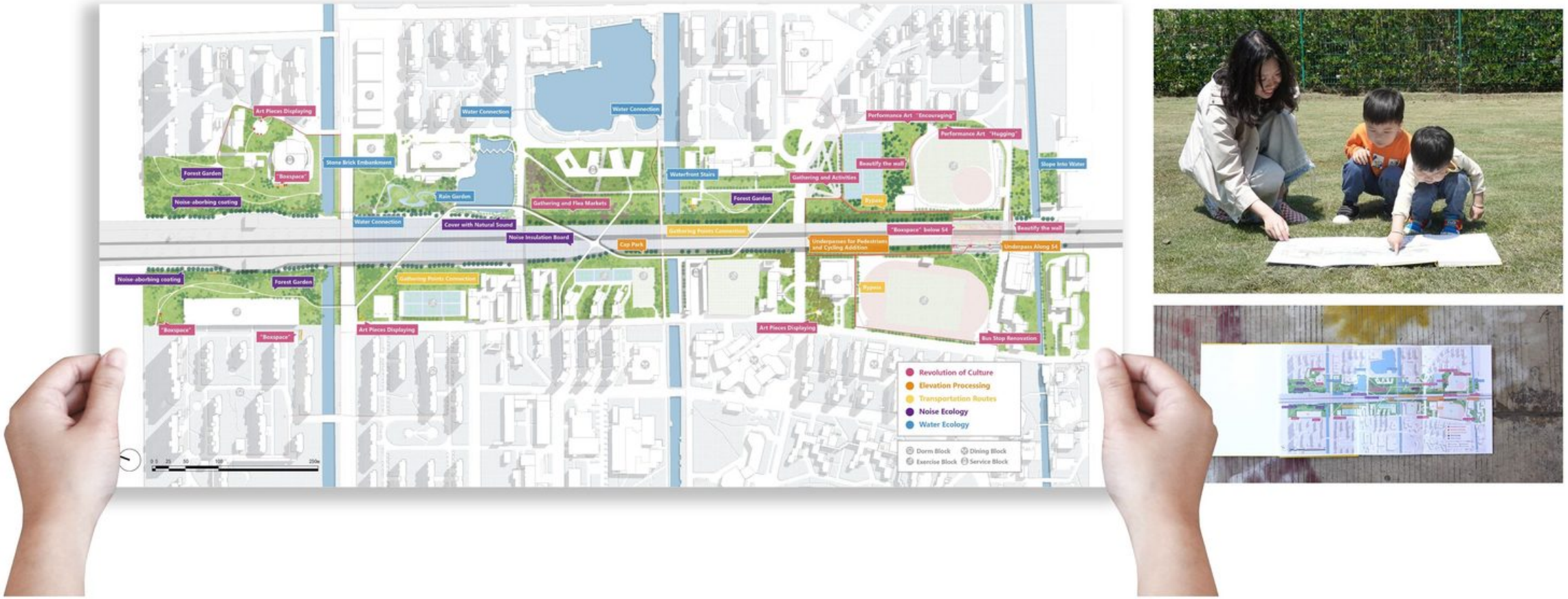
Toolkit for Different Elevation Conditions



Communication

After create different strategies, they post them on the wall under the Hwy. when users across the site, they left their comments for their strategies. They will revise the strategies based on their comments.

Communication



Based on the all revised strategies, the final master was settled down and prompt to the campus planners for them to refer and further consideration for the future design of the campus.

Student Work: STJU MLA Studio II_ Magical Foodscape

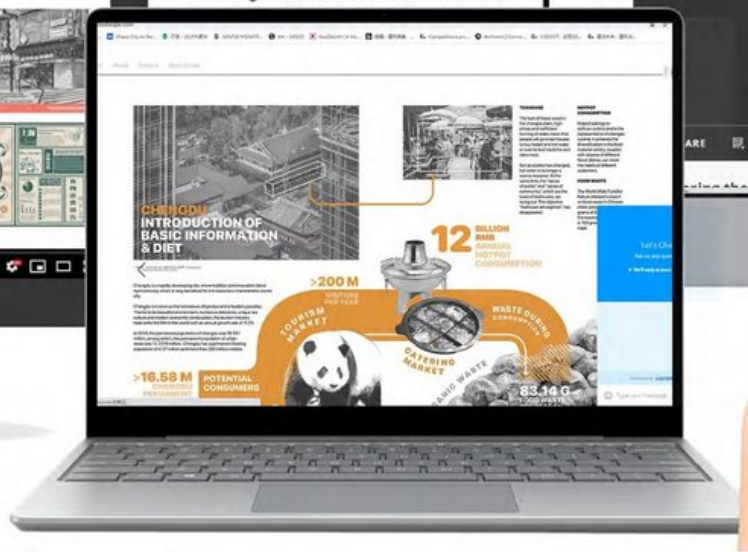
NETWORK PLATFORM EXHIBITION : MEDIA MATRIX FOR ADVOCACY AND INTERACTION

Through online exhibitions on local and international online media platforms, we promote our ideas and engage the public to participate in our plans.

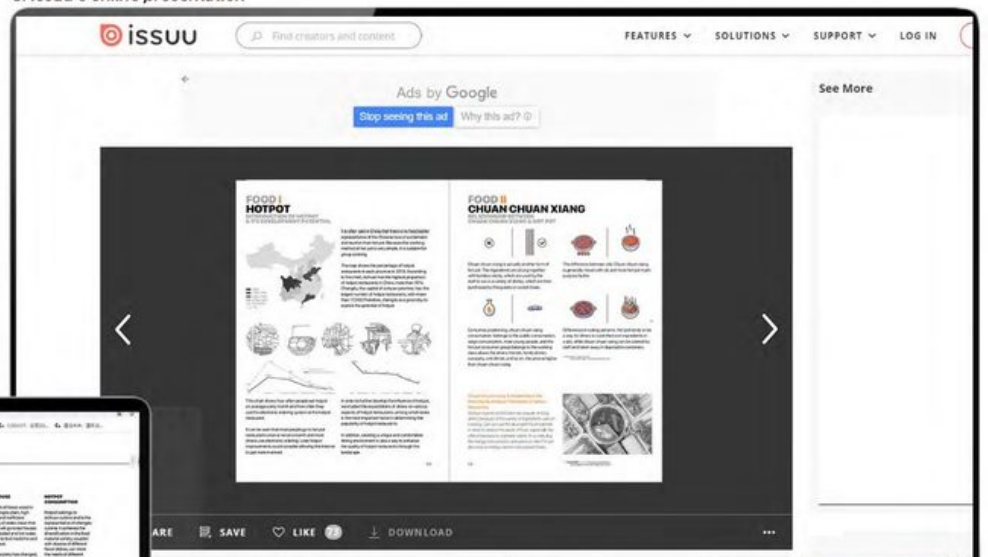
The epidemic has caused a large number of businesses to be unable to open their shops, and consumption in physical stores has decreased.Network traffic conversion and takeout has become the main means for restaurants to gain customers.We use a large number of online media tools to guide businesses to complete the offline to online transition.Multiple media matrices attracted thousands of views and gathered suggestions from hundreds of gourmet restaurateurs and city residents.

1. YouTube video promotion

2. Design a standalone websitewebsite



3. Issuu's online presentation



4. Interactive broadcast platform

5. Short video interview with Douyin

The research are focus on the Culture, Greenspace, Social. The project uses traditional Chinese food as a cut point. Nowadays, 'food' is no longer the overriding factor for urban planning while urban pattern still has its profound leverage on the long-term development of catering. Thus, the project is exploring the relationship between food, cultural and landscape planning. The research product creates several guidelines for future city renewal in an innovated theoretical way.

Theoretical Framework

doi: 10.3897/ab.e129204

THE BEGINNING OF THE PROJECT - THE RETHINGKING OF FOOD AND CITIES UNDER THE EPIDEMIC SITUATION

The crisis in the catering industry under the influence of the epidemic.

Revenue change of catering industry (unit: trillion yuan)

Distribution of raw and auxiliary material treatment paths

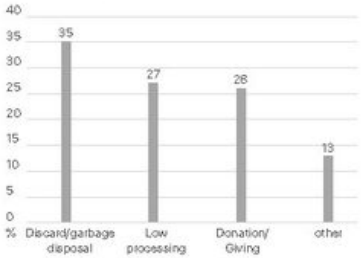
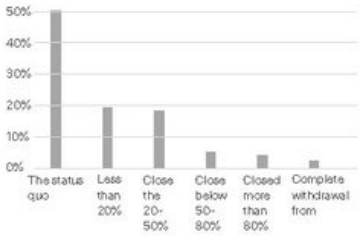


Table of expected changes in the number of restaurants

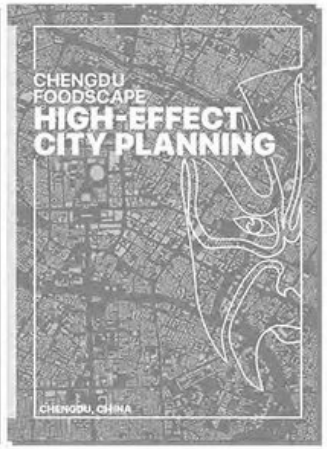


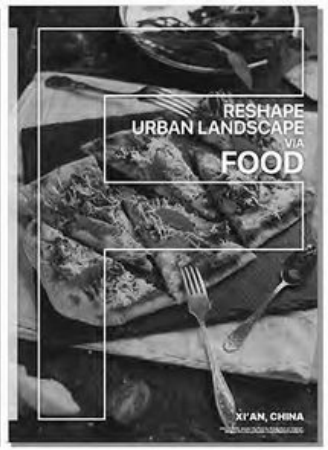
The project process

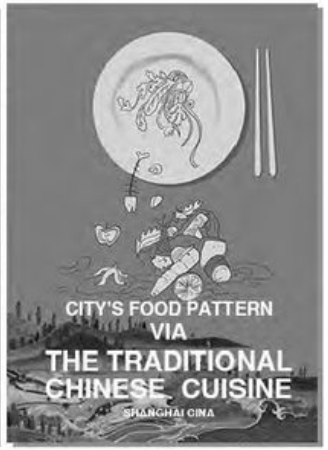
We try to link designers, government, food practitioners and citizens with strategies and feedback to guide them to participate in the research and planning of the urban food space and the role of food in the city.

Introduction of Magical Foodscape guidebook

The designers investigated the culture, traditional food and environment of four typical cities in China, explored the significance and role of food in the development of different cities, and put forward corresponding planning suggestions and guidelines as well as promotional products. The guide is divided into four main chapters:
1.Chengdu foodscape: high-effect city planning 2.Reshape urban landscape via food 3.Healthy city from the perspective of resilience 4.Shanghai traditional restaurant revival

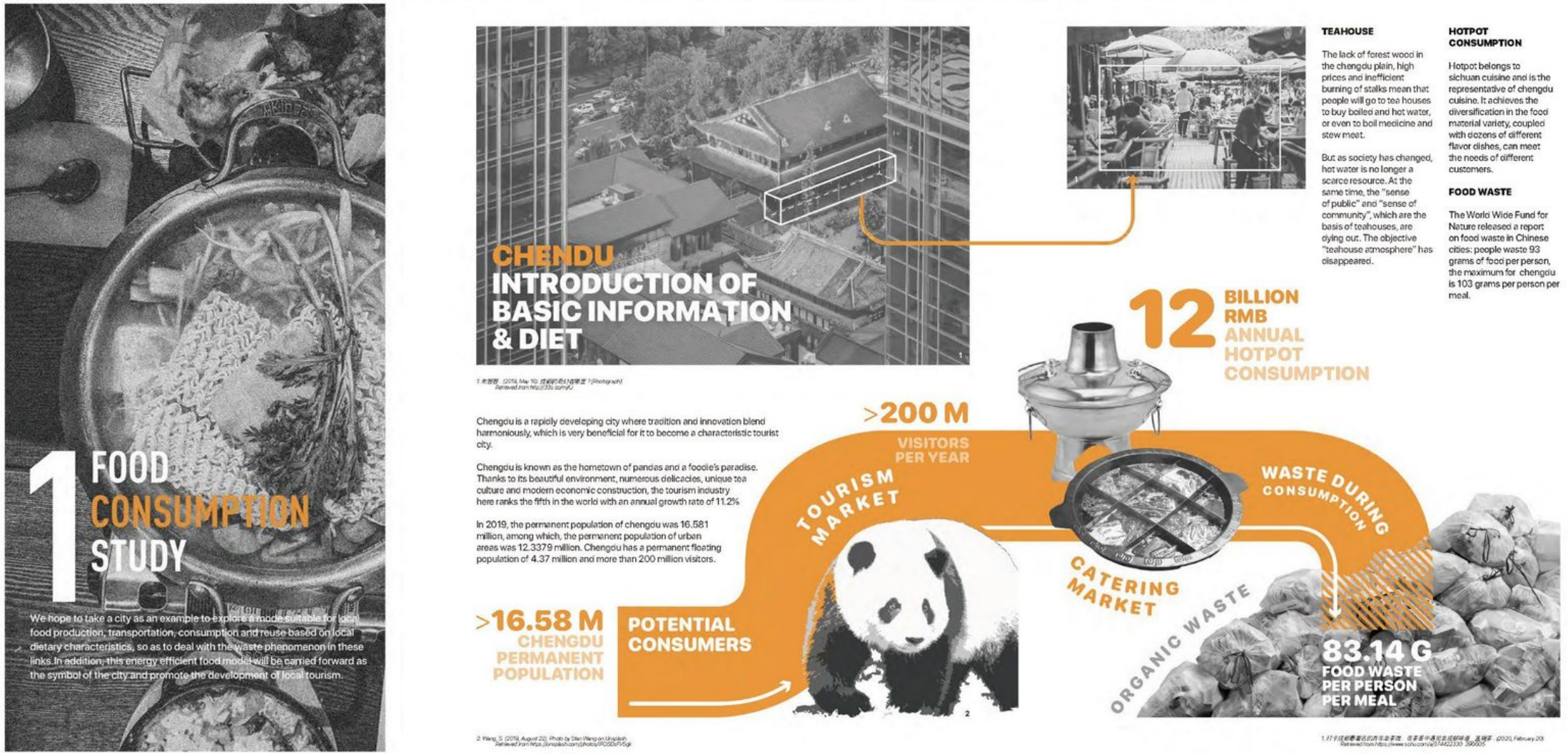






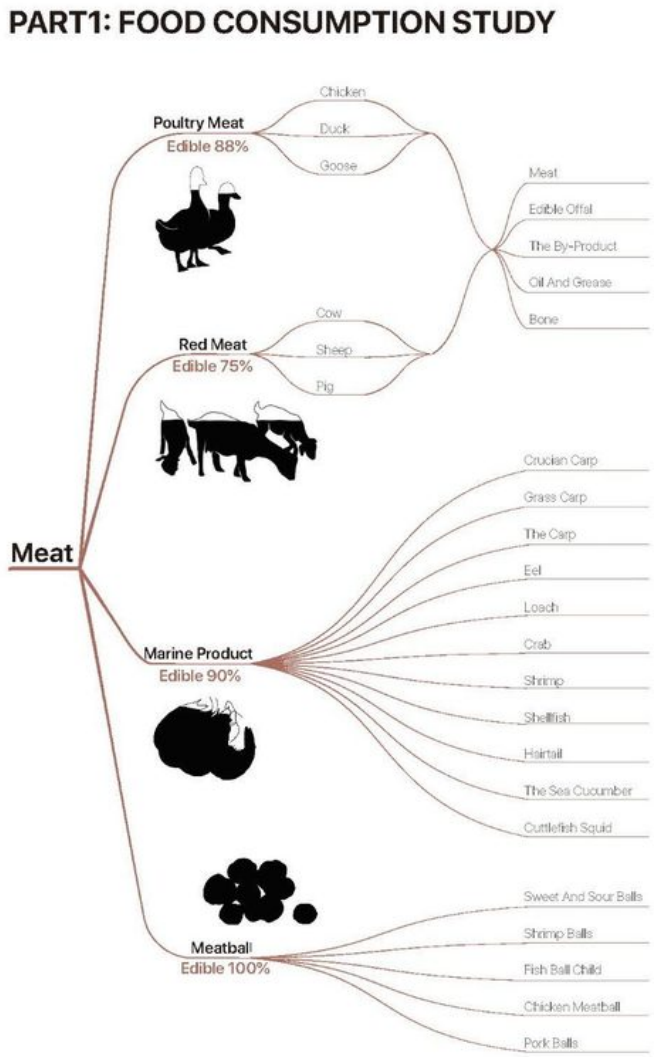
The work uses the archival research, site observation, interview to get the primary and secondary data. Three groups of students went to three cities to get the most reliable data. Each group proposed the different strategies for the cities. The final work are collected as a book and will be published this year.

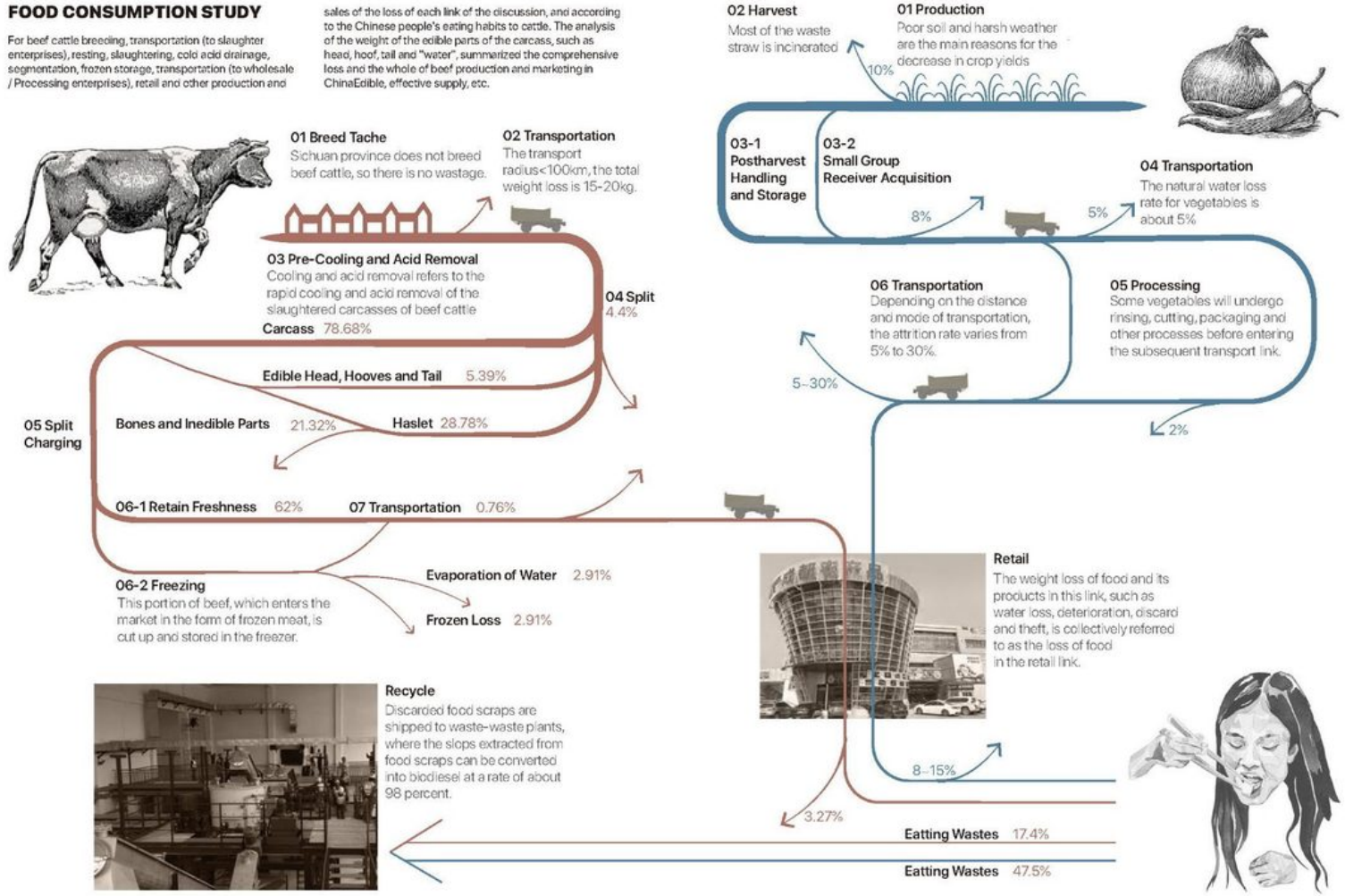
Research & Analysis



They collect many useful data and converted them into infographic for audience better understand the existing condition of the site. Eg. This group. They focus on food consumption study.

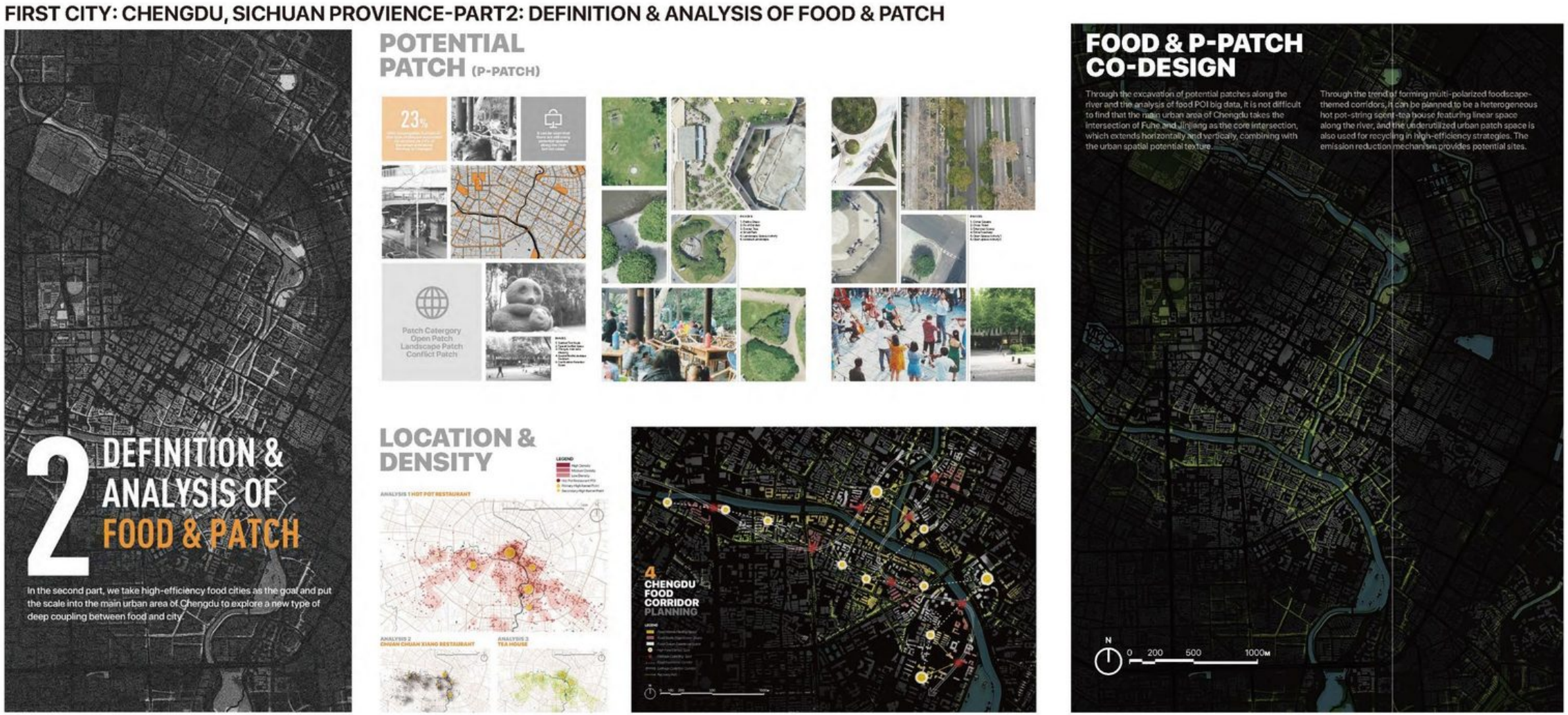
Research & Analysis



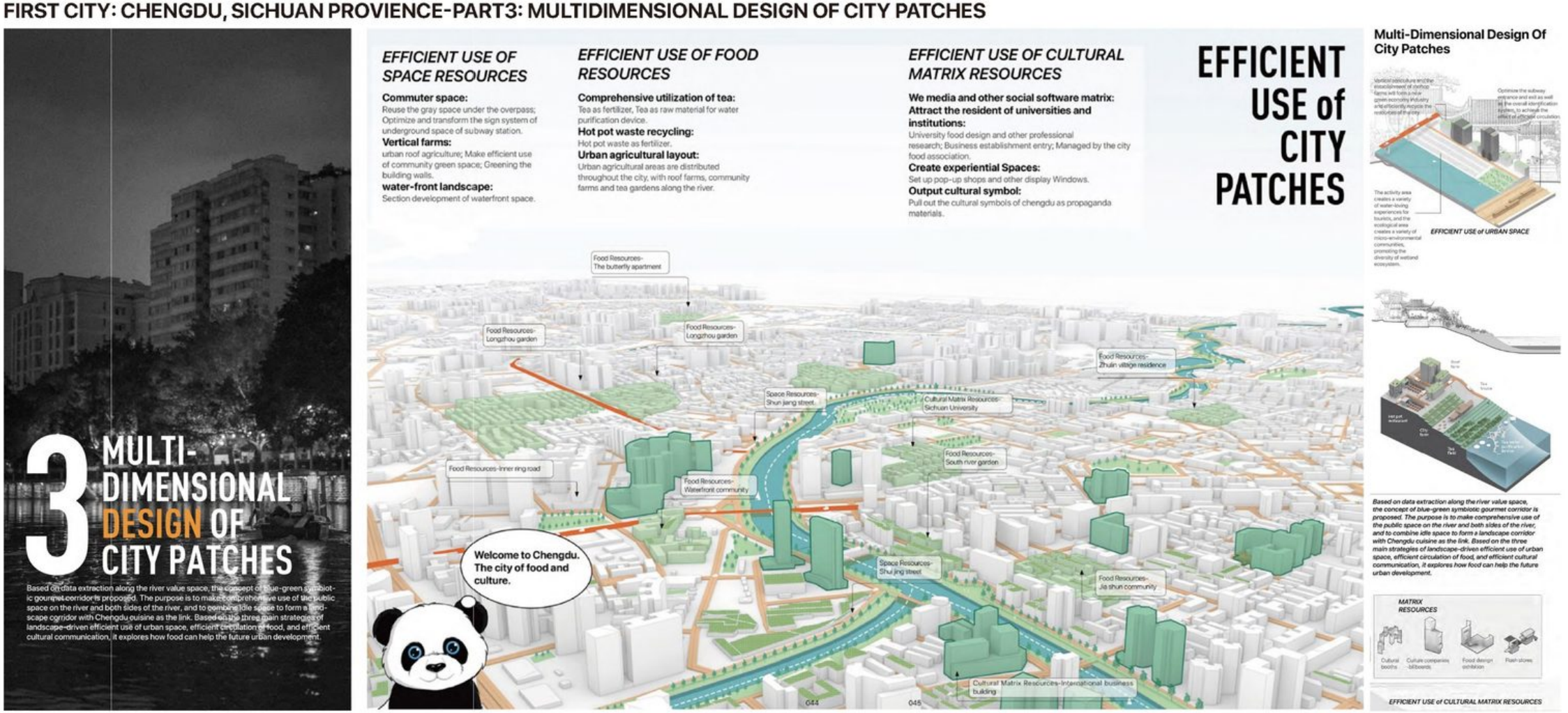


These data are not typically from peer reviewed articles. Most of the info comes from interview, news paper, report. This is a good example to show how to format the data in a readable and effective way. This also helps students deep understanding the background of the site.

Typology

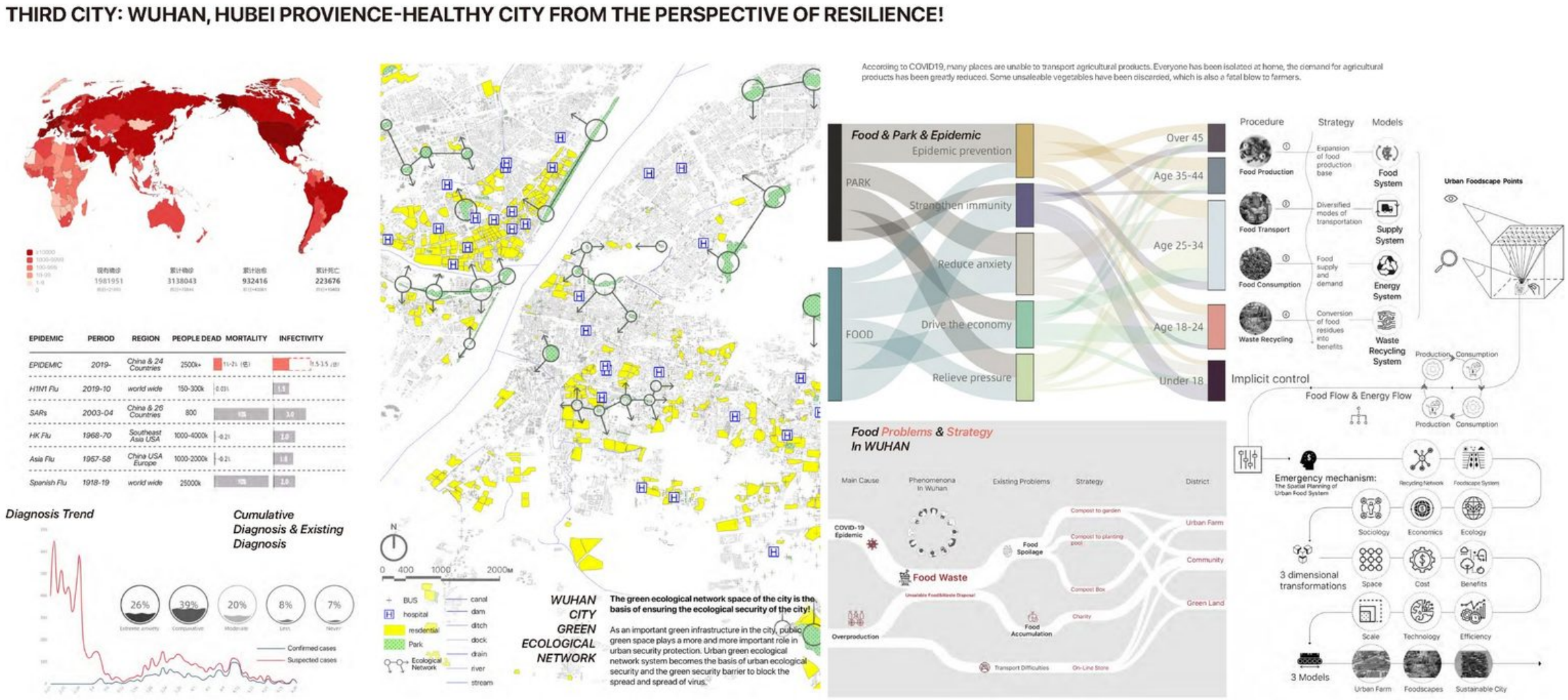

FIRST CITY: CHENGDU, SICHUAN PROVIENCE-PART2: DEFINITION & ANALYSIS OF FOOD & PATCH
2 DEFINITION & ANALYSIS OF FOOD & PATCH
In the second part, we take high-efficiency food cities as the goal and put the scale into the main urban area of Chengdu to explore a new type of deep coupling between food and city.
POTENTIAL PATCH (P-PATCH)
23%
Patch Catergory
Open Patch
Landscape Patch
Conflict Patch
LOCATION & DENSITY
ANALYSIS 1 HOT POT RESTAURANT
ANALYSIS 2
CHUAN CHUAN XIANG RESTAURANT
ANALYSIS 3
TEA HOUSE
4 CHENGDU FOOD CORRIDOR PLANNING
FOOD & P-PATCH CO-DESIGN
Through the excavation of potential patches along the river and the analysis of food POI big data, it is not difficult to find that the main urban area of Chengdu takes the intersection of Fuhe and Jinjiang as the core intersection, which extends horizontally and vertically, combining with the urban spatial potential texture.
Through the trend of forming multi-polarized foodscape-themed corridors, it can be planned to be a heterogeneous hot pot-string scent-tea house featuring linear space along the river, and the underutilized urban patch space is also used for recycling in high-efficiency strategies. The emission reduction mechanism provides potential sites.
N
0 200 500 1000M

Typology



After identifying the patches, they consider how to further develop them by considering the culture, program base on the people's need from interview.

Typology



In the covid period, the students consider the food shortage problem as well, so they decide to propose a city green ecological network by playing around the data of hospital, restaurant, traffic, and parks. Their goal is to find the best way to deliver the food to the hospital and explore the potential parks to be public social place with security distance.

Communication to Public

OFFLINE EXHIBITION AND PROMOTION GUIDANCE ACTIVITIES : MULTI-CHANNEL ACTIVITIES

Hold Magical Food SCPAE Gallery exhibition, and cooperate with communities and restaurants in Chengdu, Xi 'an, Wuhan, Shanghai and other places to collect public opinions and feedback.And provide guidance and suggestions for the store owners.

Finally, they spread the work to the urban planners, residents to share their thought.

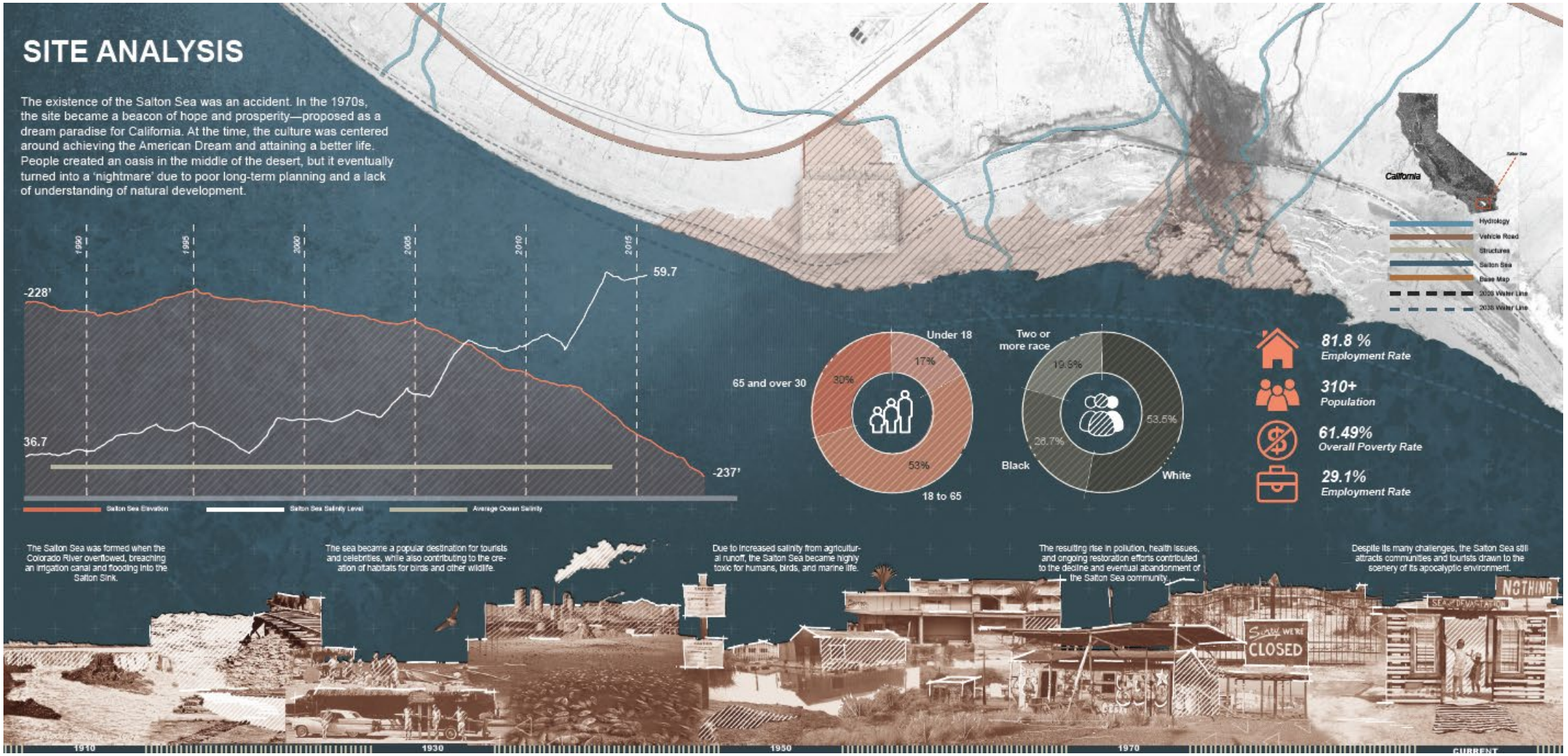
Key Concept III:

“Climate Action”

+

“Livable Urban Design”

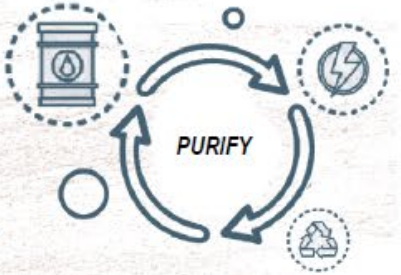
Student Work: UNLV LAND 486 Capstone Project



Credit: Castro Dawn Miguel, UNLV BLA 25'

Student Work: UNLV LAND 486 Capstone Project

DESIGN FRAMEWORK

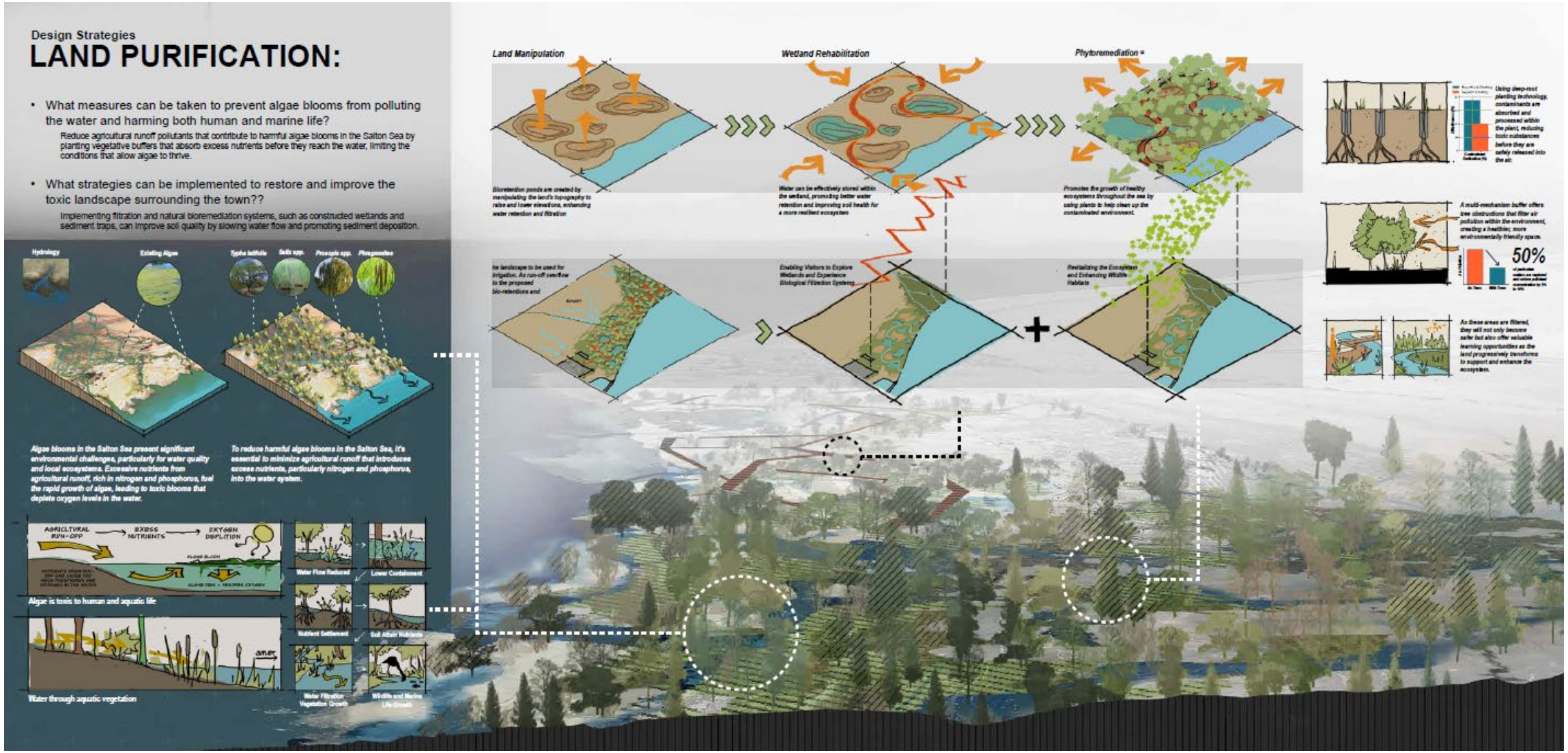
DESIGN CONCEPTS	PROBLEM SEEKING	EXISTING CONDITIONS	PROBLEM FRAMING	CONCEPT SOLUTIONS
How to provide a safe and accessible space within the Bombay Beach?	*How to distribute water for green spaces throughout the site?* *How to lower the chances of crime rates within the Bombay Beach?* *How to decrease health risks from air-born dusts to increase more outdoor activities?*		**Drainage System** Using the Sea water to irigate green space, increasing trees or vegetation for shades **Security System** Providing surveillance to lower crime rates and increase access to spaces **Land Manipulation** By changing the topography to create a barrier for dust winds and flood events	
How can the landscape be improved for wildlife habitat and well fair of people?	*How can algae be prevented from creating polluted water that can harm human and marine life?* *How to improve the current toxic land-scape that surrounds the town?* ***How to filtrate the toxic polluted water to create a sustainable ecosystem?***		**Algae Reduction** Filtering the source within river wash to prevent algae bloom **Soil Improvement** By creating wetlands to improve landscape soil and habitat restoration **Water Filtration** Existing algae and saline within water need to be filtered and improved for the ecosystem	
How can local businesses thrive in a deserted landscape town to revive its economy?	*What will attract visitors to provide **economical benefits within local** businesses?* How to provide a learning opportuanity to the sites history and prevent past mistake? *How will the site develop throughout the years as the water level decreases?*		**Land Attraction** Creating spaces for tourists attraction throughout the town **Learning Opportunity** Provide visitors access to the site's history, landscape, and Culture **Expansion Progression** As the site progress, the coastal restoration plan restores features like wetland for natural barriers to slowing water loss	

Credit: Castro Dawn Miguel, UNLV BLA 25'

Student Work: UNLV LAND 486 Capstone Project

Credit: Castro Dawn Miguel, UNLV BLA 25’

Student Work: UNLV LAND 486 Capstone Project



Credit: Castro Dawn Miguel, UNLV BLA 25’

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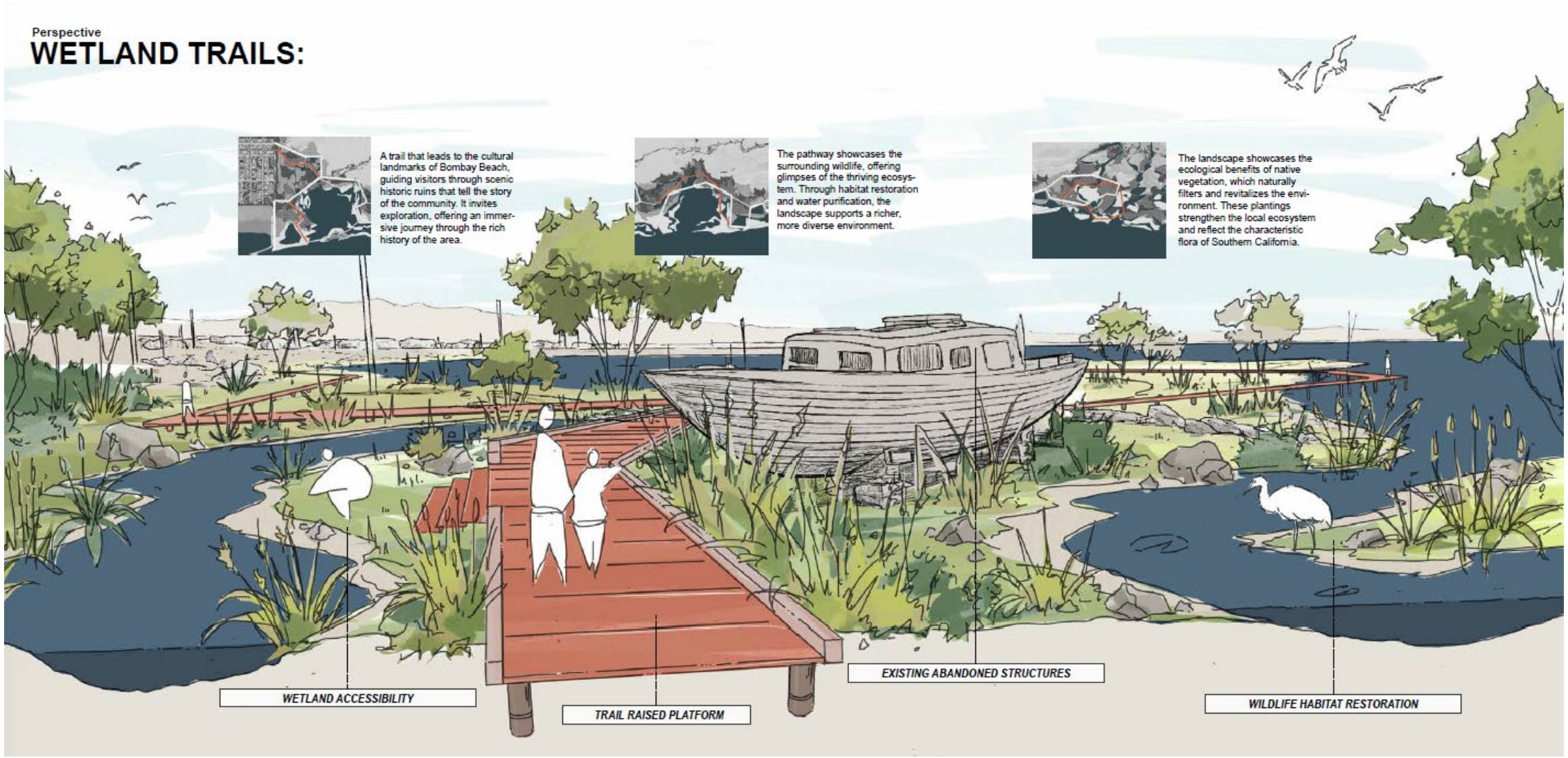

Credit: Castro Dawn Miguel, UNLV BLA 25’

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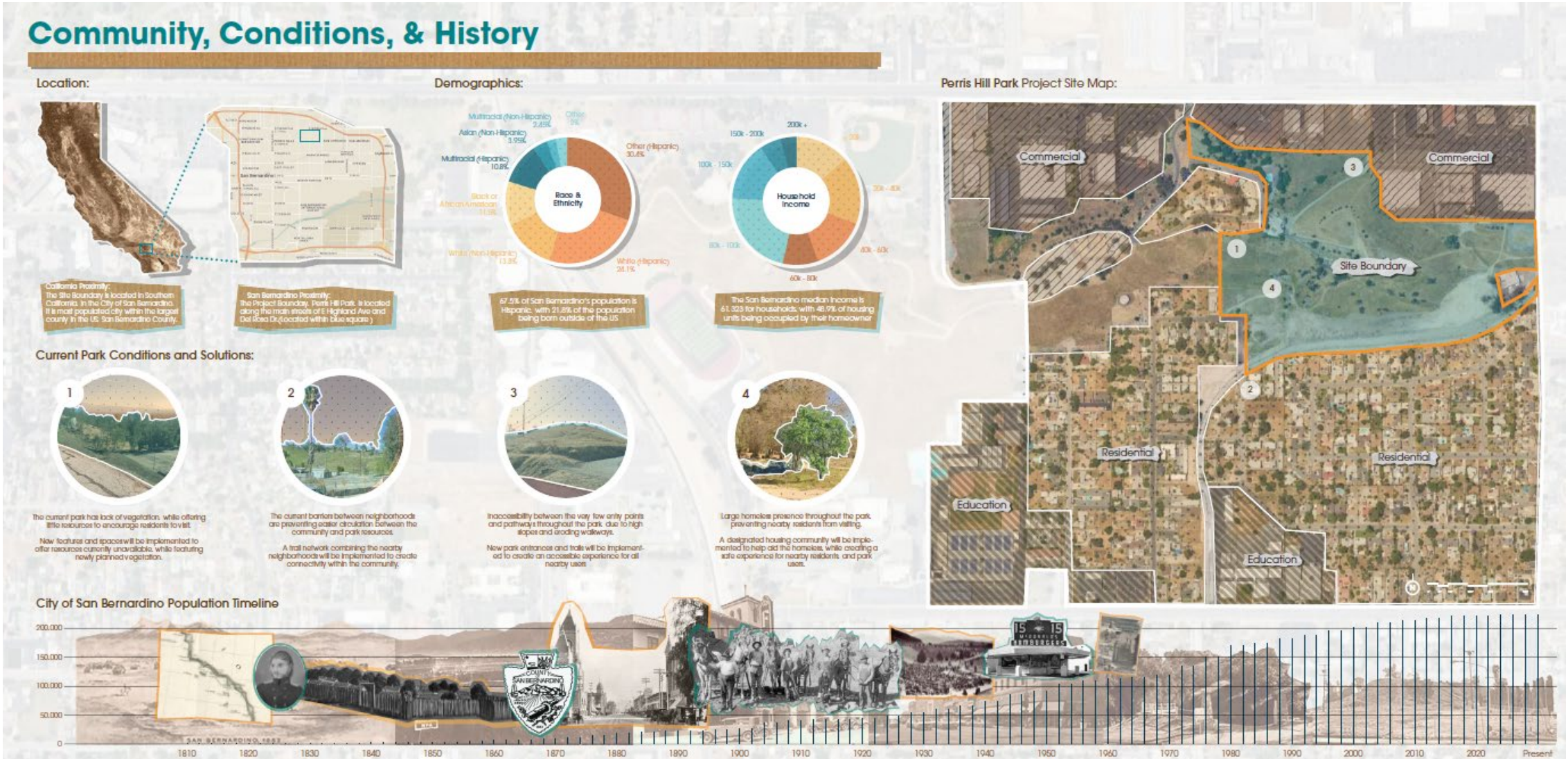
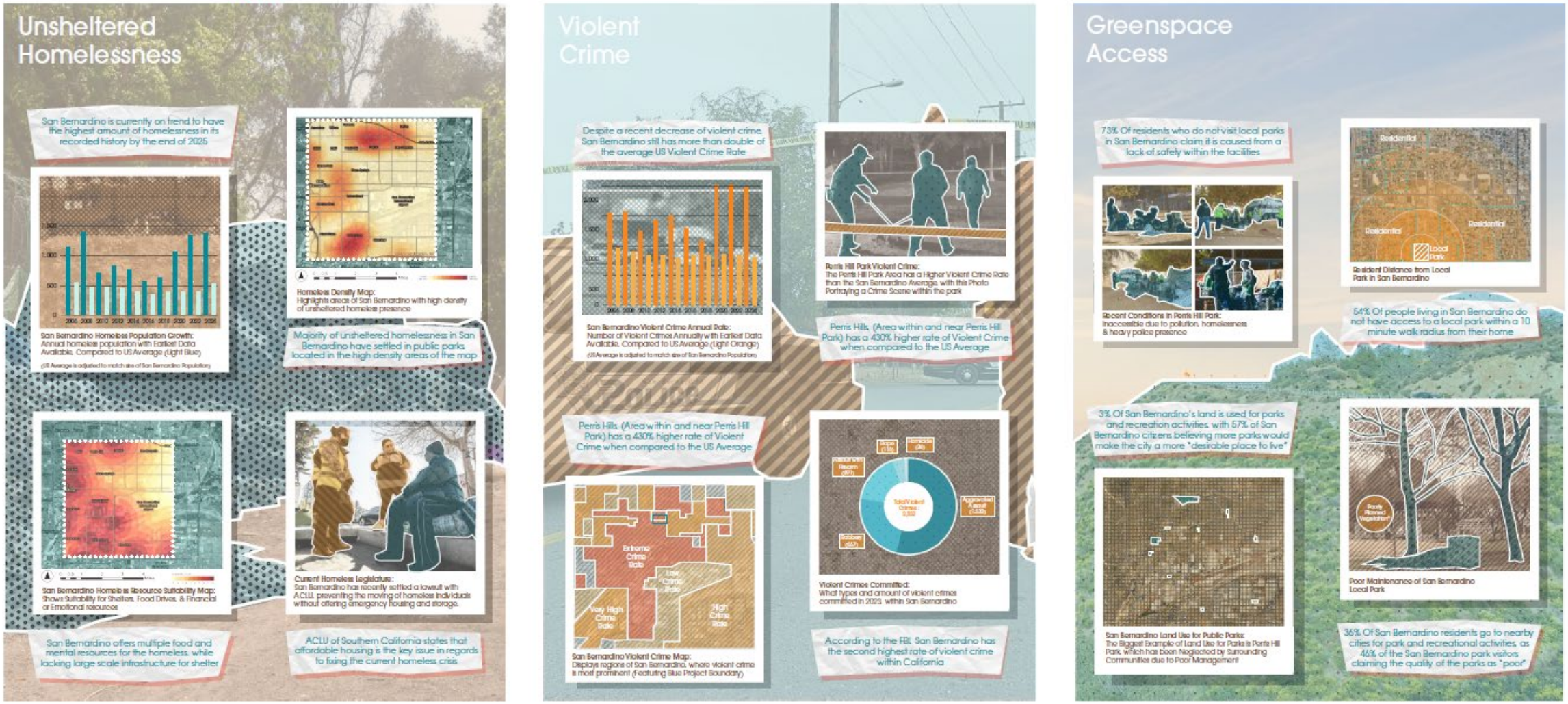


Credit: Carlos Holguin, UNLV BLA 25’

Student Work: UNLV LAND 486 Capstone Project



Credit: Carlos Holguin, UNLV BLA 25’

Student Work: UNLV LAND 486 Capstone Project

Programming Plan

Research Question 1:

How to Aid the Current Homeless Crisis, While Providing a Safe Lifestyle for Unsheltered Homeless Individuals and Nearby Residents?

Key Issues & Opportunities

- Large Presence of Homeless Camp Sites in Perris Hill Park
- Nearby Residents Unable to Visit Perris Hill Park due to Strong Homeless Presence Throughout the Entire Park
- Lack of Resources to Aid Homeless Crisis (Homeless shelters, food drives, rehabilitation, etc.)
- 68.9% Of Homelessness in San Bernardino were Surveyed as Unsheltered

Existing Conditions

Featured Programming

- Housing Community for Homeless
- Food / Resource Donation Drop Off & Collection Center
- Public Transit System For Resource Access to Aid Homeless Experience
- Public Homeless Sanitary Stations
- Housing Community Gardens

Design Goals

Self-Sustaining Community for the Homeless

Research Question 2:

What Resources Can be Provided to Avoid Possible Violent Crime and Escape the Hostile Environment?

- Perris Hills currently has Higher Violent Crime Rate than the San Bernardino City Average
- Lack of Public Safe Spaces Nearby Park Proximity, such as Libraries, Parks, and Social Service Facilities
- No Circulation Offered to Avoid Areas with High Rates of Violent Crime
- Heavy ground pollution exaggerating hostile conditions within park area

- Designated Public Safe Spaces to Aid Stressful Residents
- Trail Network Connecting to Nearby Resources to Prevent Circulation in Areas with High Rates of Violent Crime
- Landfill Cleanup Initiative to Decrease Hostility Within the Park and Community

Redesigned Trail and Transit Network to Accommodate Nearby Residents and Resources

Research Question 3:

What Changes Can Be Made to Increase Accessibility to Green-space for the Surrounding Community?

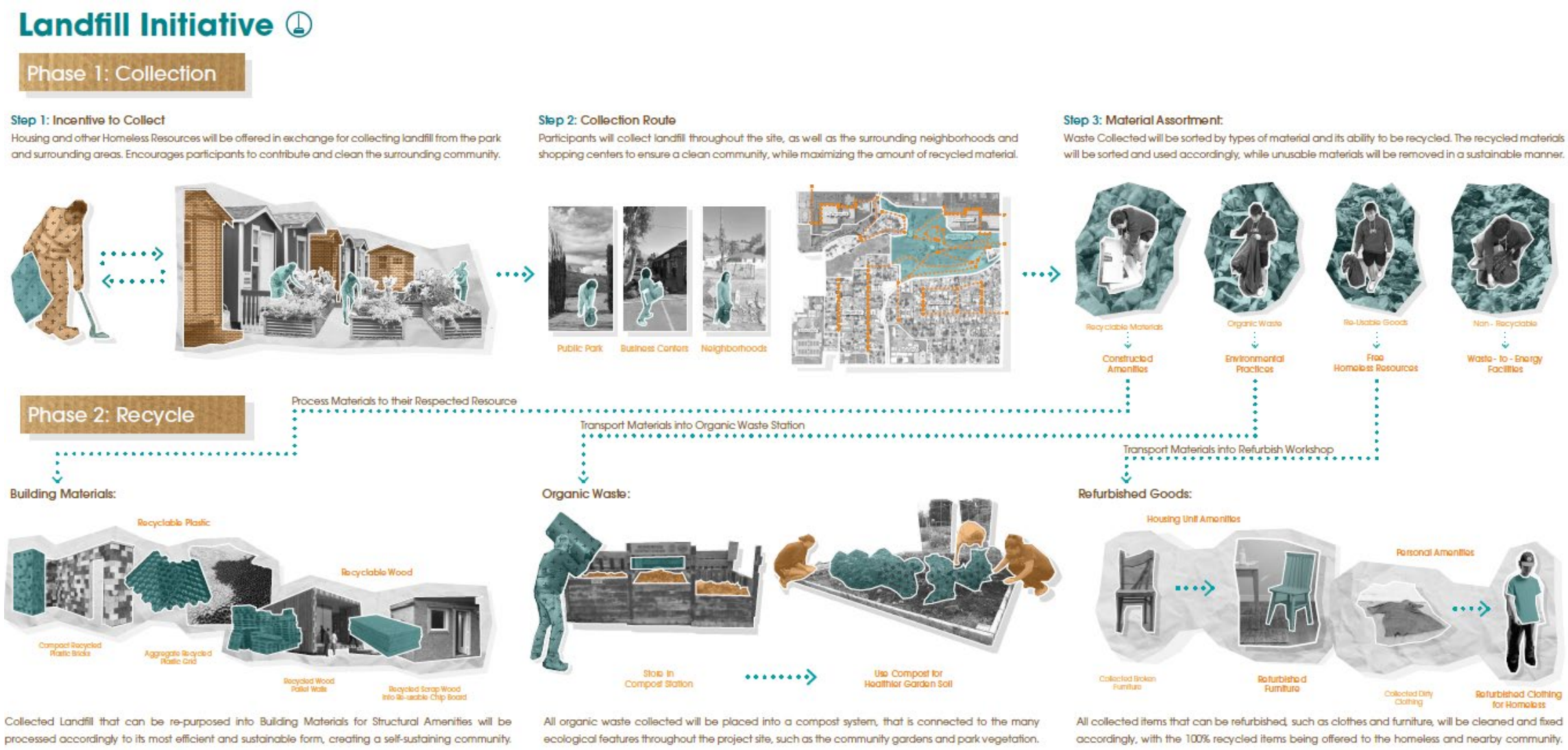
- Lack of Public Parks or Green-spaces near the Underutilized Perris Hill Park
- Current Park Trail Conditions are Preventing Easy Access, such as High Slopes and Eroding Materials
- Currently no Connectivity Between Neighborhoods, Creating Harder Accessibility for Green-spaces
- Perris Hill Park Currently Under Poor Maintenance, Creating an Inaccessible Experience

- Redesigned Park with New Vegetation to Offer a Green-space to Nearby Community
- Public Transit System Connecting Residents to Perris Hill Park to Increase Accessibility
- Trail Network System Connecting Nearby Neighborhoods for Easy Circulation towards the Park

Public Park Redesigned for Community Gatherings and Public Safe Spaces

Credit: Carlos Holguin, UNLV BLA 25’

Student Work: UNLV LAND 486 Capstone Project



Credit: Carlos Holguin, UNLV BLA 25’

Student Work: UNLV LAND 486 Capstone Project

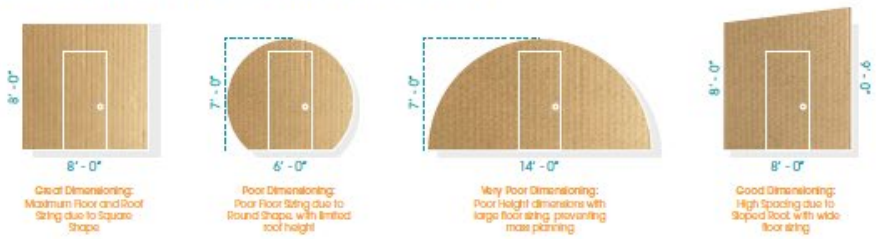
Housing Community For Homeless

Decision of Housing Shapes:

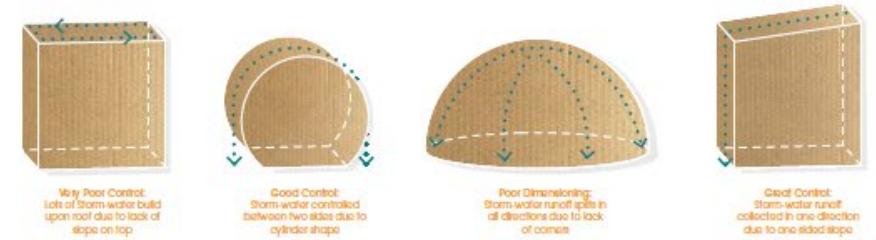
These Four Shapes for the Housing Community were chosen as they were the most commonly used within existing homeless housing case studies. The shapes are also easier for construction with a compact design, to create the maximum quantity to help aid the homeless crisis as much as possible.

The Housing Shape that Tests Best with the Chosen Factors will be selected for the Housing Community.

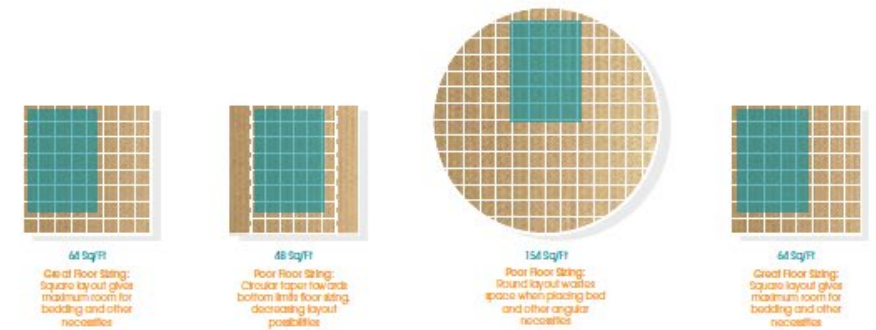
Dimensioning:
The sizing and shape of each housing unit, with conditions decided by its height, width, and compactability



Storm-water Control:
The direction of storm-water runoff from the roof, as the water can contribute to the community gardens connected with the housing



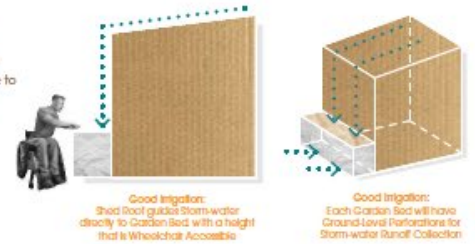
Floor Plan Sizing:
The amount of square footage per housing unit, while showing where a full size bed could be placed and the amount of room remaining



Decision of Community Garden Layout:

The Shape, Location, and Orientation of the Community Garden Beds were based off which Method Maximized the Amount of Collected Storm-water Run-off while Providing each Housing Unit with a Specific Garden Bed.

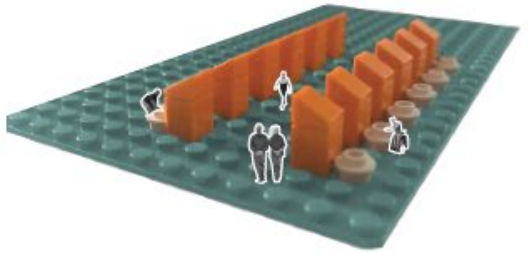
Storm-water Collection:
Each Garden Bed will Collect Storm-water from the Roof and Ground due to the Housing Slope and Ground Irrigation Holes



Housing Unit Sizing:
Each Housing Unit will have a 8' x 8' floor plan, accommodating enough space for a comfortable living condition, while being compact enough to maximize amount of former-homeless residents

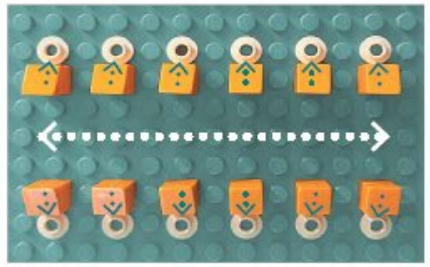
Housing Community Layout:

The Layout of the Housing Community was based on which layout maximized the amount of housing units to aid the homeless crisis as much as possible, while also creating a community space that encourages users to interact with one another



Connected Community Space:
The orientation of the housing community make each row of units face each-other, creating a greater likelihood of people interacting with one another, creating a more positive experience for the housing community.

Efficient Community Space:
The layout of the housing community maximizes the amount of housing units due to its linear orientation. The linear shape also creates a path of circulation within the housing community, with each unit having its own garden bed located behind the entrance, towards the bottom direction of the sloped roof.

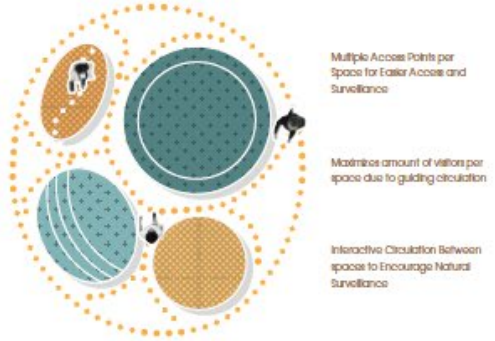


Credit: Carlos Holguin, UNLV BLA 25'

Student Work: UNLV LAND 486 Capstone Project

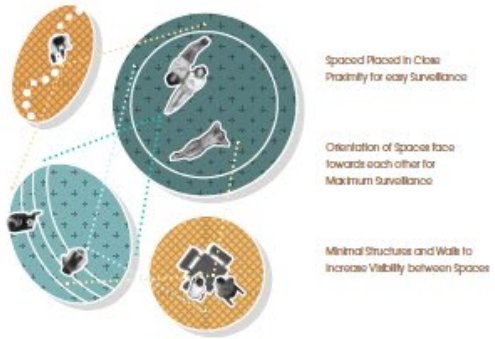
Public Safe Space Features

Natural Surveillance: Connected Circulation



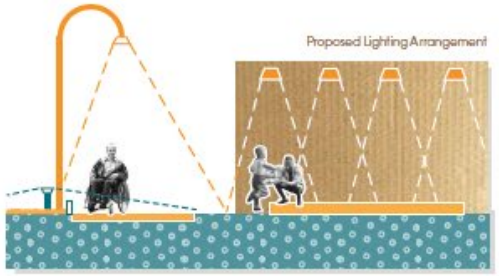
Implement constant circulation between spaces to increase the amount of users within a confined space while decreasing the amount of crime or unwanted events.

Natural Surveillance: Open Space Design



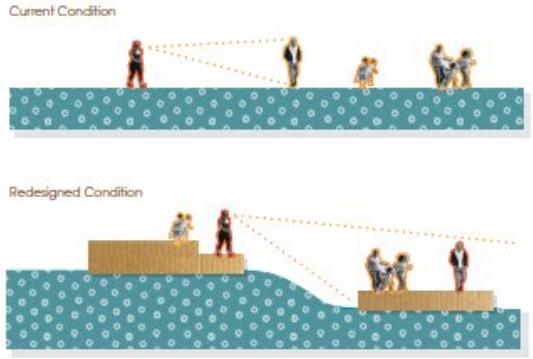
Implementing connectivity and open design between different spaces increases the field of sight for users, decreasing the amount of unwanted behavior within the park.

Lighting for Spaces and Pathways



Strategic lighting increases user safety due to a clear field of sight, guides the circulation to safe areas, increases park hours, and prevents any hidden activities.

Natural Surveillance: Tiered Amenities

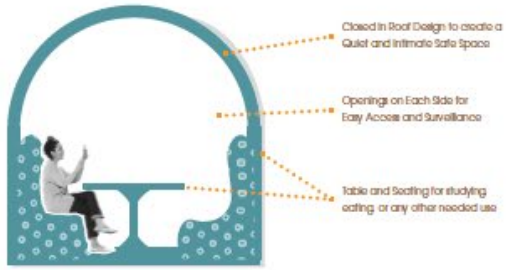


Designing spaces with different elevations increases the field of sight, as the open space view can be limited by users, with the elevation offering a way to look over any obstacles.

Public Art: Property Crime Prevention

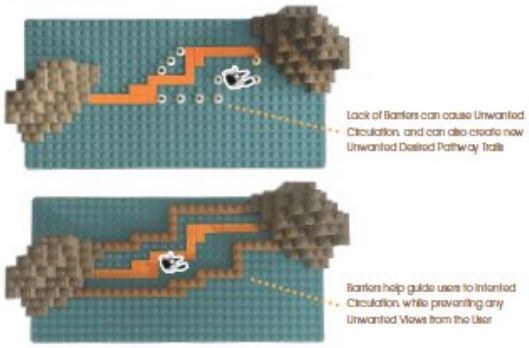
Implementing public art throughout the park decreases the likelihood of property vandalism, increases aesthetic, and offers opportunities to commission local artists.

Private Seating / Study Stations



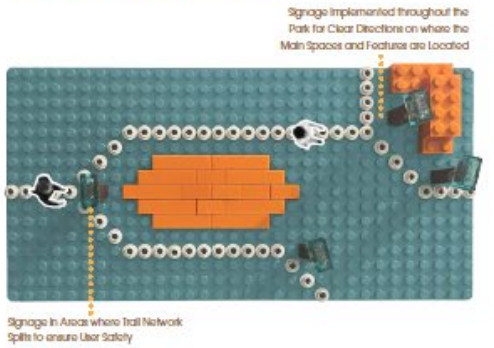
Implementing designated stations for a quiet safe space offers a solution to the stressful and unsafe environment that some users may experience.

Site Barriers



Implementing Site Barriers throughout the Park increases user safety, due to the orientation and placement guiding the circulation, preventing anyone from visiting unwanted spaces.

Signage & Emergency Stations



Implementing Signage throughout the Trail System prevents any users from circulating within an unwanted space, with Emergency Stations offered for help with any unwanted events.

Credit: Carlos Holguin, UNLV BLA 25’

Student Work: UNLV LAND 486 Capstone Project

Planting Redesign

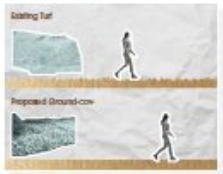
Planting Selection

- Implement Drought Tolerant vegetation for a Self Sustaining system Park Experience
- Adapted Vegetation as Drought Conditions continue to worsen in Southern California

Drought Tolerant

- Implement Biodiversity to create mini Ecosystems capable of inhabiting Wildlife
- Design a Self Sustaining system between the Plant Species and Wildlife

Wildlife Habitat



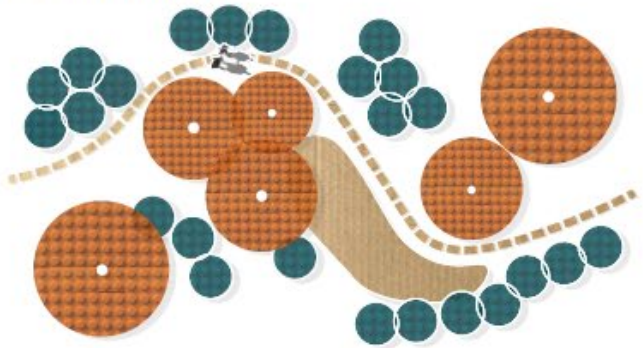
- Implement varying types of vegetation for year-round Ground-cover
- Aids the issue of current Poor Turf Conditions due to Maintenance and Water Access

Turf Alternatives

- Implement Vegetation specific to the Site Location, due to its capability of adapting to its native environment
- Create opportunities to Educate Users about their Native Plant Species

Native Planting

Spacial Defining Planting



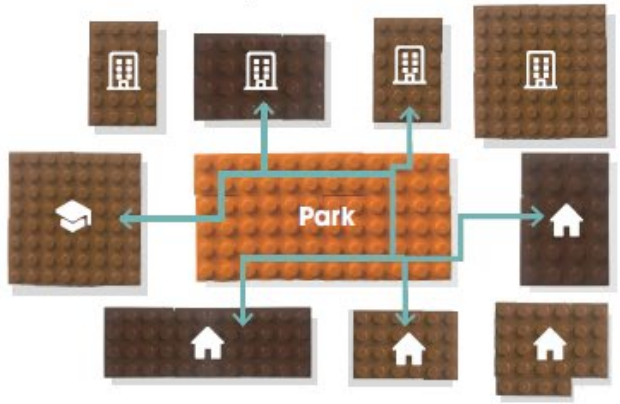
- Implement Vegetation in Specific Orientations and Locations to create Guided Pathways and Prevent Unwanted Circulation
- Create a Variety of Vegetation Density and Spacing to Create New Spaces dictated by the Vegetation Placement
- Vegetation Placement along the Intended Circulation will increase the Amount of Interaction between the Users and Nature

Tree Canopy Screening for Housing Community

- Implement Trees located on the exact elevation where the Canopy would be aligned with the Housing Community
- Creates privacy for the Housing Community, as the canopies will help enclose the space and offer shade
- Park visitors and nearby residents can enjoy the park views exclusively, without having to view the Housing Community

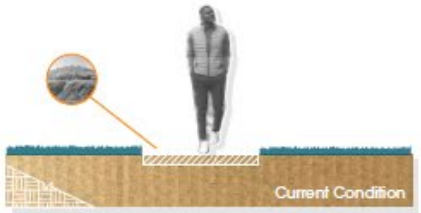
Trail Network Redesign

Trail Entrances and Connectivity



- Increase amount of trail entrances to increase connectivity between park and nearby amenities
- Increase amount of trail routes within the park to create varying trail experiences while increasing accessibility to park spaces
- Offer shorter circulation to specific spaces to aid people who can not walk for long periods of time

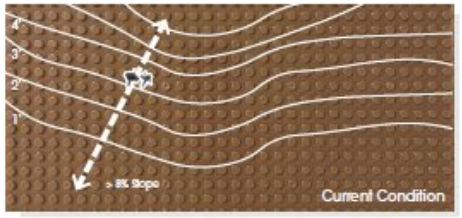
Trail Materials and Dimensions



- Trail is Flush with Surrounding Turf, discouraging users to follow trail and lead to potential danger
- Numerous "Desired Pathways" created from Unenforced Trail Design
- Compact Soil Trail in Steep Slope subject to high erosion risk, leading to possible injury
- Areas with 2' Trail Width, hard for wheelchair accessibility and multiple users at once

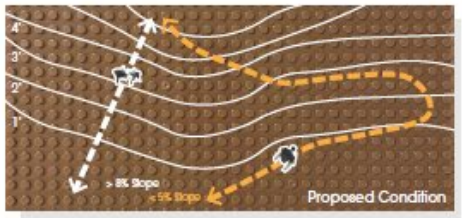
- Aggregate Paver Trail increase infiltration, preventing a slippery trail
- 8' Wide Trail for easier Wheelchair Mobility and increase amount of Users Circulating at once
- Trail Width increases Wheelchair Accessibility and Amount of Users to Circulate at once
- Trail Curb encourages users to Circulate within the Intended Trail System and Prevents external Elements from the Infiltrating Trail

Multi-Path Trail System



One Route Trail:

- Steep Trail creates Harder Access and Shorter Distance
- No Diversity of Circulation, creating a Lack of Curiosity when Visiting
- Greater than 11% slope trail, with 8.33% slope being maximum ADA Accessibility for Wheelchairs

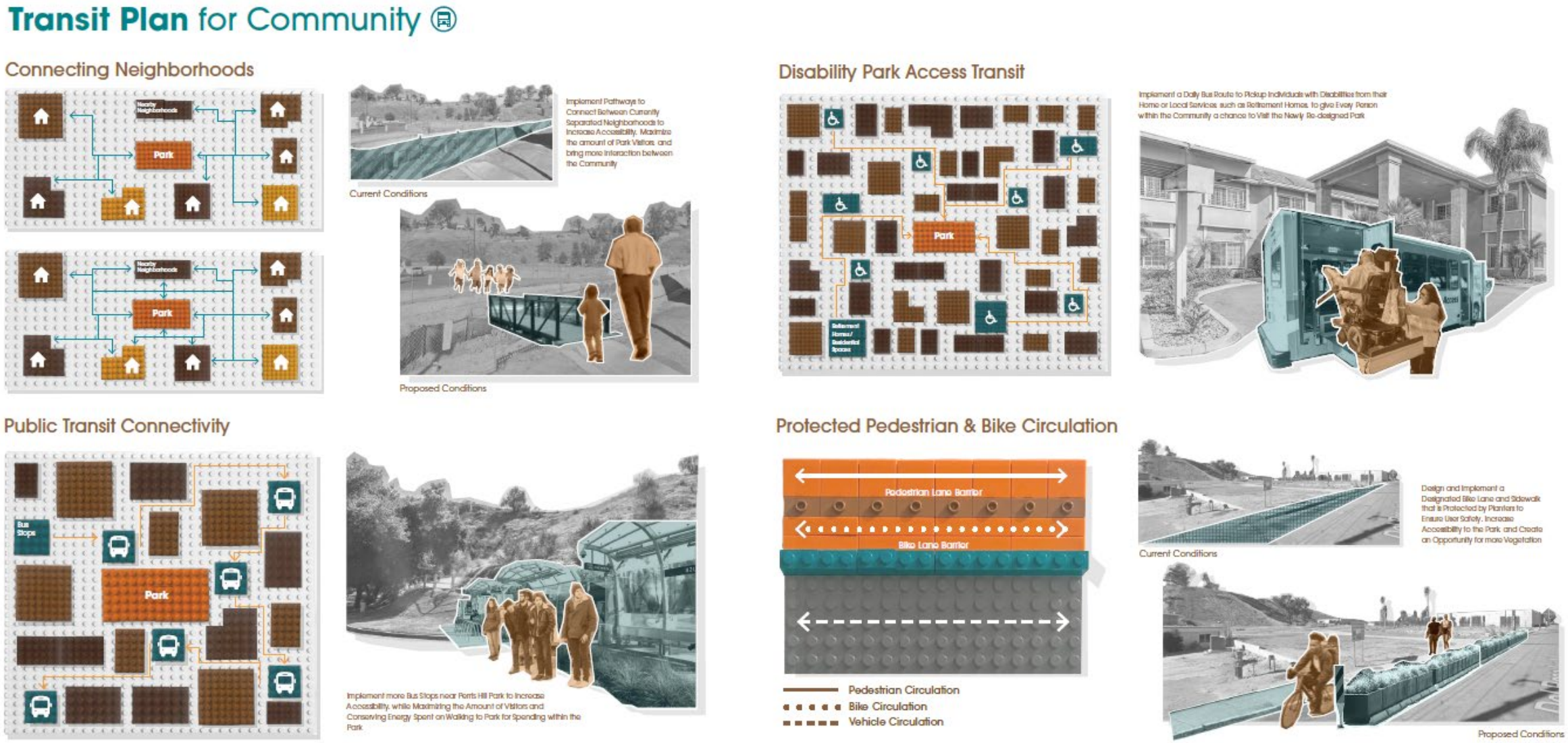


Multi-Route Trail:

- Steep Trail for Faster / Harder Circulation
- Dispersed Trail for Easier / Gradual Circulation
- More opportunity for Integrated Features and Scenic Views
- Second option with under 5% slope, making an ADA Compliant and Easily Accessible Trail Network

Credit: Carlos Holguin, UNLV BLA 25'

Student Work: UNLV LAND 486 Capstone Project



Credit: Carlos Holguin, UNLV BLA 25’

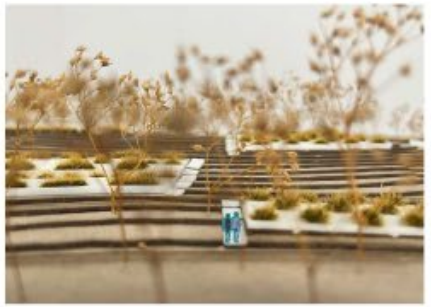
Student Work: UNLV LAND 486 Capstone Project

Credit: Carlos Holguin, UNLV BLA 25’

Student Work: UNLV LAND 486 Capstone Project

Physical Model Analysis: Pt 1

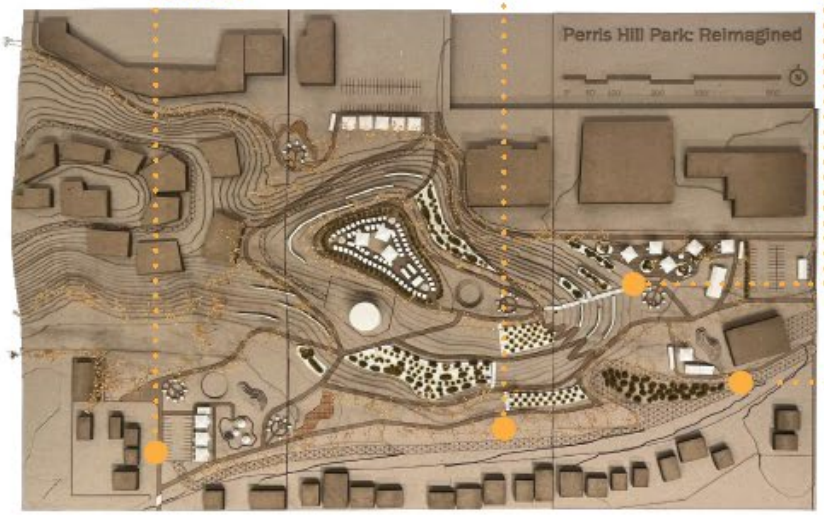
South-West Parking lot area with nearby shade structures, playground, public peace space, newly planned vegetation, and trail network system



Trail Network System with optional stair pathway with scenic views of newly planned surrounding vegetation, leading to Housing Community

Optional stair pathway connecting Public Dining Space with the Housing Community, creating an easy connection to the self sustaining community within

Public Transit Center and Landfill Collection Center with public bathroom and shaded seating nearby to accommodate any users waiting for resources



Overview displaying the self-sustaining Housing Community and the connection of the Trail Network System throughout the park

Public Dining Space and the nearby Public Peace Space, to offer resources of safety and gathering for the community

Credit: Carlos Holguin, UNLV BLA 25’

Student Work: UNLV LAND 486 Capstone Project

Physical Model Analysis: Pt 2

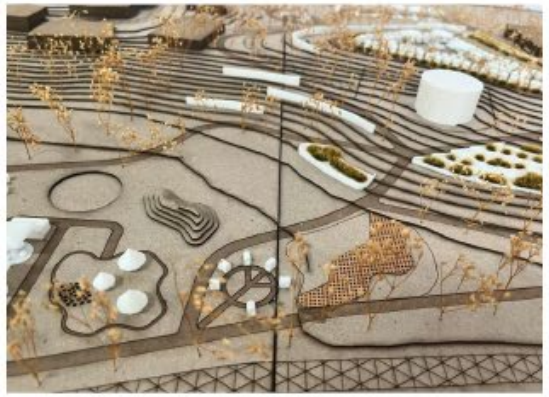
Nature Play Space to offer a variety of interactive circulation, with nearby Public Peace Space, as well as a shaded dining and seating space

Amenities offered within the Housing Community, such as the Public Peace Space and a Topographical Conversation Pit, optimal for community events as well

Public Dining Space with structural and natural canopy coverage, connected with the food preparation center, to offer resources for the Housing Community

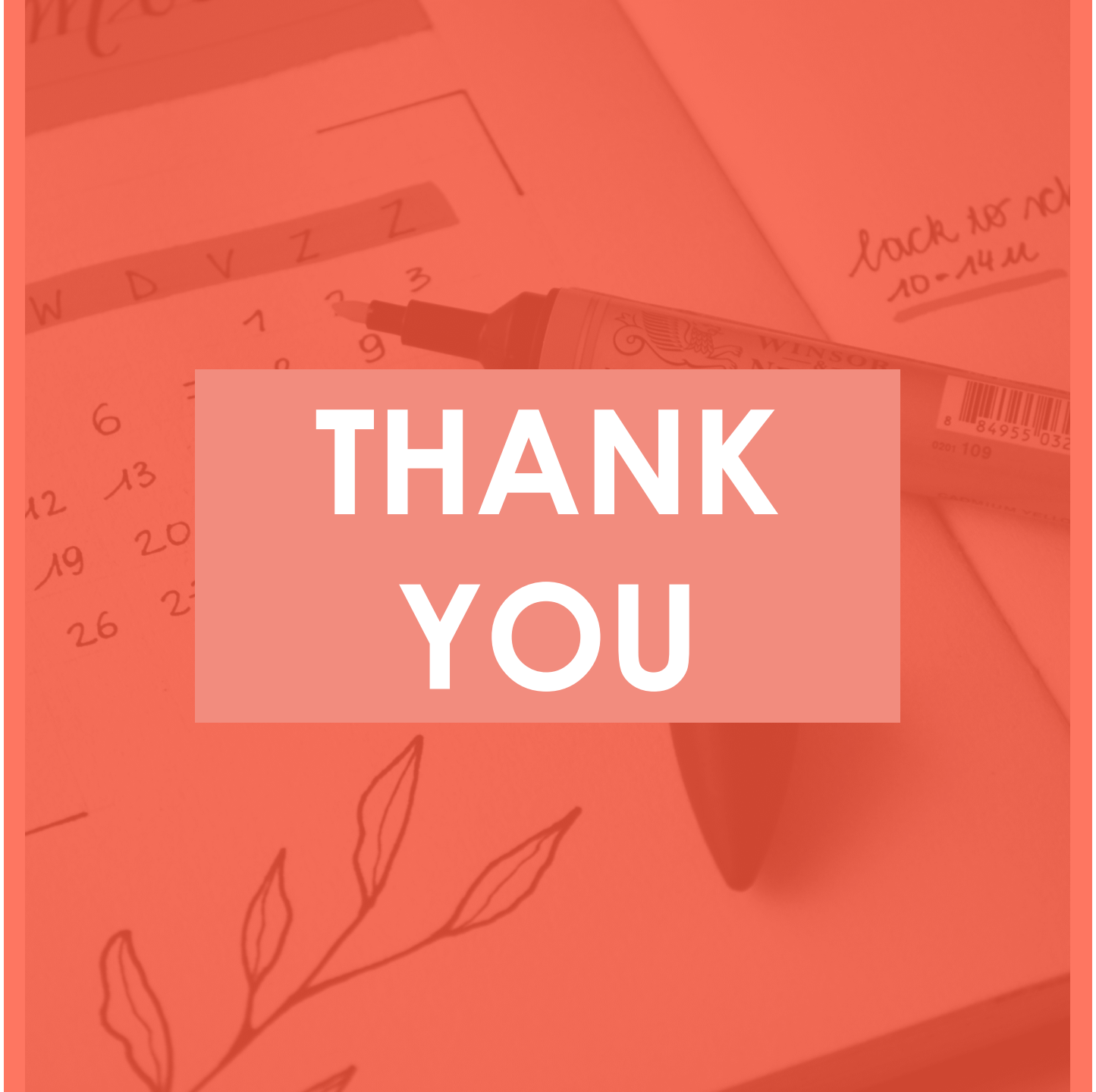
Outlook View from the Public Transit Space, displaying the tiered garden beds along the hills, surrounded by newly planned vegetation

Overview Displaying the connection between the Food Drop-Off Station, North-East Parking Lot, Public Dining Space, and the Housing Community



Overview displaying one of the community centered spaces, offering experiences for all park visitors, such as a playground, nature play space, Trail Network, Public Peace Space, scenic views, and rest spaces

Credit: Carlos Holguin, UNLV BLA 25’


THANK
YOU